
Vector Bundles and K-Theory

— *EYNTKA* Series —

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August 22, 2026

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0.0.1 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7★; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

K -theory begins with a deceptively simple idea: rather than study a space X directly, study the vector bundles that can be laid over it. A vector bundle is a continuously varying family of vector spaces parametrized by X , and the collection of all of them, taken up to isomorphism, already remembers a great deal about the shape of X . Under direct sum these bundles form a commutative monoid; completing that monoid to a group, exactly as the integers are built from the natural numbers, produces the Grothendieck group $K^0(X)$. This single construction turns the geometry of bundles into an algebraic invariant that is computable, functorial, and sensitive to phenomena that ordinary cohomology cannot detect.

What makes topological K -theory more than a convenient bookkeeping device is *Bott periodicity*: the groups $K^n(X)$ repeat with period two, and this periodicity promotes K^0 and K^1 into a full generalized cohomology theory, one satisfying every Eilenberg–Steenrod axiom except the dimension axiom ([13]). A reader who knows ordinary cohomology will find in K -theory a close structural parallel, with its own long exact sequences, products, and computational machinery, but with answers ordinary cohomology cannot see. The same circle of ideas produces the *characteristic classes* (Stiefel–Whitney, Chern, Euler, Pontryagin), cohomology classes attached to a bundle that obstruct sections, detect orientability, and measure twisting, turning qualitative geometry into numbers.

This book develops the topological theory from vector bundles through Bott periodicity to characteristic classes. It is the topological member of a family. Its algebraic counterpart, where rings replace spaces and projective modules replace bundles, is [4]; its operator-algebraic counterpart, where the Serre–Swan theorem reincarnates bundles as projections over $C(X)$, is [5]. The non-periodicity of the algebraic theory and the 2-periodicity shared by the topological and operator theories are two sides of one story, and the reader is encouraged to read across the three.

We assume the cohomology, classifying spaces, and generalized-cohomology framework of [13], and use the connections and curvature of [15], where characteristic classes are computed by the Chern–Weil theory of the Riemannian-geometry volume `NateRiemannianGeometry`. Beyond these, the prerequisites are point-set topology and linear algebra.

Vector Bundles

1.0.1 Convention: Paracompact Includes Hausdorff Every base space of a bundle in this book is assumed *paracompact*, and throughout the word means *Hausdorff*, together with the condition that every open cover admits a locally finite open refinement. Building the separation axiom into the word is the convention of both standard references for this material, [12, §5.8] and the definition following Proposition 1.2 of [10], and it is not a formality. The constructions this book runs on paracompact bases are partitions of unity: proposition 1.1.14 builds an inner product with one and theorem 1.2.13 builds a classifying map with one. Partitions of unity subordinate to an arbitrary open cover exist on a paracompact space because it is normal, and normality is what fails without the separation axiom ([8, Paracompact Hausdorff Spaces Are Normal]).

Topology [8] defines paracompactness by the covering condition alone, so a result cited from there is applied here with the Hausdorff hypothesis supplied separately at the point of use. Manifolds are Hausdorff under the same convention. Schemes are not, and never appear here as bundle bases^a

^aSee [3] for why only trivial schemes are Hausdorff; the scheme-theoretic side of K-theory is [4].

1.1 Fiber Bundles, Principal Bundles, Vector Bundles

Definition 1.1.1: Fiber Bundle

Let B be a paracompact topological space. Then a quadruple (B, E, π, F) where E is a topological space called the *total space*, $\pi : E \rightarrow B$ is a continuous surjection called the *bundle projection*, and B is called the *base space*, and F is the fiber is called *fiber bundle* if:

1. (regular fibers) each $\pi^{-1}(b)$ for $b \in B$ is isomorphic to F , or in particular

$$\pi^{-1}(b) \cong b \times F$$

2. (local trivialization) For each $b \in B$, there exists an open set $U \subseteq B$ containing b such that the following diagram commutes

$$\begin{array}{ccc} \pi^{-1}(U) & \xrightarrow{\sim} & U \times F \\ \pi \downarrow & \swarrow \pi_1 & \\ U & & \end{array}$$

When clear from context, the fiber bundle is usually denoted as E .

Definition 1.1.2: Vector Bundle

If $F = V$ is a vector space and each fiber-morphism in the local trivialization is a linear isomorphism, then is called a *vector bundle* (over B). If the vector space is real or complex, the vector bundle can be called a *real vector bundle* *complex vector bundle*.

Definition 1.1.3: Principal Bundle

If $F = G$ is a topological and $P \rightarrow X$ is a fiber bundle over X group where there is a free and transitive continuous right group action $P \times G \rightarrow P$ that preserves the fibers of P , then it is called a *principal G -bundle* (or *principal bundle* if G is clear from context).

Each fiber is called a *G -torsor* (which is a space that where the action $G \curvearrowright F_x$ has trivial stabilizer, hence “is” the group G but does not intrinsically have a notion of identity element)

Note that these conditions make each fiber homeomorphic to G . As a consequence the Orbit space P/G is homeomorphic to X .

Example 1.1.4: Fiber Bundle

1. Given a topological B and fiber F , then $E = B \times F$ is the *trivial F -fiber bundle* or simply

the *trivial bundle*. Such a bundle has a *global trivialization*, namely by letting $U = B$:

$$\begin{array}{ccc} B \times F & \xrightarrow{\text{id}} & B \times F \\ \pi \downarrow & \swarrow \pi_1 & \\ B & & \end{array}$$

2. If M is a n -manifold, then there are various interesting vector bundles defined on it:

- (a) Tangent bundle TM
- (b) cotangent bundle T^*M
- (c) differential forms $\Omega^k(M)$ for $k \leq n$

If M has more structure, ex. if is a Riemann manifold, a Khäler manifold, has a complex manifold, etc, then we can define more vector bundles that capture the extra structure (ex. Normal bundles, holomorphic bundles, spin bundles, flat bundles, etc.)

- 3. Let $p : E \rightarrow B$ be a fiber bundles and $A \subseteq B$. Then $p : p^{-1}(A) \rightarrow A$ is certainly a fiber bundle. This is called the *restriction of E over A* . The same goes through for vector-bundles.
- 4. Sub-bundle shave compliments: if $E \rightarrow X$ is a bundle (on a paracompact topological space) and F is a subbundle, then there exists a bundle $F^\perp$ such that $E = F \oplus F^\perp$.
- 5. the Tensor product is well-defined: Given a vector bundle $E \rightarrow X$ and $F \rightarrow X$, there exits a vector-bundle $E \otimes F \rightarrow X$ whose fibers over x are $E_x \otimes F_x$. From this, we get our usual more general bundles such as the *determinant bundle* $\det(E) = \Lambda^n E$ (given E is n -dimensional).
- 6. Let $E = I \times \mathbb{R} / \sim$ where $(0, t) \sim (1, -t)$. Then the projection map $I \times \mathbb{R} \rightarrow I$ produces a quotient $p : E \rightarrow S^1$ known as the Möbius bundle
- 7. Let TS^n be the tangent bundle of S^n . Then notice that this can be taken as a sub-bundle of $T\mathbb{R}^{n+1}$.
- 8. Another important bundle on S^n is the normal bundle which is a line bundle that that is where we take the one dimensional vector space panned by the line perpendicular to each point $p \in S^n$. We shall see that this bundle is isomorphic to the trivial bundle later. If NS^n is the normal bundle, we shall see that $TS^n \oplus NS^n$ is a trivial bundle.
- 9. Let $\mathbb{RP}^n$ and $\mathbb{CP}^n$ be real and complex projective space respectively. Then there is a *canonical line bundle* $p : E \rightarrow \mathbb{RP}^n$ and $p : E \rightarrow \mathbb{CP}^n$ where E is the subspace of $\mathbb{RP}^n \times \mathbb{R}^{n+1}$ consisting of pairs (ℓ, v) where $v \in \ell$ and $p(\ell, v) = \ell$. Note that the canonical line bundle over $\mathbb{RP}^n$ has an orthogonal compliment

$$E^\perp = \{(\ell, v) \in \mathbb{RP}^n \times \mathbb{R}^{n+1} \mid v \perp \ell\}$$

- 10. An important bundle for the future is the infinite real/complex projective space $\mathbb{RP}^\infty$ and $\mathbb{CP}^\infty$ taking by $\varinjlim_n \mathbb{RP}^n$ and $\varinjlim_n \mathbb{CP}^n$.
- 11. The Grassmann manifold $G_k(\mathbb{R}^n)$ was shown in [15] to be the space of all k -dimensional planes through the origin of $\mathbb{R}^n$.

Definition 1.1.5: Morphism of Fiber Bundles

If E_1, E_2 are two vector bundles, $\varphi : E_1 \rightarrow E_2$ is a vector-bundle morphism is a continuous map where this diagram commutes:

$$\begin{array}{ccc} E_1 & \xrightarrow{\varphi} & E_2 \\ & \searrow & \swarrow \\ & B & \end{array}$$

and it respects local trivialization; this includes a linear homomorphism on each fiber. An isomorphism would have φ be a homeomorphism and is a linear isomorphism on each fiber.

For each dimension n , let $\text{Vect}_k^n(B)$ be the isomorphism classes of vector bundles over B (with field k).

For a topological group G , let $b_G(B)$ be the isomorphism classes of principal G -bundles over B .

Often, if $k = \mathbb{R}$, we shall just write $\text{Vect}^n(B)$. Note that $\text{Vect}_k^n(B)$ and $b_G(B)$ are contravariant functors by applying the pullback map, for example:

$$f : X \rightarrow Y \quad \Rightarrow \quad f^* : b_G(Y) \rightarrow b_G(X)$$

Lemma 1.1.6: Morphism is Fiber-local

Let $p_1 : E_1 \rightarrow B, p_2 : E_2 \rightarrow B$ be two vector bundles over B and $\varphi : E_1 \rightarrow E_2$ a continuous. Then φ is an isomorphism if and only if it takes each fiber $p_1^{-1}(b)$ to the corresponding fiber $p_2^{-1}(b)$ via a linear isomorphism.

Proof:
exercise.

Example 1.1.7: Morphism of Bundles

We can see that the normal bundle NS^n as a subspace of $\mathbb{R}^1$ is isomorphic to $S^n \times \mathbb{R}$ via the map:

$$(x, tx) \rightarrow (x, t)$$

We shall later see that only S^1, S^3, S^7 have trivial line bundles (this shall correspond to the existence of division algebra's over $\mathbb{R}$ existing only for dimension 2, 4, 8, see [ref:HERE](#)). For S^1 , the isomorphism is simple enough as we are comfortable with complex analysis:

$$\varphi : TS^1 \rightarrow S^1 \times \mathbb{R} \quad (e^{i\theta}, ite^{i\theta}) \mapsto (e^{i\theta}, t)$$

Another important example is that the Möbius bundle is isomorphic to the canonical line bundle over $\mathbb{RP}^1$ (note that $\mathbb{RP}^1 \cong S^1$).

Definition 1.1.8: Section

Let $p : E \rightarrow B$ be a fiber bundle. Then a *section* is a continuous map $s : B \rightarrow E$ such that $p \circ s = \text{id}$.

We shall focus on sections of vector bundles. We immediately get the following

Definition 1.1.9: Zero Section

Let $p : E \rightarrow B$ be a vector bundle. Then the *zero section* is the section $z : B \rightarrow E$ given pointwise by $z(p) = 0 \in p^{-1}(b)$.

As each vector-bundle morphism on fibers is linear, the zero section is taken to the zero section. This can be taken advantage of: for example the Möbius bundle is not isomorphic to $S^1 \times \mathbb{R}$ since the compliment of the zero section is connected in the Möbius bundle but not for the product bundle. The other type of section is a *non-vanishing section* where no elements of the section are zero. Famously, by the Hairy-ball theorem (or Hedgehog theorem), there does not exist a section of TS^2 (i.e a vector-field) that is everywhere nonzero (see [13]). We shall see in [ref:HERE](#) that S^n has a nonvanishing vector field if and only if n is odd, and so we shall see that the tangent bundle of S^n cannot be isomorphic to the trivial bundle if $n > 0$ is even. The existence of zero's shall be really important, for example it shall be key in how to interpret the Euler characteristic (see [ref:HERE](#)).

Lemma 1.1.10: Characterizing Trivial Bundles using Sections

Let $p : E \rightarrow B$ be an n -dimensional vector bundle. Then p is trivial if and only if it has n sections $s_1, \dots, s_n$ such that $(s_1(b), \dots, s_n(b))$ are linearly independent on each fiber $p^{-1}(b)$.

Proof:

A trivial n -dimensional vector bundle certainly respects these properties. Conversely, given such n section $s_1, \dots, s_n$, define:

$$h : B \times \mathbb{R}^n \rightarrow E \quad h(b, t_1, \dots, t_n) \rightarrow \sum_i t_i s_i(b)$$

this is certainly a linear isomorphism on each fiber, hence by lemma 1.1.6 it is an isomorphism, completing the proof.

Note that another way of characterizing a trivial real bundle is to say there is a projection map $E \rightarrow \mathbb{R}^n$ that is a linear isomorphism on each fiber.

Using this lemma, we can show that S^3 has a trivial line bundle by relying on the Quaternions. Recall that any quaternion $z \in \mathbb{H}$ is given by:

$$z = x_1 + ix_2 + jx_3 + kx_4$$

$$i^2 = j^2 = k^2 = -1, \quad ij = k, \quad jk = i, \quad ki = j, \quad ji = -k, \quad kj = -i, \quad ik = -j$$

By identifying $\mathbb{H}$ with $\mathbb{R}^4$ via the coordinates (x_1, x_2, x_3, x_4) , then the unit sphere S^3 can be given

by the three sections of the tangent bundle:

$$\begin{aligned} z \mapsto iz & \quad \text{or} \quad (x_1, x_2, x_3, x_4) \mapsto (-x_2, x_1, -x_4, x_3) \\ z \mapsto jz & \quad \text{or} \quad (x_1, x_2, x_3, x_4) \mapsto (-x_3, x_4, x_1, -x_2) \\ z \mapsto kz & \quad \text{or} \quad (x_1, x_2, x_3, x_4) \mapsto (-x_4, -x_3, x_2, x_1) \end{aligned}$$

The key idea is that the quaternions $1, i, j, k$ form the standard orthonormal basis for $\mathbb{R}^3$, multiplying by an arbitrary unit quaternion $z \in S^3$ gets another orthonormal basis z, iz, jz, kz , and since $|zw| = |z| \cdot |w|$ (where $|\cdot|$ is the usual norm in $\mathbb{R}^3$), we get that multiplication by a quaternion in the unit sphere S^3 is an isometry of $\mathbb{H}$. A similar process can be done for the Cayley Octonions by thinking of $\mathbb{R}^8$ as $\mathbb{H} \times \mathbb{H}$ and multiplying the Octonions by:

$$(z_1, z_2)(w_1, w_2) = (z_1 w_2 - \overline{w_2} z_2, z_2 \overline{w_1} + w_2 z_1)$$

This also defines a division algebra, and also satisfies $|zw| = |z||w|$, giving 7 orthogonal tangent vectors fields on S^7 that are nowhere vanishing, and hence S^7 is also trivial. We shall show in [ref:HERE](#) that these are the only ones.

Definition 1.1.11: Direct Sum of Vector Bundles

Let $p_1 : E_1 \rightarrow B$, $p_2 : E_2 \rightarrow B$ be vector bundles over B . Then the *direct sum* or *coproduct* (historically called *Whitney sum*) of E_1, E_2 is:

$$E_1 \oplus E_2 = \{(v_1, v_2) \in E_1 \times E_2 \mid p_1(v_1) = p_2(v_2)\}$$

Certainly the fibers of the direct sum is the direct sum of the fibers. An interesting way of thinking of the direct sum is by taking the product of the two vector bundles (including the base space)

$$E_1 \times E_2 \rightarrow B \times B$$

Then the restriction to the product to the diagonal $\{(b, b) \in B \times B\}$ is $E_1 \times E_2$. Note that just like for vector spaces, the (categorical) product and coproduct are isomorphic, however the notation $E_1 \times E_2$ is sometimes be used in literature to also mean the multiplication of the base-space, a convention we shall void.

Definition 1.1.12: Stably Trivial

Let E be a vector bundle. Then E is *stably trivial* if there exists a vector bundle F such that $E \oplus F$ is the trivial bundle.

We saw that the tangent bundle S^n is stably trivial, with NS^n being the compliment. Similarly, $E \oplus E^\perp$ where E is the canonical line bundle with its orthogonal compliment is isomorphic to $\mathbb{R}P^n \times \mathbb{R}^{n+1}$ via

$$(\ell, v, w) \mapsto (\ell, v + w)$$

In the $n = 1$ case, $E^\perp$ is isomorphic to E by rotating each vector in the plane by $\pi/2$ degrees. As mention earlier, the canonical line bundle E in the $n = 1$ case is isomorphic to the Möbius bundle. Hence, the direct sum of the Möbius bundle with itself is the trivial bundle! Take a moment to think about this one as the “untwisting” is not an isometry and will have self-intersection.

Example 1.1.13: Stable Triviality of $\mathbb{RP}^n$

By the above discussion, let $S^n \times \mathbb{R}^{n+1} \cong TS^n \oplus NS^n$. Define the equivalence (i.e. quotient by)

$$(x, v) \sim (-x, -v)$$

on both sides of the isomorphism. Then when applied to TS^n , we get $T\mathbb{RP}^n$ (see [15], or even [3]). This is not preserved by the isomorphism (general quotient usually doesn't), however we do get the closest next thing: what we will show is a special case of the *Direct Sum decomposition for compact space* that will show that $T\mathbb{RP}^n$ is a direct sum of nontrivial line bundles. We will show this by simply directly computing the two sides.

In NS^n , the identification $(x, v) \sim (-x, -v)$ is explicitly given by $(x, tx) \sim (-x, t(-x))$. This yields the product bundle $\mathbb{RP}^n \times \mathbb{R}$ as $x \mapsto (-x, -x)$ is well-defined in the quotient. Now the identification $(x, v) \sim (-x, -v)$ in $S^n \times \mathbb{R}^{n+1}$ respects the coordinate macros of $\mathbb{R}^{n+1}$, so the quotient is the direct sum of $n+1$ copies of line bundles E over $\mathbb{RP}^n$ given by $(x, t) \sim (-x, -t)$ in $S^n \times \mathbb{R}$. It is now easy to see that E is the canonical line bundle over $\mathbb{RP}^n$: identifying $S^n \times \mathbb{R}$ via the isomorphism $(x, t) \rightarrow (x, tx)$, we have $(-x, -t) \mapsto ((-x, (-t)(-x)) = (-x, tx)$ and so we have the gluing in NS^n that is the canonical line bundle over $\mathbb{RP}^n$.

Hence, the direct sum of the tangent bundle $T\mathbb{RP}^n$ with a trivial line bundle is isomorphic to $n+1$ copies of the canonical bundle over $\mathbb{RP}^n$. When $n=3$, $T\mathbb{RP}^3$ is trivial (as shown earlier), and so we get four copies of the canonical line bundle over $\mathbb{RP}^3$ is trivial (and similarly for the canonical line bundle over $\mathbb{RP}^7$). We shall fully explore these properties in section [ref:HERE](#).

As we are working with vector bundles, we can define an inner product:

$$\langle -, - \rangle_{E \oplus E} \rightarrow \mathbb{R}$$

where E is a vector bundle over B . As we are over a paracompact space, we always have such an inner product

Proposition 1.1.14: Existence of Inner Product

Let $p : E \rightarrow B$ be a (real) vector bundle (over a paracompact space). Then there exists an inner product $\langle -, - \rangle$ on E

the complex inner product (also named Hermitian inner product) can be defined similarly.

Proof:

This is the same proof as for the existence of a Riemann metric on a differentiable manifold. Let $\{U_\alpha\}$ be an open cover of B given by local trivializations. These can then pull-back the standard inner product on $\mathbb{R}^n$, and then by setting:

$$\langle v, w \rangle = \sum_{\beta} \varphi_{\beta} p(v) \langle v, w \rangle_{\alpha(\beta)}$$

where $\{\varphi_{\beta}\}$ is a partition of unity subordinate to $\{U_{\alpha}\}$ with support of φ_{β} contained in $U_{\alpha(\beta)}$.

Vector bundles are nice enough so satisfy the complementary property:

Proposition 1.1.15: Vector Bundles Satisfy Complimentary Property

Let $E \rightarrow B$ be a vector bundle (over a paracompact space B) and let $E_0 \subseteq E$ be a vector subbundle. Then there exists a vector subbundle $E_0^\perp \subseteq E$ such that

$$E_0 \oplus E_0^\perp \cong E$$

Proof:

Let E have dimension n . Choose an inner product on E . Let $E_0^\perp$ be the subspace of E which in each fiber is the orthogonal to vectors in E_0 . First, the natural projection $E_0^\perp \rightarrow B$ is a vector bundle. If this is so, then $E_0 \oplus E_0^\perp$ is isomorphic to E via the map $(v, w) \mapsto v + w$ by lemma 1.1.10.

To see that $E_0^\perp$ satisfies the local triviality condition, note that we may assume E is the product $B \times \mathbb{R}^n$ as the question is local in B . As E_0 is a vector bundle, it has some $m \leq n$ independent local sections $b \mapsto (b, s_i(b))$ near each point $b_0 \in B$. This set of m independent local sections of E_0 can be enlarged to a set of n independent local sections $b \mapsto (b, s_i(b))$ of E by choosing $s_{m+1}, \dots, s_n$ first in the fiber $p^{-1}(b_0)$, then taking the same vectors for all nearby fibers, since if $s_1, \dots, s_m, s_{m+1}, \dots, s_n$ are independent at b_0 , they will be independent near b by continuity of the determinant function. Applying the Gram-Schmidt orthogonalization process to $s_1, \dots, s_m, s_{m+1}, \dots, s_n$ in each fiber we obtain new sections s'_i . Note that each s'_i is continuous as they were obtained via the explicit formulas for the Gram-Schmidt process which is certainly a continuous process, and the first m of them are a basis for E_0 in each fiber. The sections s'_i allow us to define a local trivialization $h : p^{-1}(U) \rightarrow U \times \mathbb{R}^n$ with $h(b, s'_i(b))$ equal to the i^{th} standard basis vector of $\mathbb{R}^n$. Now, this h carries E_0 to $U \times \mathbb{R}^m$ and $E_0^\perp$ to $U \times \mathbb{R}^{n-m}$, so $h|_{E_0^\perp}$ is a local trivialization of $E_0^\perp$, as we sought to show.

When $E_0 = E$, then this proof also shows that local trivializations are also isometries. When B is compact, then we get that vector bundles are “projective”

Proposition 1.1.16: Vector Bundle Over Compact Are Projective

Let $E \rightarrow B$ be a vector bundle over a compact Hausdorff space B . Then there exists a vector bundle $E' \rightarrow B$ such that $E \oplus E'$ is the trivial bundle

Note this fails for non-compact spaces, we shall show this for $\mathbb{R}P^n$ in example 3.2.6(3).

Proof:

Suppose first this holds, so E is a subbundle of a trivial bundle $B \times \mathbb{R}^N$. Composing the inclusion of E into this product with the projection of the product onto $\mathbb{R}^N$ yields a map $E \rightarrow \mathbb{R}^N$ that is a linear injection on each fiber. We shall show such a map exists by first constructing a map $E \rightarrow \mathbb{R}^N$ that is a linear injection on each fiber, then showing that this gives an embedding of E in $B \times \mathbb{R}^N$ as a direct summand.

To that end, each point $x \in B$ has an open neighborhood U_x over which E is trivial. By Urysohn’s Lemma there is a map $\varphi_x : B \rightarrow [0, 1]$ that is nonzero at x and has support contained in U_x . Letting x vary, the sets $\varphi_x^{-1}(0, 1]$ form an open cover of B . By compactness, there exists finite subcover. Relabel the corresponding U_x ’s and φ_x ’s as U_i and φ_i . Define $g_i : E \rightarrow \mathbb{R}^n$ by

$g_i(v) = \varphi_i(p(v))(\pi_i h_i(v))$ where p is the projection $E \rightarrow B$ and $\pi_i h_i$ is the composition of a local trivialization $h_i : p^{-1}(U_i) \rightarrow U_i \times \mathbb{R}^n$ with the projection π_i to $\mathbb{R}^n$. Then g_i is a linear injection on each fiber over $\varphi_i^{-1}(0, 1]$, so making the various g_i 's the coordinates of a map $g : E \rightarrow \mathbb{R}^N$ with $\mathbb{R}^N$ a product of copies of $\mathbb{R}^n$, then g is a linear injection on each fiber.

The map g is the second coordinate of a map $f : E \rightarrow B \times \mathbb{R}^N$ with first coordinate p . The image of f is a subbundle of the product $B \times \mathbb{R}^N$ since projection of $\mathbb{R}^N$ onto the i th $\mathbb{R}^n$ factor gives the second coordinate of a local trivialization over $\varphi_i^{-1}(0, 1]$. Thus, E isomorphic to a subbundle of $B \times \mathbb{R}^N$ so by the preceding proposition there is a complementary subbundle E' with $E \oplus E'$ isomorphic to $B \times \mathbb{R}^N$, as we sought to show.

Next, we define the tensor product.

Definition 1.1.17: Tensor Product of Vector Bundles

Let $p_1 : E_1 \rightarrow B$, $p_2 : E_2 \rightarrow B$ be vector bundles over B . Then *tensor product* is a vector bundle where as the set

$$E_1 \otimes E_2 = \coprod_{x \in B} p_1^{-1}(x) \otimes p_2^{-1}(x)$$

and for the topology, choose an open cover $\{U_\alpha\}$ with local trivializations $h_i : p_i^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^{n_i}$, then the topology $\mathcal{T}_U$ on $p_1^{-1}(U) \otimes p_2^{-1}(U)$ is given by letting the fiberwise tensor product map:

$$h_1 \otimes h_2 : p_1^{-1}(U) \otimes p_2^{-1}(U) \rightarrow U \times (\mathbb{R}^{n_1} \times \mathbb{R}^{n_2})$$

be a homeomorphism.

This topology is independent of the choice of local trivialization cover as any other choice is will be obtained by composing with isomorphism of $U \times \mathbb{R}^{n_i}$ of the form:

$$(x, v) \mapsto (x, g_i(x)(v))$$

where $g_i : U \rightarrow \text{GL}_{n_i}(\mathbb{R})$. Thus, $h_1 \otimes h_2$ changes by composing with analogous isomorphism of $U \times (\mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2})$ whose second coordinate $g_1 \otimes g_2$ are continuous maps

$$U \rightarrow \text{GL}_{n_1 n_2}(\mathbb{R})$$

If the reader is familiar with the fibered product of schemes, or is comfortable with the more “categorical” approach, then we may also define the tensor product by first defining the vector-bundle as the gluing of $\coprod_\alpha (U_\alpha \times \mathbb{R}^n)$ where on intersections we have the gluing condition:

$$g_{\beta\alpha} : U_\alpha \cap U_\beta \rightarrow \text{GL}_n(\mathbb{R})$$

that satisfy the cocycle condition on $U_\alpha \cap U_\beta \cap U_\gamma$

$$g_{\gamma\beta} g_{\beta\alpha} = g_{\gamma\alpha}$$

Then the tensor product between $E_1 \rightarrow B$ and $E_2 \rightarrow B$ is given by choosing an open cover $\{U_\alpha\}$ that is locally trivial for both E_1, E_2 and getting gluing functions $g_{\beta\alpha}^i : U_\alpha \cap U_\beta \rightarrow \text{GL}_{n_i}(\mathbb{R})$ for each E_i . Then the tensor bundle $E_1 \otimes E_2$ is the tensor product of the functions $g_{\beta\alpha}^1 \otimes g_{\beta\alpha}^2$ assigning to each $x \in U_\alpha \cap U_\beta$ the tensor product of the matrices.

The key is that it is a fiberwise product. Hence, over a fixed base we get that it is commutative, associate, has an identity element (is the trivial bundle), and distributive with direct sum. If we

only look at line bundles, then the tensor product of two line bundles is a line bundle. In that case, $\text{Vect}^1(B)$ (the isomorphism classes of line bundles) is an abelian group with respect to the tensor product; the inverse is giving by inverses matrices $g_{\beta\alpha}(x) \in \text{GL}_1(\mathbb{R})$ and the cocycle condition is preserved as 1 by 1 matrices commute. Giving E an inner product, we can re-scale the local trivialization h_α to be isometries, which then the vectors in fibers of E go to vectors in $\mathbb{R}^1$ of the same length. Thus, all values of the gluing functions $g_{\beta\alpha}$ are ± 1 . Then the gluing function for $E \otimes E$ are the squares of these $g_{\beta\alpha}$, and hence are identically 1; thus every element of $\text{Vect}^1(B)$ is its own inverse. This should remind the reader of the cup product in algebraic topology, and indeed in [ref:HERE](#) we shall see that $\text{Vect}^1(B)$ is isomorphic to the cohomology group $H^1(B; \mathbb{Z})$ when B is homotopy equivalent to a CW complex.

The same idea goes through for complex vector bundles but with some different behaviour: the gluing functions $g_{\beta\alpha}$ will have norm 1 (not just ± 1 , so it is not $E \otimes E$ need not be trivial. We shall see that $\text{Vect}_{\mathbb{C}}^1(B) \cong H^2(B; \mathbb{Z})$ when B is homotopy equivalent to a CW complex. However, if we take the *conjugate bundle* $\bar{E} \rightarrow B$ defined in the natural way, then as $z\bar{z} = 1$ when $|z| = 1$, $E \otimes \bar{E}$ is the trivial line bundle.

With the tensor product, we can then also define the symmetric and alternating bundle, which also gives us the determinant bundle.

1.2 Classifying Vector Bundles

Proposition 1.2.1: Pullback of Vector bundle

Let $f : A \rightarrow B$ be a continuous map and $p : E \rightarrow B$ a vector bundle. Then there exists a vector bundle $p' : E' \rightarrow A$ with a map $f' : E' \rightarrow E$ taking the fibers of E' over each $a \in A$ isomorphically onto the fibers of E over $f(a)$. The vector bundle E' is unique up to isomorphism.

Thus, we get the pullback map $f^* : \text{Vect}(B) \rightarrow \text{Vect}(A)$ mapping isomorphism classes of E to isomorphism classes of $f^*(E) = E'$.

Proof:

Let

$$E' = \{(a, v) \in A \times E \mid f(a) = p(v)\}$$

This subspace of $A \times E$ fits into the following commutative diagram

$$\begin{array}{ccc} E' & \xrightarrow{f'} & E \\ p' \downarrow & & \downarrow p \\ A & \xrightarrow{f} & B \end{array}$$

where $p'(a, v) = a$ and $f'(a, v) = v$. Let Γ_f denote the graph of f in $A \times B$. Then p' factors as the composition $E' \rightarrow \Gamma_f \rightarrow A$, $(a, v) \mapsto (a, p(v)) = (a, f(a)) \mapsto a$. The first of these two maps is the restriction of the vector bundle $\mathbb{K} \times p : A \times E \rightarrow A \times B$ over the graph Γ_f , so it is a vector bundle, and the second map is a homeomorphism, so their composition $p' : E' \rightarrow A$ is a vector bundle. The map f' obviously takes the fiber E' over a isomorphically onto the fiber of E over $f(a)$, showing existence.

For uniqueness, construct an isomorphism from an arbitrary E' satisfying the conditions in the proposition to the particular one just constructed by sending $v' \in E'$ to the pair $(p'(v'), f'(v'))$. This map takes each fiber of E' to the corresponding fiber of $f^*(E)$ by a vector space isomorphism, so by lemma 1.1.10 it is an isomorphism of vector bundles, completing the proof.

When restricting to Vect^1 , notice the trivial bundle pulls back to the trivial bundle and the tensor product is well-behaved; hence this pullback becomes a group homomorphism.

Example 1.2.2: Vector Bundle Pullbacks

1. Let $A \subseteq B$ so that $\iota : A \hookrightarrow B$. Then the restriction of E over B to A is the pullback $\iota^*(E)$.
2. Let $f : A \rightarrow B$ be a constant map: $f(A) = b$. Then

$$f^*(E) = A \times p^{-1}(b)$$

3. The tangent bundle TS^n is the pullback of $T\mathbb{RP}^n$ via the map $S^n \rightarrow \mathbb{RP}^n$.
4. Consider the Möbius bundle $E \rightarrow S^1$, and take $f : S^1 \rightarrow S^1$ given by $z \mapsto z^2$. By thinking of the Möbius bundle as $S^1 \times \mathbb{R}$ under $(z, t) \sim (-z, -t)$, then $f^*(E)$ is the trivial bundle (think about this for awhile!). If $f(z) = z^n$, then E_n is the trivial bundle if n is even, and the Möbius bundle if n is odd.
5. Let $F(E)$ be the flag bundle of n -dimensions (rest here; [10, p.20]).

Proposition 1.2.3: Properties of Vector-bundle Pullback

Let E be a vector bundle and f, g continuous maps, everything over the appropriate domains. Then:

1. $(fg)^*(E) \cong g^*(f^*(E))$
2. $\text{id}^*(E) \cong E$
3. $f^*(E_1 \oplus E_2) \cong f^*(E_1) \oplus f^*(E_2)$
4. $f^*(E_1 \otimes E_2) \cong f^*(E_1) \otimes f^*(E_2)$

Proof:
exercise

Let us next show that isomorphism classes of vector bundles are invariant under homotopy. We first need a technical lemma:

Lemma 1.2.4: Homotopy Invariant Vector Bundle

Let $E \rightarrow X \times I$ be a vector bundle where X is paracompact. Then the restriction over $X \times \{0\}$ and $X \times \{1\}$ is isomorphic.

Proof:

First two smaller proofs:

1. A vector bundle $p : E \rightarrow X \times [a, b]$ is trivial if its restrictions over $X \times [a, c]$ and $X \times [c, b]$ are both trivial for some $c \in (a, b)$. Indeed, let these restrictions be $E_1 = p^{-1}(X \times [a, c])$ and $E_2 = p^{-1}(X \times [c, b])$, and let $h_1 : E_1 \rightarrow X \times [a, c] \times \mathbb{R}^n$ and $h_2 : E_2 \rightarrow X \times [c, b] \times \mathbb{R}^n$ be isomorphisms. These isomorphisms may not agree on $p^{-1}(X \times \{c\})$, but they can be made to agree by replacing h_2 by its composition with the isomorphism $X \times [c, b] \times \mathbb{R}^n \rightarrow X \times [c, b] \times \mathbb{R}^n$ which on each slice $X \times \{x\} \times \mathbb{R}^n$ is given by $h_1 h_2^{-1} : X \times \{c\} \times \mathbb{R}^n \rightarrow X \times \{c\} \times \mathbb{R}^n$. Once h_1 and h_2 agree on $E_1 \cap E_2$, they define a trivialization of E .
2. For a vector bundle $p : E \rightarrow X \times I$, there exists an open cover $\{U_\alpha\}$ of X where each restriction $p^{-1}(U_\alpha \times I) \rightarrow U_\alpha \times I$ is trivial. Indeed, for each $x \in X$ find open neighborhoods $U_{x,1}, \dots, U_{x,k}$ in X and a partition $0 = t_0 < t_1 < \dots < t_k = 1$ of $[0, 1]$ such that the bundle is trivial over $U_{x,i} \times [t_{i-1}, t_i]$, using compactness of $[0, 1]$. Then by (1) the bundle is trivial over $U_\alpha \times I$ where $U_\alpha = U_{x,1} \cap \dots \cap U_{x,k}$.

Now for the proposition, by (2), choose an open cover $\{U_\alpha\}$ of X so that E is trivial over each $U_\alpha \times I$. Let's assume for now the simpler case that X is compact Hausdorff. Then a finite number of U_α 's cover X . Relabel these as U_i for $i = 1, 2, \dots, m$. As shown in proposition 1.1.14, there is a corresponding partition of unity by functions φ_i with the support of φ_i contained in U_i . For $i \geq 0$, let $\psi_i = \varphi_1 + \dots + \varphi_i$, so in particular $\psi_0 = 0$ and $\psi_m = 1$. Let X_i be the graph of ψ_i , the subspace of $X \times I$ consisting of points of the form $(x, \psi_i(x))$, and let $p_i : E_i \rightarrow X_i$ be the restriction of the bundle E over X_i . Since E is trivial over $U_i \times I$, the natural projection homeomorphism $X_i \rightarrow X_{i-1}$ lifts to a homeomorphism $h_i : E_i \rightarrow E_{i-1}$ which is the identity outside $p^{-1}(U_i \times I)$ and which takes each fiber of E_i isomorphically onto the corresponding fiber of E_{i-1} . Explicitly, on points in $p^{-1}(U_i \times I) = U_i \times I \times \mathbb{R}^n$ let $h_i(x, \psi_i(x), v) = (x, \psi_{i-1}(x), v)$. The composition $h = h_1 h_2 \dots h_m$ is then an isomorphism from the restriction of E over $X \times \{1\}$ to the restriction over $X \times \{0\}$.

In the general case that X is only paracompact, there is a countable cover $\{V_i\}_{i \geq 1}$ of X and a partition of unity $\{\varphi_i\}$ with φ_i supported in V_i , such that each V_i is a disjoint union of open sets each contained in some U_α (see [15]). Then E is trivial over each $V_i \times I$. As before we let $\psi_i = \varphi_1 + \dots + \varphi_i$ and let $p_i : E_i \rightarrow X_i$ be the restriction of E over the graph of ψ_i . Also as before we construct $h_i : E_i \rightarrow E_{i-1}$ using the fact that E is trivial over $V_i \times I$. The infinite composition $h = h_1 h_2 \dots$ is then a well-defined isomorphism from the restriction of E over $X \times \{1\}$ to the restriction over $X \times \{0\}$ since near each point $x \in X$ only finitely many φ_i 's are nonzero, so there is a neighborhood of x in which all but finitely many h_i 's are the identity.

Theorem 1.2.5: Homotopy Invariance Vector Bundles

Let $p : E \rightarrow B$ a vector bundle and $f_0, f_1 : A \rightarrow B$ be homotopic maps. Then if A is paracompact,

$$f_0^*(E) \cong f_1^*(E)$$

Note this result holds for fiber bundles more generally.

Proof:

Let $F : A \times I \rightarrow B$ be a homotopy from f_0 to f_1 . Then the restriction $f^*(E)$ over $A \times \{0\}$ and $A \times \{1\}$ are $f_0^*(E)$ and $f_1^*(E)$, and we get the result by the above lemma.

Corollary 1.2.6: Homotopy For N-vector Bundles

Let $f : A \rightarrow B$ be a homotopy equivalence of paracompact spaces. Then

$$f^* : \text{Vect}^n(B) \rightarrow \text{Vect}^n(A)$$

is a bijection. In particular, vector bundles over a contractible paracompact base is trivial.

Proof:

Let g be the homotopy inverse of f . Then

$$f^*g^* = \text{id}^* = \text{id} \quad g^*f^* = \text{id}^* = \text{id}$$

completing the proof.

1.2.1 Classifying Vector Bundles on Spheres

Using the tensor product, we can construct some non-trivial bundles on the sphere S^k . The clutching function is a way of creating a bundle on the sphere S^k by gluing a bundle on the upper and lower hemisphere with a compatibility condition on the “equator subspace” (this is the “clutch”). In particular, let D^k_+ , D^k_- be the upper and lower hemisphere so that $D^k_+ \cap D^k_- \cong S^{k-1}$. Given a map $f : S^{k-1} \rightarrow \text{GL}_n(\mathbb{R})$, we can define:

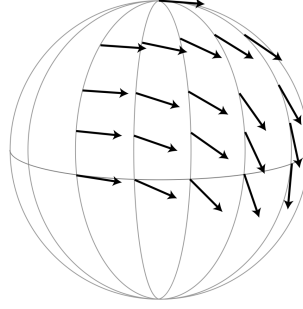
$$E_f = ((D^k_+ \times \mathbb{R}^n) \sqcup (D^k_- \times \mathbb{R}^n)) / \sim$$

$$\partial D^k_+ \times \mathbb{R}^n \ni (x, v) \sim (x, f(x)(v)) \in \partial D^k_- \times \mathbb{R}^n$$

Then we get a natural n -vector bundle map $f : E_f \rightarrow S^k$. Naturally, replacing $\mathbb{R}$ with $\mathbb{C}$ would require $f : S^{k-1} \rightarrow \text{GL}_n(\mathbb{C})$.

Example 1.2.7: Clutching Function

1. Let us create TS^2 using a clutching function. Let v_+ be a vector field on D^2_+ given by choosing a tangent vector at the north pole and parallel-transporting it along each meridian circle (see [15] for parallel-transport). Here is a visual on only a quarter of the upper hemisphere:

Figure 1.1: v_+ on D_+^2 (from Hatcher [10])

Reflect across the equator to create the vector field v_- on D_-^2 . Then take w_+ and w_- to be v_+ and v_- respectively each rotate by $\pi/2$ degree when viewed from outside the sphere. Then these for vector fields give a trivialization of TS^2 over $D_\pm^2$: the clutching function f read off the coordinates of v_-, w_- in the v_+, w_+ coordinate system, in particular f rotates (v_+, w_+) to (v_-, w_-) . As we go around the equator S^1 counter-clockwise, the angle will go from 0 to 4π .

2. Can you think of how do do this for $\mathbb{CP}^1 = S^2$ (hint: we should get a clutching function $f : S^1 \rightarrow \text{GL}_1(\mathbb{C})$ where $f(z) = (z^{-1})$).

The advantage of working with clutching functions is that we get that if f, g are clutching functions for E_f, E_g we have $E_f \cong E_g$ if and only if f, g are homotopic. When $k = \mathbb{C}$, we get a bijection (we shall show what we need to modify for the bijection when our $\mathbb{R}$, particularly it will have to be dealt with that $\mathbb{R} \setminus \{0\}$ is no longer connected)

Proposition 1.2.8: Clutching Function Vector Bundle Correspondence

The map:

$$\Phi : [S^{k-1}, \text{GL}_n(\mathbb{C})] \rightarrow \text{Vect}_{\mathbb{C}}^n(S^k)$$

sending homotopy classes of clutching functions $f : S^{k-1} \rightarrow \text{GL}_n(\mathbb{C})$ to complex vector bundles E_f is a bijection

Proof:

Let $[X, Y]$ denote the set of homotopy equivalences of functions. If $f \simeq g$, then there is a homotopy $F : S^{k-1} \times I \rightarrow \text{GL}_n(\mathbb{C})$. Then the same sort of clutching construction can be used to produce a vector bundle $E_F \rightarrow S^k \times I$ that restricts to E_f over $S^k \times \{-1\}$ and E_g over $S^k \times \{1\}$, hence $E_f \cong E_g$ by theorem 1.2.5. We thus have a map:

$$\Phi : [S^{k-1}, \text{GL}_n(\mathbb{C})] \rightarrow \text{Vect}_{\mathbb{C}}^n(S^k)$$

Let us find an inverse Ψ . Let $p : E \rightarrow S^k$ be an n -vector bundle, and let E_+, E_- be its restriction to D_+^k and D_-^k . $D_\pm^k$ is contractible, by theorem 1.2.5 these bundles are trivial. Choose trivializations $h_\pm : E_\pm \rightarrow D_\pm^k \times \mathbb{C}^n$. Then we get that $h_+ h_-^{-1}$ defines a map $S^{k-1} \rightarrow \text{GL}_n(\mathbb{C})$ whose homotopy class is by construction

$$\Psi(E) \in [S^{k-1}, \text{GL}_n(\mathbb{C})]$$

Let us make sure $\Psi(E)$ is well-defined, that is the trivialization is unique up to homotopy. First off, any two choices of $h_{\pm}$ differ by a map $D_{\pm}^k \rightarrow \mathrm{GL}_n(\mathbb{C})$, as $D_{\pm}^k$ is contractible, these maps are homotopic to a constant map. As $\mathrm{GL}_n(\mathbb{C})$ is path connected, elementary row operations in $\mathrm{GL}_n(\mathbb{C})$ can be realized by a path in $\mathrm{GL}_n(\mathbb{C})$ (insert a factor of $t \in [0, 1]$ in front of the scalar multiples). As $\mathbb{C}$ is algebraically closed, $M \in \mathrm{GL}_n(\mathbb{C})$ can be diagonalized. As the set of diagonal matrices in $\mathrm{GL}_n(\mathbb{C})$ is path-connected (since it is homeomorphic to the product of n copies of $\mathbb{C} - \{0\}$), we see that h_+ and h_- is unique up to homotopy. Thus $h_+ h_-^{-1} : S^{k-1} \rightarrow \mathrm{GL}_n(\mathbb{C})$ is unique up to homotopy, hence $\Psi : \mathrm{Vect}_{\mathbb{C}}^n(S^k) \rightarrow [S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$ is well-defined, and by construction it is clear it is the inverse, completing the proof.

This correspondence gives some immediate results. For example, every complex vector bundles over S^1 is trivial since every class $\mathrm{Vect}_{\mathbb{C}}^n(S^1)$ corresponds to $[s^0, \mathrm{GL}_n(\mathbb{C})]$ which as a single element (as $\mathrm{GL}_n(\mathbb{C})$ is path connected). Another important result is that the canonical line bundle $H \rightarrow \mathbb{CP}^1$ satisfies the relation:

$$(H \otimes H) \oplus 1 \cong H \oplus H$$

where 1 is the trivial one-dimensional bundle (see [10, p.24]).

For the real case, we have that $\mathrm{GL}_n(\mathbb{R})$ has two homeomorphic path-connected components given by the pre-image of $\det$. However, everything passes through if we consider *oriented (real) vector bundles* (where all the transition maps have the same sign) as we only need the fact the component is path connect:

Proposition 1.2.9: Clutching Function Vector Bundle Correspondence

The map:

$$\Phi : [S^{k-1}, \mathrm{GL}_n^+(\mathbb{R})] \rightarrow \mathrm{Vect}_+^n(S^k)$$

sending homotopy classes of clutching functions $f : S^{k-1} \rightarrow \mathrm{GL}_n^+(\mathbb{R})$ to oriented real vector bundles E_f is a bijection.

Proof:

exercise.

From this, let us figure out some information about $\mathrm{Vect}_{\mathbb{R}}^n(S^k)$ for various n, k . To understand this structure better, let us first define a map $\mathrm{Vect}_{x_0, \mathbb{R}}^n(S^k) \rightarrow \mathrm{Vect}_{\mathbb{R}}^n(S^k)$. The set $\mathrm{Vect}_{x_0, \mathbb{R}}^n(S^k)$ is all isomorphism classes with an orientation specified by the point $x_0 \in S^{k-1}$; by our above results we get that $\mathrm{Vect}_{x_0, \mathbb{R}}^n(S^k)$ is in bijective correspondence with the homotopy classes of maps $S^{k-1} \rightarrow \mathrm{GL}_n(\mathbb{R})$ taking x_0 to $\mathrm{GL}_n^+(\mathbb{R})$. Then we get the map

$$\mathrm{Vect}_{x_0, \mathbb{R}}^n(S^k) \rightarrow \mathrm{Vect}_{\mathbb{R}}^n(S^k)$$

that forgets the orientation over x_0 . This map is certainly surjective. It is 2-to-1 at all points except where a bundle has an automorphism that reverses the orientation of the fiber over x_0 ; on such points it is 1-to-1.

When $k = 1$ and $n \in \mathbb{N}_{>0}$, there is two homotopy classes $S^0 \rightarrow \mathrm{GL}_n(\mathbb{R})$: x_0 goes to $\mathrm{id} \in \mathrm{GL}_n^+(\mathbb{R})$, and then the other point goes to either the identity or the reflection. When $n = 1$, we get the trivial bundle and the Möbius bundle. When $n > 1$, we get that these bundles all have automorphisms reversing the orientations of the fibers. Hence, $\mathrm{Vect}_{\mathbb{R}}^n(S^1)$ always have two elements for each $n \geq 1$, with the nontrivial bundle being nonorientable.

When $k > 1$, then S^{k-1} is path-connected, and hence $S^{k-1} \rightarrow \mathrm{GL}_n(\mathbb{R})$ taking x_0 to $\mathrm{GL}_n^+(\mathbb{R})$ must take all of S^{k-1} to $\mathrm{GL}_n^+(\mathbb{R})$, implying that we get a bijection

$$\mathrm{Vect}_+^n(S^k) \xrightarrow{\sim} \mathrm{Vect}_{x_0, \mathbb{R}}^n(S^k)$$

and so every vector bundle over S^k is orientable and has (exactly) two different orientations. Thus, the map:

$$\mathrm{Vect}_+^n(S^k) \rightarrow \mathrm{Vect}^n(S^k)$$

is surjective, being 2-to-1 on bundles not having an orientation-reversing automorphism, and is injective otherwise.

As we are working over paracompact spaces, and have access to an inner product, we get that we can deformation-retract $\mathrm{GL}_n(\mathbb{R})$ to the orthogonal subgroup $O(n)$ where $O(n) \leq \mathrm{GL}_n(\mathbb{R})$ where:

$$O(n) = \{A \in \mathrm{GL}_n(\mathbb{R}) \mid AA^T = I\}$$

The deformation retract is easy by keeping track the Gram-Schmidt process and applying “paths” to each process (as has been done earlier). Similarly, $\mathrm{GL}_n^+(\mathbb{R})$ is a deformation retract of the special orthogonal group $SO(n)$ where

$$SO(n) = \{A \in O(n) \mid \det(A) = 1\}$$

In the complex case, we get a deformation retract onto the unitary group $U(n)$ where

$$U(n) = \{A \in \mathrm{GL}_n(\mathbb{C}) \mid AA^* = I\}$$

and can be characterized by having column forming an orthonormal basis.

Now, as we have a deformation retract $SO(n) \hookrightarrow \mathrm{GL}_n^+(\mathbb{R})$ and $U(n) \hookrightarrow \mathrm{GL}_n(\mathbb{C})$ we get natural bijections:

$$[S^{k-1}, SO(n)] \xrightarrow{\sim} [S^{k-1}, \mathrm{GL}_n^+(\mathbb{R})] \quad \text{and} \quad [S^{k-1}, U(n)] \xrightarrow{\sim} [S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$$

This can greatly simplify classification efforts as $SO(n)$ is a much smaller Lie groups. For example, $SO(2)$ is a circle (orientation-preserving isometric of $\mathbb{R}^2$ are rotations). Letting $k = 2$ and $n = 2$, we get for each integer $n \in \mathbb{Z}$ a map $f_n : S^1 \rightarrow S^1$ given by $z \mapsto z^n$ which is a clutching function for an oriented 2-dimensional vector bundle $E_n \rightarrow S^2$. Note that E_2 is TS^2 and E_1 is the real vector bundle underlying the canonical complex line bundle over $\mathbb{CP}^1 = S^2$. As every arbitrary continuous map $f : S^1 \rightarrow S^1$ is homotopic to a map $z \mapsto z^n$ with a unique $n \in \mathbb{Z}$ (see [13]), we get:

$$\mathrm{Vect}_+^2(S^2) \cong \mathbb{Z}$$

Note that reversing the orientation of E_n gives E_{-n} , and hence:

$$\mathrm{Vect}_{\mathbb{R}}^2(S^2) \cong \mathbb{N}$$

As $U(1)$ is homeomorphic to S^1 , we similarly get:

$$\mathrm{Vect}_{\mathbb{C}}^2(S^2) \cong \mathbb{Z}$$

For $k > 2$, recall that any two maps $S^{k-1} \rightarrow S^1$ are homotopic if $k > 2$ (such a map lifts to a covering space $\mathbb{R}$ of S^1 and $\mathbb{R}$ is contractible, see [13]). Hence, every 2-dimensional vector bundle over S^k for $k \geq 3$ is trivial:

$$\mathrm{Vect}_{\mathbb{R}}^2(S^k) \cong \{0\} \quad k \geq 3$$

The same logic applies to complex vector bundles:

$$\text{Vect}_{\mathbb{C}}^2(S^k) \cong \{0\} \quad k \geq 3$$

We shall give some more examples after the next proposition. We first point out a couple more important facts. First, for any space X , the set $[X, SO(n)]$ and $[X, U(n)]$ has a natural group structure given by point-wise multiplication. Using this fact, we have that:

$$[S^i, SO(n)] \cong_{\text{Grp}} \pi_i(SO(n)) \quad \text{and} \quad [S^i, U(n)] \cong_{\text{Grp}} \pi(U(n))$$

where homotopies preserve base points.

Proposition 1.2.10: Vector bundles and Suspension

et SX denote the suspension of a paracompact space X . Then given a choice of base-point $x \in X$, there is a 1-1 correspondence between the set $\text{Vect}^n(SX)$ of isomorphism classes of n -dimensional (resp. real, complex, or quaternion) vector bundles over SX and the respective set of based homotopy classes of maps

$$[X, O_n]_*, \quad [X, U_n]_*, \quad \text{or} \quad [X, Sp_n]_*$$

from X to the orthogonal group O_n , unitary group U_n , or symplectic group Sp_n .

Proof:

The proof is a complicated proof, the following is a sketch given from [2]. From [13], we know that SX is the union of two contractible cones C_1 and C_2 whose intersection is X . As every vector bundle on the cones C_i is trivial as it is contractible, and every vector bundle on SX is obtained from an isomorphism of trivial bundles over X via the clutching construction (see [2]; it is the Mayer-Vietoris sequence for this suspension construction). Such an isomorphism is given by a homotopy class of maps from X to GL_n , or equivalently, to the appropriate deformation retract (O_n , U_n , or Sp_n) of GL_n . Then the indeterminacy in the resulting map from $[X, GL_n]$ to classes of vector bundles is eliminated by insisting that the base point of X map to $1 \in GL_n$.

Now, from [17] or [13], we know the computations of the homotopy groups classical Lie groups, which we now take advantage of to find the isomorphism classes of vector bundles on spheres

Example 1.2.11: Vector Bundles On Spheres

1. Let $X = S^1$, and recall that $|\pi_0(O_n)| = 2$ for real vector bundles of dimension n for all $n \geq 1$. This corresponds to the trivial bundle and the möbius bundle. The Whitney direct sum of these two shall give all non-trivial real vector bundles on S^1 .

As $|\pi_0(U_n)| = 1$ for all n , all complex vector bundles are trivial!

2. Let $X = S^2$. Then $\pi_1(O_1) = 0$, and so there is only the trivial real line bundle on S^1 . As $\pi_1(O_2) = \mathbb{Z}$, there are infinitely many 2-dimensional real line bundles of dimension 2 (indexed by the degree of the clutching map mentioned in the above proof). As $\pi_1(O_n) = \mathbb{Z}/2\mathbb{Z}$ for $n \geq 2$, there is only two real line bundles for dimension $n \geq 3$.

For complex line bundles on S^3 , we get the same phenomena as $\pi_1(U_2) = \mathbb{Z}$

3. Recall from [17] that $\pi_2(G) = 0$ for all compact Lie groups. Thus, all vector bundles on S^3 are trivial!
4. From the above, we can conclude that all (real and complex) line bundles on S^4 trivial. as:

$$\begin{array}{lll} \pi_3(O_n) = 3 & \pi_3(U_n) = \mathbb{Z} & \pi_3(Sp_n) = \mathbb{Z} \\ \text{for } n \geq 5 & \text{for } n \geq 2 & \text{for } n \geq 1 \end{array}$$

We get the corresponding infinite family of n -dimensional Real/Complex/Quaternion vector bundles.

1.2.2 Universal Bundles

In this section, we shall show that every vector bundles is the pullback of a canonical vector bundle on the Grassmannian G_n over $\mathbb{R}, \mathbb{C}$, or $\mathbb{H}$. As a quick reminder of the Grassmannian, let V be a finite dimensional vector space and let $\text{Grass}_k(V)$ be all k -dimensional subspaces of V ; i.e. the *Grassmannian manifold* of all sub-spaces of V . Let's define a topology on $\text{Grass}_n(\mathbb{R}^k)$ by first defining the *Stiefel Manifold* $V_n(\mathbb{R}^k)$ which is the space of orthonormal n -frames in $\mathbb{R}^k$. This is a subspace of the product of n copies of the unit sphere S^{k-1} , that is the subspace of orthonormal n -tuples. It is a closed subspace as orthogonality is given by an algebraic equation. Hence, $V_n(\mathbb{R}^k)$ is compact, being the product of spheres. Then there is a natural surjection $V_n(\mathbb{R}^k) \rightarrow \text{Grass}_n(\mathbb{R}^k)$ sending n -frames to the subspace it spans. This gives the quotient topology on $G_n(\mathbb{R}^k)$, and it is easy to verify it is also compact. To show its a manifold, see [15]. The *infinite Grassmannian* Grass_n is the union all $\text{Grass}_n(V)$ as V ranges over all finite dimensional subspace of a fixed infinite dimensional vector space (usually $\mathbb{R}^\infty$, $\mathbb{C}^\infty$, or $\mathbb{H}^\infty$). With a bit of work, we see that it is an infinite-dimensional CW complex (it is the limit of CW complexes respecting the appropriate directed mappings, see 1.2.3). These are rather natural, for example if taking $n = 1$, we get that Grass_1 is $\mathbb{R}P^\infty$, $\mathbb{C}P^\infty$, or $\mathbb{H}P^\infty$ respectively.

Just like for $\mathbb{R}P^n$, $\mathbb{C}P^n$, and $\mathbb{H}P^n$, there a canonical line bundle for Grassmannian: let:

$$E_n(\mathbb{R}^k) = \{(\ell, v) \in \text{Grass}_n(\mathbb{R}^k) \times \mathbb{R}^k \mid v \in \ell\}$$

and similarly for $\mathbb{C}^k, \mathbb{H}^k$. Then the inclusion $\mathbb{R}^k \subseteq \mathbb{R}^{k+1} \subseteq \dots$ induces inclusions $E_n(\mathbb{R}^k) \subseteq E_n(\mathbb{R}^{k+1}) \subseteq \dots$, and hence

$$E_n(\mathbb{R}^\infty) = \cup_k E_n(\mathbb{R}^k)$$

where it is given the colimit topology, and similarly for the complex and quaternion case. Then:

Proposition 1.2.12: Grassmannian Bundle is Vector Bundle

The projection $p : E_n(\mathbb{R}^k) \rightarrow \text{Grass}_n(\mathbb{R}^k)$ where $p(\ell, v) = \ell$ is a vector bundle for finite and infinite k .

Proof:

By proving the finite case, the infinite case follows by local trivializations working over every $k \in \mathbb{N}$, hence we prove the finite case. For $\ell \in G_n(\mathbb{R}^k)$, let $\pi_\ell : \mathbb{R}^k \rightarrow \ell$ be orthogonal projection and let

$$U_\ell = \{\ell' \in G_n(\mathbb{R}^k) \mid \pi_\ell(\ell') \text{ has dimension } n\}.$$

so that $\ell \in U_\ell$. We will show that U_ℓ is open in $G_n(\mathbb{R}^k)$ and that the map $h : p^{-1}(U_\ell) \rightarrow U_\ell \times \ell \cong U_\ell \times \mathbb{R}^n$ defined by $h(\ell', v) = (\ell', \pi_\ell(v))$ is a local trivialization of $E_n(\mathbb{R}^k)$.

For U_ℓ to be open is equivalent to its preimage in $V_n(\mathbb{R}^k)$ being open. This preimage consists of orthonormal frames v_i , $1 \leq i \leq n$, such that $\pi_\ell(v_1), \dots, \pi_\ell(v_n)$ are independent. Let A be the matrix of π_ℓ with respect to the standard basis in the domain $\mathbb{R}^k$ and any fixed basis in the range ℓ . The condition on v_i is then that the $n \times n$ matrix with columns Av_i has nonzero determinant. Since the value of this determinant is a continuous function of v_i , it follows that the frames v_i yielding a nonzero determinant form an open set in $V_n(\mathbb{R}^k)$.

Then it's clear that h is a bijection which is a linear isomorphism on each fiber. Furthermore h is continuous since its two coordinate functions are continuous. To see that h^{-1} is continuous, for $\ell' \in U_\ell$ there is a unique invertible linear map $L_{\ell'} : \mathbb{R}^k \rightarrow \mathbb{R}^k$ restricting to π_ℓ on ℓ' and the identity on $\ell^\perp = \text{Ker } \pi_\ell$. Then $L_{\ell'}$, regarded as a $k \times k$ matrix, depends continuously on ℓ' . Indeed, we can write L_ℓ as a product AB^{-1} where:

- B sends the standard basis to $v_1, \dots, v_n, v_{n+1}, \dots, v_k$ with $v_1, \dots, v_n$ an orthonormal basis for ℓ' and $v_{n+1}, \dots, v_k$ a fixed basis for $\ell^\perp$.
- A sends the standard basis to $\pi_\ell(v_1), \dots, \pi_\ell(v_n), v_{n+1}, \dots, v_k$.

Both A and B depend continuously on $v_1, \dots, v_n$. Since matrix multiplication and matrix inversion are continuous operations, it follows that the product $L_{\ell'} = AB^{-1}$ depends continuously on $v_1, \dots, v_n$. But since $L_{\ell'}$ depends only on ℓ' , not on the basis $v_1, \dots, v_n$ for ℓ' , it follows that $L_{\ell'}$ depends continuously on ℓ' since $G_n(\mathbb{R}^k)$ has the quotient topology from $V_n(\mathbb{R}^k)$. We have $h^{-1}(\ell', w) = (\ell', L_{\ell'}^{-1}(w))$ with $L_{\ell'}^{-1}$ depending continuously on ℓ' since matrix inversion is continuous, so h^{-1} is continuous, and so we have a local trivialization, as we sought to show.

Theorem 1.2.13: Classifying Vector Bundles

Let X be a paracompact space. Then $\text{Vect}^n(X)$ is in correspondence with the set of homotopy classes of maps $X \rightarrow \text{Grass}_n$, denoted $[X, \text{Grass}_n]$. In other words:

$$\text{Vect}^n(X) \cong [X, \text{Grass}_n]$$

In particular, if $E \rightarrow X$ is an n -dimensional vector bundle, it is isomorphic to $f^*(E_n)$ for some $f : X \rightarrow \text{Grass}_n$ where E determines f up to homotopy.

Proof:

We shall prove it for the real case, the complex, quaternion, and orientable real case follow similarly. To shorten notation, let G_n denote the real Grassmannian for n -dimensional subspaces.

The key observation is the following: For an n -dimensional vector bundle $p : E \rightarrow X$, an isomorphism $E \cong f^*(E_n)$ is equivalent to a map $g : E \rightarrow \mathbb{R}^\infty$ that is a linear injection on each fiber. To see this, suppose that we have a map $f : X \rightarrow G_n$ and an isomorphism $E \cong f^*(E_n)$.

Then we have a commutative diagram

$$\begin{array}{ccccccc}
 E & \xrightarrow{\cong} & f^*(E_n) & \xrightarrow{\tilde{f}} & E_n & \xrightarrow{\pi} & \mathbb{R}^\infty \\
 & \searrow p & \downarrow & & \downarrow & & \\
 & & X & \xrightarrow{f} & G_n & &
 \end{array}$$

where $\pi(\ell, v) = v$. The composition across the top row is a map $g : E \rightarrow \mathbb{R}^\infty$ that is a linear injection on each fiber, since both $\tilde{f}$ and π have this property. Conversely, given a map $g : E \rightarrow \mathbb{R}^\infty$ that is a linear injection on each fiber, define $f : X \rightarrow G_n$ by letting $f(x)$ be the n -plane $g(p^{-1}(x))$. This clearly yields the above commutative diagram.

To show surjectivity of the map $[X, G_n] \rightarrow \text{Vect}^n(X)$, suppose $p : E \rightarrow X$ is an n -dimensional vector bundle. Let $\{U_\alpha\}$ be an open cover of X such that E is trivial over each U_α . As X is paracompact, there a countable open cover $\{U_i\}$ of X such that E is trivial over each U_i , and there is a partition of unity $\{\varphi_i\}$ with φ_i supported in U_i . Let $g_i : p^{-1}(U_i) \rightarrow \mathbb{R}^n$ be the composition of a trivialization $p^{-1}(U_i) \rightarrow U_i \times \mathbb{R}^n$ with projection onto $\mathbb{R}^n$. The map $(\varphi_i p)g_i$, $v \mapsto \varphi_i(p(v))g_i(v)$, extends to a map $E \rightarrow \mathbb{R}^n$ that is zero outside $p^{-1}(U_i)$. Near each point of X only finitely many φ_i 's are nonzero, and at least one φ_i is nonzero, so these extended $(\varphi_i p)g_i$'s are the coordinates of a map $g : E \rightarrow (\mathbb{R}^n)^\infty = \mathbb{R}^\infty$ that is a linear injection on each fiber.

For injectivity, if we have isomorphisms $E \cong f_0^*(E_n)$ and $E \cong f_1^*(E_n)$ for two maps $f_0, f_1 : X \rightarrow G_n$, then these give maps $g_0, g_1 : E \rightarrow \mathbb{R}^\infty$ that are linear injections on fibers, as in the first paragraph of the proof. Then g_0 and g_1 are homotopic through maps g_t that are linear injections on fibers. If this is so, then f_0 and f_1 will be homotopic via $f_t(x) = g_t(p^{-1}(x))$.

To show this, a homotopy g_t is constructed. The first step is to compose g_0 with the homotopy $L_t : \mathbb{R}^\infty \rightarrow \mathbb{R}^\infty$ defined by $L_t(x_1, x_2, \dots) = (1-t)(x_1, x_2, \dots) + t(x_1, 0, x_2, 0, \dots)$. For each t this is a linear map whose kernel is easily computed to be 0, so L_t is injective. Composing the homotopy L_t with g_0 moves the image of g_0 into the odd-numbered coordinates. Similarly we can homotope g_1 into the even-numbered coordinates. Still calling the new g 's g_0 and g_1 , let $g_t = (1-t)g_0 + tg_1$. This is linear and injective on fibers for each t since g_0 and g_1 are linear and injective on fibers.

Definition 1.2.14: Universal Bundle

The Grassmannians Grass_n are called the *universal bundles*; for $\mathbb{R}, \mathbb{C}, \mathbb{H}$ they are often denoted BO_n, BU_n , and BSp_n respectively.

As we saw we can put a natural structure group on vector bundles (i.e. O_n, U_n or Sp_n depending on whether the vector bundle is a real, complex, or quaternion bundle respectively). Hence, the infinite Grassmannian is called “classifying space” of O_n, U_n or Sp_n . For this reason, the infinite Grassmannian is denoted BO_n, BU_n , or BSp_n respectively.

1.2.3 *Cell-Structure on Grassmannian

HERE

1.3 The Stabilization Theorem

The classification of theorem 1.2.13 reduces rank- n bundles over X to homotopy classes of maps into the Grassmannian Grass_n , one classification for each rank separately. That is not yet a single invariant of the space: the sets $[X, \text{Grass}_n]$ are indexed by n , and bundles of different rank have no common home. The remedy is to let the rank grow. Adding a trivial line embeds $\mathbf{VB}_n(X)$ in $\mathbf{VB}_{n+1}(X)$, and the colimit along these inclusions discards the rank while keeping everything about a bundle that survives stabilization. The theorem below identifies that colimit with the reduced K -theory of chapter 2.

Write $\tilde{K}(X)$ for that reduced group (the kernel of the rank map); it is captured by maps into the stable classifying space, in each field at once.

Theorem 1.3.1: Stabilization Theorem of Classifying Spaces

Let X be a compact space or a finite-dimensional connected CW complex. Then

$$\tilde{K}(X) \cong \varinjlim_n \mathbf{VB}_n(X) \cong \varinjlim_n [X, \text{Grass}_n]$$

and in particular $\widetilde{KO}(X) \cong \varinjlim_n [X, BO_n]$, $\widetilde{KU}(X) \cong \varinjlim_n [X, BU_n]$, and $\widetilde{KSp}(X) \cong \varinjlim_n [X, BSp_n]$.

Proof:

Over a compact (or finite-dimensional connected CW) base the trivial bundles are cofinal in $\mathbf{VB}(X)$: every bundle E is a summand of a trivial bundle (proposition 1.1.16), so $E \oplus E' \cong T^N$ for some complement E' and some N . Cofinality lets every reduced class be written in the normal form $[E] - n$, and $[E] - n = [F] - n$ holds in $\tilde{K}(X)$ exactly when $E \oplus T^k \cong F \oplus T^k$ for some k , that is, when E and F are stably isomorphic. Hence $\tilde{K}(X) \cong \varinjlim_n \mathbf{VB}_n(X)$, the colimit taken along the stabilization $E \mapsto E \oplus T^1$. The classification of rank- n bundles by homotopy classes into the Grassmannian (theorem 1.2.13) identifies $\mathbf{VB}_n(X) \cong [X, \text{Grass}_n]$ compatibly with stabilization, giving $\varinjlim_n [X, \text{Grass}_n]$; specializing the field to $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ gives BO_n , BU_n , BSp_n , completing the proof.

The stabilization theorem is what makes K -theory representable: the colimit over n replaces the unstable Grassmannians by the single space BO , BU , or BSp , and chapter 2 restores the rank component to obtain $\tilde{K}(X) \cong [X, \mathbb{Z} \times BU]$.

K-Theory for Vector Bundles

The previous chapter solved the classification problem for vector bundles over a paracompact base: by theorem 1.2.13, the isomorphism classes of n -dimensional bundles over X are the homotopy classes $[X, \text{Grass}_n]$, and by theorem 1.2.5 this set is a homotopy invariant of X . Yet the collection of all vector bundles over X , taken together under the direct sum of definition 1.1.11, carries more structure than a bare set of homotopy classes: it is a commutative monoid. The single defect of this monoid is that it has no inverses, since a nonzero bundle can never sum to the zero bundle. K-theory is what results from repairing exactly this defect. One formally adjoins inverses to the monoid of vector bundles, in the same way that the integers are built from the natural numbers, and the resulting group $K^0(X)$ becomes a powerful and computable invariant of the space.

This chapter develops topological K-theory from that single idea. The group $K^0(X)$ is introduced here as the group completion of the monoid of bundles, with its reduced companion $\tilde{K}^0(X)$ recording bundles up to the addition of trivial summands. The stabilization theorem of the previous chapter (theorem 1.3.1) already hints that this group is representable: stable equivalence classes of bundles are homotopy classes of maps into the infinite Grassmannian. Making this precise identifies $K^0(X)$ with the homotopy functor $[X, \mathbb{Z} \times BU]$ in the complex case, and the tensor product of definition 1.1.17 upgrades the group to a ring. The chapter then turns these structural results into a genuine cohomology theory: Bott periodicity (theorem 2.2.16) supplies the periodicity isomorphism that lets one define negative-degree groups, and the resulting theory satisfies the generalized cohomology axioms. We close with explicit computations, including the K-theory of spheres and projective spaces.

2.1 The Group $K^0(X)$

Throughout this section X denotes a compact Hausdorff space, and $\text{Vect}(X)$ denotes the set of isomorphism classes of (finite-dimensional, complex) vector bundles over X . The direct sum of definition 1.1.11 descends to isomorphism classes: if $E \cong E'$ and $F \cong F'$, then $E \oplus F \cong E' \oplus F'$, so $\oplus$

is a well-defined binary operation on $\text{Vect}(X)$. It is commutative and associative up to isomorphism, and the zero bundle (the rank-0 bundle, whose total space is X itself) is a two-sided identity. Thus $(\text{Vect}(X), \oplus)$ is a commutative monoid. The obstruction to it being a group is total: the only invertible element is 0, because the rank of a direct sum is the sum of the ranks, and ranks are non-negative integers. The construction that follows forces inverses to exist by a purely formal device, exactly the device that produces $\mathbb{Z}$ from $\mathbb{N}$.

Definition 2.1.1: The Grothendieck Group $K^0(X)$

Let X be a compact Hausdorff space and let $(\text{Vect}(X), \oplus)$ be the commutative monoid of isomorphism classes of complex vector bundles over X . The *Grothendieck group* (or *topological K-theory group*) $K^0(X)$ is the abelian group with one generator $[E]$ for each isomorphism class of bundle E and one relation

$$[E \oplus F] = [E] + [F]$$

for each pair of bundles E, F . Equivalently, $K^0(X)$ is the quotient of the free abelian group on isomorphism classes of bundles by the subgroup generated by all elements $[E \oplus F] - [E] - [F]$.

The elements of $K^0(X)$ are *formal differences* $[E] - [F]$ of bundles, called *virtual bundles*, and two such differences $[E] - [F]$ and $[E'] - [F']$ are equal precisely when $E \oplus F' \oplus G \cong E' \oplus F \oplus G$ for some bundle G . The auxiliary bundle G is unavoidable: it encodes the equivalence relation built into the Grothendieck construction, where one identifies pairs that become equal only after adding a common summand. The class $[E]$ of a bundle is the image of the monoid element E under the canonical map $\text{Vect}(X) \rightarrow K^0(X)$, and every element of $K^0(X)$ is a difference of two such classes. The universal property that pins down $K^0(X)$ is that any monoid homomorphism from $\text{Vect}(X)$ into an abelian group factors uniquely through $K^0(X)$; this is what makes K-theory the receptacle for every additive invariant of vector bundles.

Example 2.1.2: K^0 of a Point

Let $X = \{*\}$ be the one-point space. A vector bundle over a point is the same data as its single fiber, a finite-dimensional complex vector space, so isomorphism classes of bundles are classified by dimension. Hence $\text{Vect}(\{*\}) \cong \mathbb{N}$ as monoids, with $\oplus$ corresponding to addition of dimensions and $[\mathbb{C}^n]$ corresponding to n . The Grothendieck group of $(\mathbb{N}, +)$ is $\mathbb{Z}$: the formal differences $m - n$ of natural numbers are exactly the integers, with $m - n = m' - n'$ in $\mathbb{Z}$ iff $m + n' = m' + n$, which is the Grothendieck relation with trivial auxiliary term. Therefore

$$K^0(\{*\}) \cong \mathbb{Z}$$

with the class $[\mathbb{C}^n] - [\mathbb{C}^m]$ corresponding to the integer $n - m$. The isomorphism sends a virtual bundle to its rank, the difference of the dimensions of its positive and negative parts.

The rank computation in this example is the universal one. Over a connected base, every bundle has a well-defined rank (a single non-negative integer), and rank is additive under direct sum, so it defines a monoid homomorphism $\text{Vect}(X) \rightarrow \mathbb{N} \hookrightarrow \mathbb{Z}$. By the universal property this extends to a group homomorphism, which is the central tool for splitting off the trivial part of $K^0(X)$.

Proposition 2.1.3: The Rank Homomorphism

Let X be a compact Hausdorff space with connected components $X_1, \dots, X_r$. Assigning to a bundle its rank on each component defines a surjective group homomorphism

$$\mathrm{rk}: K^0(X) \longrightarrow \mathbb{Z}^r, \quad [E] \longmapsto (\mathrm{rk}E|_{X_1}, \dots, \mathrm{rk}E|_{X_r})$$

which is split by the homomorphism $\mathbb{Z}^r \rightarrow K^0(X)$ sending the i -th standard generator to the class of the line bundle that is trivial of rank 1 on X_i and rank 0 elsewhere. In particular, when X is connected, there is a split surjection $\mathrm{rk}: K^0(X) \rightarrow \mathbb{Z}$.

Proof:

On each component X_i the rank of a bundle is a single non-negative integer, because the rank function $X_i \rightarrow \mathbb{N}$ is locally constant by the local triviality of definition 1.1.2 and X_i is connected. Hence $E \mapsto (\mathrm{rk}E|_{X_1}, \dots, \mathrm{rk}E|_{X_r})$ is a well-defined map $\mathrm{Vect}(X) \rightarrow \mathbb{N}^r$, and it is a monoid homomorphism since rank is additive under $\oplus$. By the universal property of definition 2.1.1 it extends uniquely to a group homomorphism $\mathrm{rk}: K^0(X) \rightarrow \mathbb{Z}^r$.

For the splitting, let T_i denote the bundle that restricts to the trivial line bundle on X_i and to the rank-0 bundle on the remaining components; this is a genuine bundle over X since the components are open and closed. Then $\mathrm{rk}[T_i]$ is the i -th standard generator $e_i \in \mathbb{Z}^r$, so the homomorphism $s: \mathbb{Z}^r \rightarrow K^0(X)$ defined on generators by $s(e_i) = [T_i]$ satisfies $\mathrm{rk} \circ s = \mathrm{id}$. This exhibits rk as a split surjection, and surjectivity is immediate from the existence of the splitting, completing the proof.

The kernel of the rank homomorphism is the part of $K^0(X)$ that sees genuine twisting rather than the dimension count. Isolating it gives the reduced theory. For the rest of the section it is convenient to assume X is connected so that rank is a single integer; the general statement over r components differs only by replacing $\mathbb{Z}$ with $\mathbb{Z}^r$.

Definition 2.1.4: Reduced K-Theory $\tilde{K}^0(X)$

Let X be a connected compact Hausdorff space. The *reduced K-theory* of X is the kernel of the rank homomorphism,

$$\tilde{K}^0(X) = \ker(\mathrm{rk}: K^0(X) \rightarrow \mathbb{Z})$$

the subgroup of virtual bundles of rank zero.

Reduced K-theory discards the information that any space, however trivial, already carries: the integer recording the dimension of a bundle. What remains in $\tilde{K}^0(X)$ is the obstruction to a virtual bundle being a difference of trivial bundles. The splitting of proposition 2.1.3 shows that this discarding loses nothing beyond the dimension count, since the full group decomposes as the reduced group together with a copy of the integers.

Corollary 2.1.5: The Splitting $K^0 = \tilde{K}^0 \oplus \mathbb{Z}$

For X a connected compact Hausdorff space there is a natural isomorphism

$$K^0(X) \cong \tilde{K}^0(X) \oplus \mathbb{Z}$$

where the $\mathbb{Z}$ summand is generated by the class $[\mathbb{C}]$ of the trivial line bundle and the projection to $\mathbb{Z}$ is the rank.

Proof:

The rank homomorphism $\text{rk}: K^0(X) \rightarrow \mathbb{Z}$ is split surjective by proposition 2.1.3, with splitting $s: \mathbb{Z} \rightarrow K^0(X)$, $s(1) = [\mathbb{C}]$. For a split short exact sequence of abelian groups

$$0 \rightarrow \tilde{K}^0(X) \rightarrow K^0(X) \xrightarrow{\text{rk}} \mathbb{Z} \rightarrow 0$$

the splitting lemma gives $K^0(X) \cong \ker(\text{rk}) \oplus \mathbb{Z} = \tilde{K}^0(X) \oplus \mathbb{Z}$, the isomorphism being $\xi \mapsto (\xi - s(\text{rk } \xi), \text{rk } \xi)$ with inverse $(\eta, n) \mapsto \eta + n[\mathbb{C}]$. Naturality in X holds because rank, the trivial bundle, and the splitting are all preserved by pullback along continuous maps, completing the proof.

To make $\tilde{K}^0(X)$ usable one needs a concrete description of its elements, and this is where the compactness of X enters decisively. The Grothendieck construction in general produces formal differences $[E] - [F]$ with two genuine bundles in play, and the equivalence relation involves an unknown auxiliary summand G . Over a compact base both of these complications collapse. The mechanism is that trivial bundles are *cofinal* in $\text{Vect}(X)$: every bundle embeds in a trivial one as a direct summand, by proposition 1.1.16. The next two results extract from cofinality first a normal form for elements of $K^0(X)$ and then a clean criterion for two bundles to be equal in the reduced group.

Lemma 2.1.6: Cofinality and the Normal Form $[E] - n$

Let X be a compact Hausdorff space and write $\underline{\mathbb{C}}^n$ for the trivial bundle of rank n .

1. For every bundle F over X there is a bundle F' and an integer n with $F \oplus F' \cong \underline{\mathbb{C}}^n$.
2. Every element of $K^0(X)$ can be written in the form $[E] - n := [E] - [\underline{\mathbb{C}}^n]$ for some bundle E and some $n \geq 0$.

Proof:

Part (1): cofinality. By proposition 1.1.16, for the bundle F over the compact Hausdorff space X there is a complementary bundle F' with $F \oplus F'$ isomorphic to a trivial bundle, say $F \oplus F' \cong \underline{\mathbb{C}}^n$ where $n = \text{rk } F + \text{rk } F'$. This is the cofinality statement: every bundle is a summand of a trivial bundle.

Part (2): normal form. A general element of $K^0(X)$ is a formal difference $[E_0] - [F]$ of two bundle classes, since by definition 2.1.1 the group is generated by the classes $[E]$ under addition and subtraction, and any such combination collapses to a single difference using $[E] + [E'] = [E \oplus E']$.

Apply part (1) to F : choose F' with $F \oplus F' \cong \mathbb{C}^n$. Then in $K^0(X)$,

$$[E_0] - [F] = [E_0] - [F] + ([F'] - [F']) = [E_0] + [F'] - ([F] + [F']) = [E_0 \oplus F'] - [\mathbb{C}^n]$$

using $[F] + [F'] = [F \oplus F'] = [\mathbb{C}^n]$. Setting $E = E_0 \oplus F'$ gives the normal form $[E_0] - [F] = [E] - n$, completing the proof.

The normal form $[E] - n$ replaces a formal difference of two unknown bundles by a single honest bundle minus a trivial summand, which is a substantial simplification: it says every virtual bundle is, up to a shift by trivial bundles, an actual bundle. The same cofinality argument now eliminates the auxiliary summand G from the equivalence relation, pinning down exactly when two normal forms agree. This is the telescoping argument, and it is the heart of the reduced theory.

Definition 2.1.7: Stable Isomorphism and Stable Equivalence

Two vector bundles E and F over X are *stably isomorphic*, written $E \cong_s F$, if there is an integer $k \geq 0$ with

$$E \oplus \mathbb{C}^k \cong F \oplus \mathbb{C}^k$$

(the *same* trivial summand on each side). They are *stably equivalent*, written $E \approx_s F$, if there exist integers $m, n \geq 0$ with

$$E \oplus \mathbb{C}^m \cong F \oplus \mathbb{C}^n$$

(possibly *different* trivial summands). A bundle is *stably trivial* (in the sense of definition 1.1.12) precisely when it is stably equivalent to a trivial bundle.

The two notions differ by exactly the rank: stable isomorphism preserves rank ($\text{rk} E = \text{rk} F$, since adding $\mathbb{C}^k$ to both sides forces equal ranks), whereas stable equivalence allows the ranks to differ. Stable isomorphism is the relation that equality in $K^0(X)$ detects, and stable equivalence is the coarser relation that equality in $\tilde{K}^0(X)$ detects. The next theorem proves both statements, with the cofinality lemma supplying the telescoping that turns the abstract Grothendieck congruence into these concrete bundle isomorphisms.

Theorem 2.1.8: Reduced K-Theory Classifies Stable Equivalence

Let X be a connected compact Hausdorff space, and let E, F be vector bundles over X .

1. In $K^0(X)$ one has $[E] = [F]$ if and only if E and F are stably isomorphic, $E \cong_s F$.
2. Sending a bundle E to the class $[E] - \text{rk} E \in \tilde{K}^0(X)$ induces a bijection between stable equivalence classes of bundles and elements of $\tilde{K}^0(X)$. Under this bijection $\tilde{K}^0(X)$ is the set of stable equivalence classes, made into a group by direct sum.

Proof:

Part (1). Suppose first that $E \cong_s F$, so $E \oplus \mathbb{C}^k \cong F \oplus \mathbb{C}^k$ for some k . Passing to classes in $K^0(X)$ and using definition 2.1.1,

$$[E] + k[\mathbb{C}] = [E \oplus \mathbb{C}^k] = [F \oplus \mathbb{C}^k] = [F] + k[\mathbb{C}]$$

Cancelling $k[\mathbb{C}]$ in the abelian group $K^0(X)$ gives $[E] = [F]$.

For the converse, suppose $[E] = [F]$ in $K^0(X)$. By construction of the Grothendieck group, the relation $[E] = [F]$ holds exactly when E and F become isomorphic after adding a common bundle G , that is $E \oplus G \cong F \oplus G$ for some bundle G . This is the defining congruence of definition 2.1.1: equality of $[E]$ and $[F]$ in the group completion of the monoid $\text{Vect}(X)$ means precisely that there exists G with $E \oplus G \cong F \oplus G$. Now apply cofinality (lemma 2.1.6(1)) to G : choose G' with $G \oplus G' \cong \underline{\mathbb{C}}^k$. Adding G' to both sides of $E \oplus G \cong F \oplus G$,

$$E \oplus G \oplus G' \cong F \oplus G \oplus G' \implies E \oplus \underline{\mathbb{C}}^k \cong F \oplus \underline{\mathbb{C}}^k$$

This is exactly stable isomorphism, $E \cong_s F$. The telescoping step (replacing the unknown summand G by a trivial summand $\underline{\mathbb{C}}^k$) is what makes the equivalence concrete.

Part (2). Define Φ on bundles by $\Phi(E) = [E] - \text{rk} E \cdot [\underline{\mathbb{C}}] \in K^0(X)$. Since $\text{rk}([E] - \text{rk} E [\underline{\mathbb{C}}]) = \text{rk} E - \text{rk} E = 0$, the image lies in $\tilde{K}^0(X)$. First, $\Phi(E) = \Phi(F)$ if and only if $E \cong_s F$. Indeed,

$$\Phi(E) = \Phi(F) \iff [E] + \text{rk} F [\underline{\mathbb{C}}] = [F] + \text{rk} E [\underline{\mathbb{C}}] \iff [E \oplus \underline{\mathbb{C}}^{\text{rk} F}] = [F \oplus \underline{\mathbb{C}}^{\text{rk} E}]$$

and by part (1) the last equality means $E \oplus \underline{\mathbb{C}}^{\text{rk} F} \cong_s F \oplus \underline{\mathbb{C}}^{\text{rk} E}$, that is $E \oplus \underline{\mathbb{C}}^{\text{rk} F + k} \cong F \oplus \underline{\mathbb{C}}^{\text{rk} E + k}$ for some k ; setting $m = \text{rk} F + k$ and $n = \text{rk} E + k$, this is precisely $E \cong_s F$. Conversely any stable equivalence $E \oplus \underline{\mathbb{C}}^m \cong F \oplus \underline{\mathbb{C}}^n$ runs the chain backwards, since it gives $[E] + m[\underline{\mathbb{C}}] = [F] + n[\underline{\mathbb{C}}]$ and hence $\Phi(E) = [E] - \text{rk} E [\underline{\mathbb{C}}] = [F] - \text{rk} F [\underline{\mathbb{C}}] = \Phi(F)$ after using $\text{rk} E + m = \text{rk} F + n$. Thus Φ is injective on stable equivalence classes.

For surjectivity, let $\eta \in \tilde{K}^0(X)$. By lemma 2.1.6(2) write $\eta = [E] - n$ for a bundle E ; applying rank gives $0 = \text{rk} \eta = \text{rk} E - n$, so $n = \text{rk} E$ and $\eta = [E] - \text{rk} E [\underline{\mathbb{C}}] = \Phi(E)$. Hence Φ is onto. The group operation $\Phi(E) + \Phi(F) = \Phi(E \oplus F)$ holds by additivity of rank and of the bracket, so the bijection transports $\oplus$ to the group law of $\tilde{K}^0(X)$, completing the proof.

The content of theorem 2.1.8 is that the reduced group $\tilde{K}^0(X)$ is not an abstract quotient but a concrete object: its elements are the stable equivalence classes of bundles, two bundles representing the same element exactly when they agree after stabilizing by trivial summands. This is the form in which K-theory becomes computable, because stable equivalence of bundles over a compact space is governed by maps into the infinite Grassmannian. Indeed the stabilization theorem of the previous chapter (theorem 1.3.1) identifies $\tilde{K}^0(X)$ with $\varinjlim_n [X, \text{Grass}_n] = [X, BU]$, and assembling this with the rank splitting gives a representing space for the full group.

Theorem 2.1.9: Representability of K^0

Let X be a connected compact Hausdorff space. There are natural isomorphisms

$$\tilde{K}^0(X) \cong [X, BU], \quad K^0(X) \cong [X, \mathbb{Z} \times BU]$$

where $BU = \varinjlim_n BU_n$ is the infinite complex Grassmannian and $[X, \mathbb{Z} \times BU]$ denotes homotopy classes of maps into $\mathbb{Z} \times BU$. In the real case the same holds with BU replaced by BO , giving $\widetilde{KO}^0(X) \cong [X, BO]$ and $KO^0(X) \cong [X, \mathbb{Z} \times BO]$.

Proof:

Step 1: the reduced group as maps into BU . By theorem 2.1.8, $\tilde{K}^0(X)$ is the set of stable equivalence classes of bundles over X . By the stabilization theorem (theorem 1.3.1), for X compact the natural maps assemble into an isomorphism

$$\tilde{K}^0(X) \cong \varinjlim_n \text{Vect}^n(X) \cong \varinjlim_n [X, BU_n]$$

where the colimit is taken along the stabilization maps $\text{Vect}^n(X) \rightarrow \text{Vect}^{n+1}(X)$, $E \mapsto E \oplus \mathbb{C}$, which on classifying maps correspond to the inclusions $BU_n \hookrightarrow BU_{n+1}$ given by $V \mapsto V \oplus \mathbb{C}$. Each identification $\text{Vect}^n(X) \cong [X, BU_n]$ is theorem 1.2.13 (in its complex form, with $\text{Grass}_n = BU_n$ the complex Grassmannian of definition 1.2.14). Since X is compact, every map $X \rightarrow BU = \varinjlim_n BU_n$ has image in some BU_n (a compact subset of the colimit $\varinjlim_n BU_n$ of the closed inclusions $BU_n \hookrightarrow BU_{n+1}$ meets only finitely many of the cells adjoined at each stage, hence lies in a single finite stage BU_n), and likewise the compact image of any homotopy $X \times [0, 1] \rightarrow BU$ between two such maps lands in some BU_N . Therefore

$$\varinjlim_n [X, BU_n] \cong [X, \varinjlim_n BU_n] = [X, BU]$$

Combining the displays gives $\tilde{K}^0(X) \cong [X, BU]$. This isomorphism is natural in X : a map $f: X' \rightarrow X$ induces $f^*: \text{Vect}^n(X) \rightarrow \text{Vect}^n(X')$ by pullback (proposition 1.2.1), which corresponds to precomposition with f on classifying maps, hence to $f^*: [X, BU] \rightarrow [X', BU]$.

Step 2: the full group as maps into $\mathbb{Z} \times BU$. Since X is connected, $[X, \mathbb{Z}] \cong \mathbb{Z}$, the homotopy classes of maps into the discrete space $\mathbb{Z}$ being the constant maps. The splitting of corollary 2.1.5 reads $K^0(X) \cong \tilde{K}^0(X) \oplus \mathbb{Z}$. Combining with Step 1,

$$K^0(X) \cong [X, BU] \oplus \mathbb{Z} \cong [X, BU] \times [X, \mathbb{Z}] \cong [X, BU \times \mathbb{Z}] = [X, \mathbb{Z} \times BU]$$

where the last identification uses that homotopy classes of maps into a product are pairs of homotopy classes, $[X, Y \times Z] \cong [X, Y] \times [X, Z]$, and that for connected X the factor $[X, \mathbb{Z}]$ recovers the rank. Naturality is inherited from Step 1 together with naturality of the rank splitting. The real case is identical, replacing BU_n by $BO_n = \text{Grass}_n(\mathbb{R}^\infty)$ throughout and invoking the real form of theorem 1.2.13, completing the proof.

The representability theorem reframes K-theory entirely in homotopy-theoretic terms: computing $K^0(X)$ is computing homotopy classes of maps from X into the fixed space $\mathbb{Z} \times BU$. This is the first sign that K-theory will be a generalized cohomology theory, since cohomology theories are exactly the functors represented by spaces (or spectra). The space $\mathbb{Z} \times BU$ is the degree-zero representing space; Bott periodicity, taken up in the next section, will show that the degree-one space is U and that these two spaces suffice to represent the whole theory. For now the value of the theorem is computational, as the closing examples illustrate.

So far $K^0(X)$ has only an additive structure. The tensor product of definition 1.1.17 supplies a multiplication, turning $K^0(X)$ into a commutative ring. The verification that the tensor product descends to the Grothendieck group is a matter of checking that the bilinearity and distributivity of $\otimes$ over $\oplus$ at the level of bundles survive the passage to formal differences.

Proposition 2.1.10: The Ring Structure on $K^0(X)$

Let X be a compact Hausdorff space. The tensor product of vector bundles (definition 1.1.17) induces on $K^0(X)$ a well-defined product

$$([E_1] - [F_1]) \cdot ([E_2] - [F_2]) = [E_1 \otimes E_2] + [F_1 \otimes F_2] - [E_1 \otimes F_2] - [F_1 \otimes E_2]$$

making $K^0(X)$ a commutative ring with unit $[\mathbb{C}]$, the class of the trivial line bundle. The reduced group $\tilde{K}^0(X)$ is the augmentation ideal of this ring, the kernel of the ring homomorphism $\text{rk}: K^0(X) \rightarrow \mathbb{Z}$. Pullback $f^*: K^0(X) \rightarrow K^0(X')$ along a continuous map is a ring homomorphism.

Proof:

Step 1: the product is well-defined. The assignment $(E_1, E_2) \mapsto [E_1 \otimes E_2]$ is a map $\text{Vect}(X) \times \text{Vect}(X) \rightarrow K^0(X)$. It is additive in each variable: by the distributivity of tensor over direct sum at the bundle level, $(E_1 \oplus E'_1) \otimes E_2 \cong (E_1 \otimes E_2) \oplus (E'_1 \otimes E_2)$ and symmetrically in the second variable (these isomorphisms are recorded in the discussion following definition 1.1.17, where the tensor product is shown to be commutative, associative, unital, and distributive over $\oplus$ fiberwise). Passing to classes,

$$[(E_1 \oplus E'_1) \otimes E_2] = [E_1 \otimes E_2] + [E'_1 \otimes E_2]$$

so the map is biadditive. A biadditive map from $\text{Vect}(X) \times \text{Vect}(X)$ into an abelian group factors through the tensor product of the Grothendieck groups, equivalently extends uniquely to a $\mathbb{Z}$ -bilinear map $K^0(X) \times K^0(X) \rightarrow K^0(X)$. Concretely, this forces the formula in the statement: expanding $([E_1] - [F_1])([E_2] - [F_2])$ by bilinearity gives $[E_1][E_2] - [E_1][F_2] - [F_1][E_2] + [F_1][F_2]$, which is the displayed expression. Independence of the representatives of each virtual bundle is exactly the assertion that the formula respects the relation $[E \oplus F] = [E] + [F]$, which biadditivity guarantees.

Step 2: ring axioms. Commutativity, associativity, and the unit law for the product on $K^0(X)$ follow from the corresponding properties of $\otimes$ on bundles, since the product is determined by its values $[E_1] \cdot [E_2] = [E_1 \otimes E_2]$ on the generating classes and these properties are stated for the bundle-level tensor product. Explicitly, $E_1 \otimes E_2 \cong E_2 \otimes E_1$ gives $[E_1][E_2] = [E_2][E_1]$; $(E_1 \otimes E_2) \otimes E_3 \cong E_1 \otimes (E_2 \otimes E_3)$ gives associativity; and $\mathbb{C} \otimes E \cong E$ gives $[\mathbb{C}][E] = [E]$, so $[\mathbb{C}]$ is the unit. Distributivity of the product over addition in $K^0(X)$ is the bilinearity established in Step 1. Hence $(K^0(X), +, \cdot)$ is a commutative ring with unit $[\mathbb{C}]$.

Step 3: rank is a ring map and $\tilde{K}^0$ is its kernel. For bundles over a connected base, $\text{rk}(E_1 \otimes E_2) = \text{rk} E_1 \cdot \text{rk} E_2$ since the fiber of $E_1 \otimes E_2$ over a point is the tensor product of the fibers, of dimension the product of the dimensions. Thus rk is multiplicative as well as additive, and $\text{rk}[\mathbb{C}] = 1$, so $\text{rk}: K^0(X) \rightarrow \mathbb{Z}$ is a unital ring homomorphism. Its kernel is $\tilde{K}^0(X)$ by definition 2.1.4, which is therefore an ideal, the augmentation ideal of the ring. (Over a base with r components, rk lands in the product ring $\mathbb{Z}^r$.)

Step 4: naturality. For a continuous $f: X' \rightarrow X$, the pullback satisfies $f^*(E_1 \otimes E_2) \cong f^*E_1 \otimes f^*E_2$ and $f^*(E_1 \oplus E_2) \cong f^*E_1 \oplus f^*E_2$ by proposition 1.2.3, and $f^*\mathbb{C} \cong \mathbb{C}$. Hence the induced map $f^*: K^0(X) \rightarrow K^0(X')$ preserves products, sums, and the unit, so it is a ring homomorphism, completing the proof.

With the ring structure in hand, K^0 is a contravariant functor from compact Hausdorff spaces to commutative rings, refining the homotopy-classification of bundles into an algebraic invariant that multiplies. The reduced group sits inside as the augmentation ideal, and one checks readily that the product of two reduced classes is again reduced (the product of two elements of an ideal lies in the ideal), so $\tilde{K}^0(X)$ is a non-unital ring in its own right. The two closing computations below show the smallest nontrivial values of the theory and confirm that the formal machinery agrees with direct calculation.

Example 2.1.11: K^0 of a Contractible Space

Let X be a contractible compact Hausdorff space, for instance a disk D^n or the interval $[0, 1]$. By corollary 1.2.6, every vector bundle over a contractible paracompact base is trivial, so $\text{Vect}^n(X)$ has a single element for each n and $\text{Vect}(X) \cong \mathbb{N}$ as monoids, with $E \mapsto \text{rk} E$. Exactly as in example 2.1.2, the Grothendieck group is

$$K^0(X) \cong \mathbb{Z}, \quad \tilde{K}^0(X) = 0$$

with the rank map an isomorphism onto $\mathbb{Z}$ and the reduced group trivial. This also follows from theorem 2.1.9: a contractible X admits only nullhomotopic maps into BU , so $\tilde{K}^0(X) = [X, BU] = 0$, and $K^0(X) = [X, \mathbb{Z} \times BU] \cong [X, \mathbb{Z}] = \mathbb{Z}$. The ring structure is the unique unital ring structure on $\mathbb{Z}$. Thus K-theory cannot distinguish a contractible space from a point, as a homotopy invariant must.

The contractible case shows that all the content of K-theory is in the reduced group, which vanishes whenever X has the homotopy type of a point. The smallest space with nontrivial reduced K-theory is the 2-sphere, where the canonical line bundle over $\mathbb{CP}^1 = S^2$ generates an infinite cyclic group. This computation is the prototype for every later calculation and the first concrete appearance of Bott periodicity.

Example 2.1.12: Reduced K-Theory of S^2

Let $X = S^2$, identified with the complex projective line $\mathbb{CP}^1$. By theorem 2.1.9,

$$\tilde{K}^0(S^2) \cong [S^2, BU] = \pi_2(BU)$$

the last equality because S^2 is the 2-sphere and homotopy classes of based maps from S^2 agree with free homotopy classes here, BU being simply connected (it is a colimit of the simply connected BU_n , and $\pi_1(BU_n) = \pi_0(U_n) = 0$). The space BU classifies stable bundles, and its homotopy is computed from that of the unitary group by the standard fibration relation $\pi_k(BU) \cong \pi_{k-1}(U)$, valid because BU is the classifying space of the stable unitary group $U = \varinjlim_n U(n)$. Hence

$$\pi_2(BU) \cong \pi_1(U) \cong \mathbb{Z}$$

where $\pi_1(U) \cong \mathbb{Z}$ is computed with no appeal to periodicity: the determinant $\det: U(n) \rightarrow U(1)$ is a surjective homomorphism with kernel $\text{SU}(n)$, which is simply connected for every $n \geq 2$ ([17, The Fundamental Groups of $\text{SU}(2)$, $\text{SO}(3)$, and $\text{SU}(n)$]), so the long exact sequence of the fibration $\text{SU}(n) \rightarrow U(n) \rightarrow U(1)$ gives $\pi_1(U(n)) \cong \pi_1(U(1)) = \pi_1(S^1) \cong \mathbb{Z}$. These isomorphisms are compatible with the inclusions $U(n) \hookrightarrow U(n+1)$, since $\det$ is, so $\pi_1(U) \cong \mathbb{Z}$ in the colimit. Therefore

$$\tilde{K}^0(S^2) \cong \mathbb{Z}$$

An explicit generator is the reduced class of the canonical (tautological) complex line bundle $H \rightarrow \mathbb{CP}^1$ introduced in section 1.2: the element $[H] - [\mathbb{C}] \in \tilde{K}^0(S^2)$ generates. To see that it

is a generator and not a proper multiple of one, compute its clutching function. Cover $\mathbb{CP}^1$ by $U_0 = \{z_0 \neq 0\}$ and $U_1 = \{z_1 \neq 0\}$ with coordinates $w = z_1/z_0$ and $v = z_0/z_1$, and trivialize H over them by the nowhere-zero sections $s_0[1 : w] = (1, w)$ and $s_1[v : 1] = (v, 1)$. On the overlap $v = w^{-1}$, so $s_1 = (w^{-1}, 1) = w^{-1}s_0$, and the transition function from the U_1 -trivialization to the U_0 -trivialization is $w \mapsto w^{-1}$. With $D_+ = \{|w| \leq 1\}$ and $D_- = \{|w| \geq 1\} \cup \{\infty\}$, the convention fixed in section 2.2, this says $H = [\mathbb{C}, w^{-1}]$.

Now $\text{Vect}_{\mathbb{C}}^1(S^2) \cong [S^1, \text{GL}_1(\mathbb{C})]$ by proposition 1.2.8, and since $\text{GL}_1(\mathbb{C}) = \mathbb{C}^\times$ is path-connected with abelian fundamental group, free homotopy classes agree with $\pi_1(\mathbb{C}^\times) \cong \mathbb{Z}$, detected by winding number. The class of H is the loop $w \mapsto w^{-1}$, of winding number -1 , hence a generator. Passing between $\text{GL}_n(\mathbb{C})$ and $U(n)$ is harmless: polar decomposition writes every invertible complex matrix uniquely as $g = up$ with u unitary and p positive-definite Hermitian ([17, The Classical Polar Decomposition]), and scaling p to the identity deformation retracts $\text{GL}_n(\mathbb{C})$ onto $U(n)$, so the inclusion induces an isomorphism on π_1 . Stabilizing then preserves the generator: writing $\iota : U(1) \rightarrow U(n)$ for $u \mapsto \text{diag}(u, 1, \dots, 1)$, one has $\det \circ \iota = \text{id}_{U(1)}$, so $\det_* \iota_* = \text{id}$ on π_1 , and since $\det_*$ is an isomorphism so is ι_* . Therefore the class of H generates $\pi_1(U(n))$ for every n , and by theorem 1.3.1 it generates the colimit $\tilde{K}^0(S^2)$. So $[H] - [\underline{\mathbb{C}}]$ generates. By corollary 2.1.5 the full group is

$$K^0(S^2) \cong \tilde{K}^0(S^2) \oplus \mathbb{Z} \cong \mathbb{Z} \oplus \mathbb{Z}$$

with the rank accounting for the second summand. As a ring, $K^0(S^2) \cong \mathbb{Z}[H]/(H-1)^2$, where H is shorthand for the class $[H]$ and the relation $(H-1)^2 = 0$ records that the reduced generator $h = [H] - 1$ squares to zero. This relation comes from an isomorphism of bundles: the canonical line bundle over $\mathbb{CP}^1$ satisfies $(H \otimes H) \oplus \underline{\mathbb{C}} \cong H \oplus H$, recorded in section 1.2. Passing to classes in the ring $K^0(S^2)$ and using proposition 2.1.10, this reads $[H]^2 + 1 = 2[H]$, that is $[H]^2 - 2[H] + 1 = ([H] - 1)^2 = 0$, so $h^2 = 0$. Since $\{1, h\}$ generates $K^0(S^2) \cong \mathbb{Z} \oplus \mathbb{Z}$ additively and $h^2 = 0$, the ring is exactly $\mathbb{Z}[h]/(h^2)$.

The relation $([H] - 1)^2 = 0$ in the last example is worth isolating, because it is the K-theoretic form of the geometric identity $(H \otimes H) \oplus \underline{\mathbb{C}} \cong H \oplus H$ over $\mathbb{CP}^1$. In ordinary cohomology the analogous generator squares to zero for dimensional reasons; here it squares to zero because of an isomorphism of vector bundles. This pattern, where a relation among bundles produces a relation in the K-theory ring, recurs in the computation of $K^0(\mathbb{CP}^n)$ in section 2.4, and the squaring-to-zero phenomenon over S^2 is the seed of the Bott periodicity isomorphism studied next.

Exercise 2.1.1

1. Let X be a compact Hausdorff space with r connected components. Show that $K^0(X) \cong \tilde{K}^0(X) \oplus \mathbb{Z}^r$, where $\tilde{K}^0(X)$ is the kernel of the total rank homomorphism $\text{rk} : K^0(X) \rightarrow \mathbb{Z}^r$. Deduce that for the two-point space $X = \{p, q\}$ one has $K^0(X) \cong \mathbb{Z}^2$ and $\tilde{K}^0(X) = 0$.
2. Let A be a commutative monoid. Its Grothendieck group $G(A)$ is the universal abelian group receiving a monoid homomorphism from A . Prove that the canonical map $A \rightarrow G(A)$ is injective if and only if A is cancellative (that is, $a + c = b + c$ implies $a = b$). Using lemma 2.1.6 and theorem 2.1.8, explain why the monoid $\text{Vect}(X)$ need not be cancellative, and identify exactly when two bundle classes $[E], [F]$ become equal in $K^0(X)$.
3. Show that the tangent bundle TS^2 represents the trivial element of the reduced real K-theory $\tilde{KO}^0(S^2)$, even though TS^2 is not the trivial bundle. (Use that $TS^2 \oplus NS^2 \cong \mathbb{R}^3$ for the normal bundle NS^2 , which is trivial, and pass to the appropriate reduced group.)

4. Let $f: X \rightarrow Y$ be a homotopy equivalence of compact Hausdorff spaces. Prove directly from theorem 2.1.9 that $f^*: K^0(Y) \rightarrow K^0(X)$ is a ring isomorphism. Conclude that K^0 is a homotopy-invariant functor.
5. Verify the ring isomorphism $K^0(S^2) \cong \mathbb{Z}[h]/(h^2)$, where $h = [H] - 1$ is the reduced class of the canonical line bundle. Specifically, show that $\{1, h\}$ is a $\mathbb{Z}$ -basis of $K^0(S^2)$ as an abelian group, that $h^2 = 0$, and that the rank homomorphism is the quotient $\mathbb{Z}[h]/(h^2) \rightarrow \mathbb{Z}$ sending $h \mapsto 0$.
6. Let $X = \{*\}$ and recompute $K^0(\{*\})$ as a ring using proposition 2.1.10. Then explain why, for any nonempty compact Hausdorff X , the unique map $X \rightarrow \{*\}$ and any choice of point $* \rightarrow X$ exhibit $\mathbb{Z} = K^0(\{*\})$ as a direct summand (indeed a unital subring) of $K^0(X)$, recovering the rank splitting of corollary 2.1.5 functorially.

2.2 Bott Periodicity

The group $\tilde{K}^0(X)$ of section 2.1 is one contravariant homotopy functor. What turns it into a cohomology theory is a periodicity: smashing a space with S^2 leaves its reduced K-theory unchanged. This section proves that statement and derives from it the periodicity of the stable homotopy groups of the unitary group.

Two formulations are in circulation. Bott's original theorem is a statement about homotopy groups, $\pi_k(U) \cong \pi_{k+2}(U)$, proved by Morse theory for the energy functional on the loop space of a symmetric space. The form K-theory consumes is multiplicative: external multiplication by a single class β on S^2 is an isomorphism $\tilde{K}^0(X) \rightarrow \tilde{K}^0(\Sigma^2 X)$ for every compact X . The argument below establishes the multiplicative form directly, running on the clutching-function description of bundles over a sphere from section 1.2, and obtains the homotopy-group form from it as a corollary.

This section works throughout in complex K-theory, writing $K = KU$ and $\tilde{K} = \widetilde{KU}$ unless stated otherwise, and restricts to compact Hausdorff spaces, where every bundle has a complement (proposition 1.1.16) and the representability theorem (theorem 2.1.9) applies. Throughout, X_+ denotes X with a disjoint basepoint adjoined, $\Sigma X = S^1 \wedge X$ is the reduced suspension of a pointed space, and X/A is the quotient of X by a closed subspace A , basepointed at the image of A . The smash product $X \wedge Y = (X \times Y)/(X \vee Y)$ is the reduced product. For a pointed space the reduced group $\tilde{K}^0(X)$ is the kernel of restriction to the basepoint, which agrees with the rank kernel of definition 2.1.4 when X is connected.

2.2.1 Convention: Where a CW Hypothesis Is Needed Two different hypotheses are in play from here on, and it is worth separating them once.

Everything proved by the clutching construction or by the bundle argument of lemma 2.2.3 needs only that the spaces are *compact Hausdorff* and the subspaces closed. This covers the whole of the product theorem theorem 2.2.15 in its unreduced ring form, and Axioms 1 to 3 of theorem 2.3.6.

Extending a cofiber sequence to the left is different: it needs the Puppe sequence, which [13, The Puppe Sequence and its Coexactness] establishes for *pointed CW complexes*. Everything resting on that extension is stated for finite CW complexes and finite CW pairs, whose basepoints are taken to be 0-cells. The complete list is: the reduced half of definition 2.2.6, hence definition 2.2.7; proposition 2.2.5; the reduced Bott-map clause of theorem 2.2.15; box 2.3.5 and with it the whole $\mathbb{Z}/2$ -graded family; Axiom 4 of theorem 2.3.6; and theorems 2.3.7 and 2.4.6 and corollaries 2.3.8 and 2.3.9; and within theorem 2.3.6, the periodicity clause of Axiom 5 and the multiplicativity claim, both of which consume periodicity. Axioms 1 to 3, and Axiom 5 in degrees ≤ 0 , need only compact Hausdorff.

Spheres, projective spaces and every space in section 2.4 are finite CW complexes, with one exception: box 2.4.4 concerns $\mathbb{C}P^\infty$, which is neither compact nor finite, so this book's K^0 is not defined on it. That remark records what the theory gives once it is extended to infinite complexes by representability, which is [13, The K-theory Spectrum KU], together with [13, The Milnor Exact Sequence]; the computation there is a report on that extension rather than a consequence of anything proved in this book.

The restriction is to the hypothesis the cited theorem actually carries. A weaker condition would do, since what the Puppe argument uses is the homotopy extension property ([13, Homotopy Extension Property]), which a CW pair has ([13, Homotopy Extension For CW Pair]); the CW form is taken because it is the hypothesis under which the Puppe sequence is stated there.

Before the product can be defined, one point about reduced groups has to be settled. The external product of two reduced classes will be a class on $X \times Y$ that vanishes on the wedge $X \vee Y$, and the natural home for such a class is the smash product $X \wedge Y = (X \times Y)/(X \vee Y)$. That the reduced K-theory of the smash is exactly the group of such classes is a statement about the collapse map, and it is proved below. No statement in this group of lemmas uses periodicity, and the general exact sequences of section 2.3 rest on lemma 2.2.3.

Lemma 2.2.2: Reduced K-Theory of a Wedge

Let A and B be compact Hausdorff pointed spaces. Restriction along the two inclusions gives an isomorphism

$$\tilde{K}^0(A \vee B) \xrightarrow{\cong} \tilde{K}^0(A) \oplus \tilde{K}^0(B)$$

Proof:

A vector bundle on $A \vee B$ is a bundle E_A on A , a bundle E_B on B , and an identification of their fibres over the common basepoint, since $A \vee B$ is the pushout of $A \leftarrow \{*\} \rightarrow B$ and a bundle on a pushout of compact Hausdorff spaces along a closed inclusion is exactly this gluing data. Reduced classes are those trivial at the basepoint, for which the gluing datum carries no information, so the assignment $E \mapsto (E|_A, E|_B)$ is a bijection on reduced classes. It is additive, so it induces the displayed isomorphism of groups. The inverse sends (a, b) to the class glued

from representatives trivialized at the basepoint, completing the proof.

Lemma 2.2.3: Exactness at the Middle Term

Let X be compact Hausdorff and $A \subseteq X$ a closed subspace, with $i: A \hookrightarrow X$ the inclusion and $q: X \rightarrow X/A$ the collapse. Then

$$\tilde{K}^0(X/A) \xrightarrow{q^*} \tilde{K}^0(X) \xrightarrow{i^*} \tilde{K}^0(A)$$

is exact at $\tilde{K}^0(X)$.

Proof:

Step 1: $\text{im } q^* \subseteq \ker i^*$. The composite $q \circ i$ sends all of A to the basepoint of X/A , so it factors through a point. Since $\tilde{K}^0$ of a point vanishes (definition 2.1.4), $i^* q^* = 0$.

Step 2: *A class trivial on A is trivial near A .* Let $\alpha \in \tilde{K}^0(X)$ with $i^* \alpha = 0$. Writing $\alpha = [E] - \text{rk } E$ and adding a trivial bundle, one may assume $E|_A$ is trivial. Fix a trivialization, equivalently sections $s_1, \dots, s_n$ of $E|_A$ forming a basis of each fibre over A . Cover A by finitely many open sets $U_1, \dots, U_r$ of X over which E is trivial. Shrink the cover first: $\{U_1, \dots, U_r, X \setminus A\}$ is an open cover of the compact Hausdorff, hence normal, space X , so it admits a shrinking ([8, Shrinking Lemma]); let $V_1, \dots, V_r$ be the shrunk members with $\bar{V}_j \subseteq U_j$. Since the last member is unchanged on A , the V_j still cover A . In a trivialization over U_j , the restriction $s_i|_{A \cap \bar{V}_j}$ is a continuous $\mathbb{C}^n$ -valued function on a subspace closed in X , so the Tietze extension theorem ([8, Tietze Extension Theorem]), whose hypothesis of normality is met by X itself, extends it coordinatewise in real and imaginary parts to a continuous $\mathbb{C}^n$ -valued function on X ; multiplying by a continuous bump function that is 1 on $\bar{V}_j$ and vanishing outside U_j ([8, Urysohn]) and reading the result back through the trivialization gives a section s_{ij} of E over U_j agreeing with s_i on $A \cap \bar{V}_j$. Let $\{\varphi_1, \dots, \varphi_r, \varphi\}$ be a partition of unity subordinate to the cover $\{V_1, \dots, V_r, X \setminus A\}$ of X , which exists because a compact Hausdorff space is paracompact ([8, Existence of Subordinate Partitions of Unity]), and set $\tilde{s}_i = \sum_j \varphi_j s_{ij}$, a global section of E restricting to s_i on A , because φ vanishes on A , the φ_j sum to 1 there, and each s_{ij} agrees with s_i on the support of φ_j .

Over each U_j the sections $\tilde{s}_1, \dots, \tilde{s}_n$ form a matrix-valued function whose determinant is nonzero at every point of $A \cap U_j$, hence nonzero on an open neighbourhood of $A \cap U_j$. The union of these neighbourhoods is an open $U \supseteq A$ over which $\tilde{s}_1, \dots, \tilde{s}_n$ trivialize E .

Step 3: *Descending to the quotient.* Let h denote the trivialization of $E|_A$ and let E/h be the quotient of E by the identifications $h^{-1}(x, v) \sim h^{-1}(y, v)$ for $x, y \in A$ and $v \in \mathbb{C}^n$. The projection $E \rightarrow X$ descends to $E/h \rightarrow X/A$. Away from the basepoint this is E itself, and over the image U/A of the neighbourhood U from Step 2 the trivialization of E over U descends to a trivialization of E/h , because it agrees with h along A . Hence E/h is a vector bundle on X/A . The quotient map $E \rightarrow E/h$ is an isomorphism on each fibre and covers q , so $q^*(E/h) \cong E$, giving $\alpha \in \text{im } q^*$ and completing the proof.

Lemma 2.2.4: Collapsing a Contractible Subspace

Let X be compact Hausdorff and $A \subseteq X$ a closed contractible subspace. Then $q^*: \tilde{K}^0(X/A) \rightarrow \tilde{K}^0(X)$ is an isomorphism.

Proof:

A bundle E on X is trivial over A , since A is contractible and homotopy invariance (theorem 1.2.5) makes $E|_A$ isomorphic to the pullback of a fibre. Choosing a trivialization h , Step 3 of lemma 2.2.3 produces a bundle E/h on X/A with $q^*(E/h) \cong E$, so q^* is surjective on isomorphism classes and hence on $\tilde{K}^0$. For injectivity, two trivializations of $E|_A$ differ by a map $A \rightarrow \mathrm{GL}_n(\mathbb{C})$, which is null-homotopic because A is contractible, and a null-homotopy gives an isomorphism between the two descended bundles by theorem 1.2.5 applied over $X/A \times [0, 1]$. So the descended bundle is well defined up to isomorphism, and $E \mapsto E/h$ is a two-sided inverse to q^* , completing the proof.

Exactness extends one step to the left by the standard cone construction, and that one step is what makes the smash identification an isomorphism rather than merely a surjection.

Proposition 2.2.5: The Smash Product Carries the Reduced Product

Let X and Y be finite pointed CW complexes (box 2.2.1), let $r: X \vee Y \hookrightarrow X \times Y$ be the inclusion and $q: X \times Y \rightarrow X \wedge Y$ the collapse. Then

$$q^*: \tilde{K}^0(X \wedge Y) \longrightarrow \tilde{K}^0(X \times Y)$$

is injective, with image exactly $\ker r^*$.

Proof:

The image is $\ker r^*$ by lemma 2.2.3 applied to the pair $(X \times Y, X \vee Y)$, so only injectivity is at issue.

Step 1: Extending the sequence. Let $A = X \vee Y$ and $Z = X \times Y$, and let $C_i = Z \cup CA$ be the mapping cone of $i: A \hookrightarrow Z$, formed with the *reduced* cone $CA = (A \times [0, 1]) / (A \times \{1\} \cup \{a_0\} \times [0, 1])$ glued along $A \times \{0\} = A$, where a_0 is the basepoint. The reduced cone is used rather than the unreduced one because CA/A is then the reduced suspension $\Sigma A = S^1 \wedge A$ on the nose, which is what Σ denotes throughout; the unreduced cone would give the unreduced suspension, for which the identification $\Sigma(X \vee Y) = \Sigma X \vee \Sigma Y$ used in Step 2 fails. The reduced cone is contractible by the straight-line contraction to the cone point. Every space occurring below is built from finite CW complexes by attaching cones and collapsing subcomplexes, so all are again finite CW complexes, in particular compact Hausdorff, and lemmas 2.2.3 and 2.2.4 apply throughout.

Collapsing CA gives $C_i/CA = Z/A = X \wedge Y$, so lemma 2.2.4 makes $\tilde{K}^0(X \wedge Y) \rightarrow \tilde{K}^0(C_i)$ an isomorphism. Collapsing Z gives $C_i/Z = CA/A = \Sigma A$. Applying lemma 2.2.3 to the pair (C_i, Z) and transporting along the first isomorphism yields exactness at $\tilde{K}^0(X \wedge Y)$ in

$$\tilde{K}^0(\Sigma A) \xrightarrow{p^*} \tilde{K}^0(X \wedge Y) \xrightarrow{q^*} \tilde{K}^0(Z)$$

where $p: C_i \rightarrow C_i/Z = \Sigma A$ is the collapse. So $\ker q^* = \mathrm{im} p^*$, and it suffices to show $p^* = 0$.

Set $V = C_i \cup CZ$, the mapping cone of $Z \hookrightarrow C_i$. Then $V/C_i = CZ/Z = \Sigma Z$ and $V/CZ = C_i/Z = \Sigma A$, so lemma 2.2.3 applied to (V, C_i) and lemma 2.2.4 applied to (V, CZ) give exactness at $\tilde{K}^0(\Sigma A)$ in

$$\tilde{K}^0(\Sigma Z) \xrightarrow{\alpha} \tilde{K}^0(\Sigma A) \xrightarrow{p^*} \tilde{K}^0(C_i)$$

the outgoing map being p^* because the composite $C_i \hookrightarrow V \rightarrow V/CZ$ is p . It follows that $p^* = 0$, and hence that q^* is injective, as soon as α is surjective.

The map α is the transported collapse $V \rightarrow V/C_i$, and identifying it is the one point that does not follow from the two lemmas above. It is the content of the Puppe sequence ([13, The Puppe Sequence and its Coexactness]): every three consecutive terms of $A \rightarrow Z \rightarrow C_i \xrightarrow{p} \Sigma A \xrightarrow{\Sigma i} \Sigma Z \rightarrow \dots$ form, up to based homotopy equivalence, the cofiber sequence of the first map, so the triple $(C_i, \Sigma A, \Sigma Z)$ identifies α with $(\Sigma i)^*$ up to the self-equivalence of ΣA reversing the suspension coordinate. That reversal is an isomorphism, so it does not affect surjectivity, though it does move the image. The cited proposition applies on its own hypothesis here: $A = X \vee Y$ and $Z = X \times Y$ are finite CW complexes, as are all the cones and quotients formed from them. Therefore q^* is injective as soon as $(\Sigma i)^*$ is surjective.

Step 2: The restriction is surjective. Suspension carries the wedge to a wedge, $\Sigma A = \Sigma X \vee \Sigma Y$, so by lemma 2.2.2 an element of $\tilde{K}^0(\Sigma A)$ is a pair (a, b) with $a \in \tilde{K}^0(\Sigma X)$ and $b \in \tilde{K}^0(\Sigma Y)$. The projections $p_X: Z \rightarrow X$ and $p_Y: Z \rightarrow Y$ suspend to maps $\Sigma Z \rightarrow \Sigma X$ and $\Sigma Z \rightarrow \Sigma Y$, and the class $(\Sigma p_X)^*a + (\Sigma p_Y)^*b \in \tilde{K}^0(\Sigma Z)$ restricts to (a, b) : on the ΣX summand, p_X composed with the inclusion $X \hookrightarrow Z$ is the identity, while p_Y composed with it is the constant map to the basepoint, whose suspension is constant and so pulls every reduced class back to zero. The same computation holds on the ΣY summand. Hence the restriction is surjective and q^* is injective, completing the proof.

K-theory carries a multiplication finer than the internal ring structure of proposition 2.1.10: an external product that combines classes on two different spaces into a class on their product. This is what upgrades K-theory from a sequence of abelian groups to a multiplicative cohomology theory, and it defines the Bott map.

Definition 2.2.6: External Product

Let X and Y be compact Hausdorff spaces. The *external product* is the homomorphism

$$\mu: K^0(X) \otimes K^0(Y) \longrightarrow K^0(X \times Y)$$

defined on generators by $[E] \otimes [F] \mapsto [p_X^* E \otimes p_Y^* F]$, where $p_X: X \times Y \rightarrow X$ and $p_Y: X \times Y \rightarrow Y$ are the projections and $\otimes$ on the right is the tensor product of bundles (definition 1.1.17). On reduced groups, the external product descends to a map

$$\tilde{\mu}: \tilde{K}^0(X) \otimes \tilde{K}^0(Y) \longrightarrow \tilde{K}^0(X \wedge Y)$$

when X and Y are finite pointed CW complexes: the product of two reduced classes restricts to zero on $X \vee Y$, so it is q^* of a class on the smash by lemma 2.2.3, and that class is unique by the injectivity in proposition 2.2.5, which is what needs the CW hypothesis (box 2.2.1). Then $\tilde{\mu}$ is defined to be that class.

The external product is well-defined because pullback commutes with tensor product (proposition 1.2.3)

and because both operations respect direct sums, so the assignment is additive in each variable and passes to the Grothendieck group of definition 2.1.1. Its reduction to the smash product is the algebraic form of the geometric fact that a class trivial on either factor restricts trivially to the wedge $X \vee Y \subseteq X \times Y$, hence factors through the quotient $X \wedge Y = (X \times Y)/(X \vee Y)$. The next definition extracts from this product the construction periodicity is stated in terms of.

Definition 2.2.7: The Bott Generator and Bott Map

Let $H \rightarrow \mathbb{CP}^1 = S^2$ be the canonical complex line bundle (section 1.2). The *Bott generator* is the reduced class

$$\beta = [H] - 1 \in \tilde{K}^0(S^2)$$

The *Bott map* for a finite pointed CW complex X is the homomorphism

$$\beta_X: \tilde{K}^0(X) \longrightarrow \tilde{K}^0(S^2 \wedge X) = \tilde{K}^0(\Sigma^2 X), \quad \alpha \longmapsto \tilde{\mu}(\beta \otimes \alpha)$$

given by external multiplication with β , using the homeomorphism $S^2 \wedge X \cong \Sigma^2 X$.

The Bott map is the comparison morphism whose invertibility is periodicity. It takes a virtual bundle on X and tensors it, externally, with the Bott class on S^2 . The resulting class lives on the double suspension $\Sigma^2 X$ because β is a reduced class on S^2 and the external product of reduced classes lands in the smash. That β_X is an isomorphism is the periodicity theorem, and the rest of this section proves it.

The argument computes $K^0(X \times S^2)$ outright, by presenting every bundle over $X \times S^2$ through clutching data over X and then normalizing that data until only two kinds remain. Section 1.2 carried out the unparametrized case, where the base was a sphere and a clutching function was a map $S^{k-1} \rightarrow \text{GL}_n(\mathbb{C})$. What is needed now is the version with a parameter space X carried along, where the constant group $\text{GL}_n(\mathbb{C})$ is replaced by the automorphisms of a bundle over X .

Write $S^2 = D_+ \cup D_-$ for the decomposition into closed upper and lower hemispheres, meeting in the equator $S^1 = D_+ \cap D_-$. Write $\pi_{\pm}: X \times D_{\pm} \rightarrow X$ and $\pi: X \times S^1 \rightarrow X$ for the projections to X from a hemisphere and from the equator, and $\pi_X: X \times S^2 \rightarrow X$, $\pi_{S^2}: X \times S^2 \rightarrow S^2$ for the two projections off the product.

Definition 2.2.8: Clutching Data over $X \times S^2$

Let $E \rightarrow X$ be a complex vector bundle over a compact Hausdorff space. A *clutching function* for E is a continuous section f of the bundle $\text{Aut}(\pi^* E) \rightarrow X \times S^1$, so that $f(x, z)$ is a linear automorphism of the fibre E_x for every $x \in X$ and $z \in S^1$. The *clutched bundle* is

$$[E, f] = (\pi_+^* E \sqcup \pi_-^* E) / \sim, \quad (x, z, v)_- \sim (x, z, f(x, z)v)_+ \quad (z \in S^1, v \in E_x)$$

a complex vector bundle over $X \times S^2$ of the same rank as E .

Two clutching functions for the same E are *homotopic* when they are connected by a path of clutching functions, that is by a section of $\text{Aut}(\pi^* E)$ over $X \times S^1 \times [0, 1]$. The next proposition collects everything about this construction that the proof uses.

Proposition 2.2.9: Properties of Parametrized Clutching

Let X be compact Hausdorff and let $E, E' \rightarrow X$ be complex vector bundles.

1. If $f \simeq g$ as clutching functions for E , then $[E, f] \cong [E, g]$.
2. $[E, f] \oplus [E', f'] \cong [E \oplus E', f \oplus f']$ and $[E, f] \otimes [E', f'] \cong [E \otimes E', f \otimes f']$.
3. $[E, \text{id}] \cong \pi_X^* E$, so its class in $K^0(X \times S^2)$ lies in the image of π_X^* .
4. Every complex vector bundle over $X \times S^2$ is isomorphic to $[E, f]$ for some bundle $E \rightarrow X$ and some clutching function f for E .
5. If α is a section of $\text{Aut}(\pi_+^* E)$ over $X \times D_+$ and γ a section of $\text{Aut}(\pi_-^* E)$ over $X \times D_-$, then $[E, \alpha|_{X \times S^1} f \gamma|_{X \times S^1}] \cong [E, f]$.
6. The presentation in (4) may be taken *normalized*, meaning $f(x, 1) = \text{id}_{E_x}$ for every x , where $1 \in S^1$ is the basepoint. Any two normalized clutching functions for the same E presenting isomorphic bundles are homotopic through normalized clutching functions.

Proof:

1. A homotopy F of clutching functions over $X \times S^1 \times [0, 1]$ is itself a clutching function for the bundle $E \times [0, 1]$ over $X \times [0, 1]$, so it clutches to a bundle $[E \times [0, 1], F]$ over $X \times [0, 1] \times S^2$ restricting to $[E, f]$ at $t = 0$ and to $[E, g]$ at $t = 1$. Homotopy invariance (theorem 1.2.5) applied to the two inclusions $X \times S^2 \rightrightarrows X \times [0, 1] \times S^2$, which are homotopic, gives $[E, f] \cong [E, g]$.
2. Both follow from the construction: the gluing relation for a direct sum is the direct sum of the gluing relations, and likewise for tensor products, because $\pi_\pm^*$ commutes with $\oplus$ and $\otimes$ (proposition 1.2.3).
3. With $f = \text{id}$ the relation identifies $(x, z, v)_-$ with $(x, z, v)_+$, so the quotient is the pullback of E along the projection, which is $\pi_X^* E$.
4. Let $F \rightarrow X \times S^2$ be a bundle. Each hemisphere $D_\pm$ is contractible, so the inclusion $X \times \{n_\pm\} \hookrightarrow X \times D_\pm$ at the pole $n_\pm$ is a deformation retract, and homotopy invariance (theorem 1.2.5) gives isomorphisms $h_\pm: F|_{X \times D_\pm} \xrightarrow{\cong} \pi_\pm^* E_\pm$ where $E_\pm = F|_{X \times \{n_\pm\}}$. The poles lie in the path-connected space S^2 , so $E_+ \cong E_-$ by homotopy invariance again. Fix such an isomorphism and call the common bundle E . Over the equator the two trivializations differ by $f = h_+ h_-^{-1}$, a section of $\text{Aut}(\pi^* E)$ over $X \times S^1$, and by construction $F \cong [E, f]$.
5. Over $X \times D_+$ the automorphism α changes the chosen isomorphism h_+ to αh_+ , and over $X \times D_-$ the automorphism γ^{-1} changes h_- to $\gamma^{-1} h_-$. The clutched bundle is unchanged because it is the quotient by the relation these isomorphisms transport, and the new comparison over the equator is $(\alpha h_+)(\gamma^{-1} h_-)^{-1} = \alpha f \gamma$.
6. Given any $h_\pm$ as in the proof of (4), replace $h_\pm$ by $(h_\pm|_{X \times \{1\}})^{-1} \circ h_\pm$, which is again an isomorphism onto $\pi_\pm^* E$ and restricts to the identity over $X \times \{1\}$. The resulting $f = h_+ h_-^{-1}$ then satisfies $f(x, 1) = \text{id}$, so a normalized presentation exists.

For the uniqueness statement, let $f_0 = h_+^0(h_-^0)^{-1}$ and $f_1 = h_+^1(h_-^1)^{-1}$ be normalized clutching functions for E with $[E, f_0] \cong [E, f_1]$. Transporting the trivializations $h_\pm^1$ through such an isomorphism, both arise from presentations of a single bundle, so it suffices to compare two normalized choices of $h_\pm$ for one bundle. Two such choices differ by $g_\pm = h_\pm^1(h_\pm^0)^{-1}$, a section of $\text{Aut}(\pi_\pm^* E)$ over $X \times D_\pm$ with $g_\pm|_{X \times \{1\}} = \text{id}$. Since $D_\pm$ is a disk and 1 lies on its boundary circle, there is a deformation retraction $r_s: D_\pm \rightarrow D_\pm$ of $D_\pm$ onto $\{1\}$ with $r_0 = \text{id}$ and $r_1 \equiv 1$. Then $s \mapsto g_\pm \circ (\text{id}_X \times r_s)$ is a homotopy from $g_\pm$ to the constant id , through sections of $\text{Aut}(\pi_\pm^* E)$ that restrict to id over $X \times \{1\}$ at every stage. Restricting these homotopies to the equator and composing as in (5) joins f_0 to f_1 through clutching functions, each normalized because each $g_\pm$ stage is the identity at $z = 1$.

establishing each of the six claims in turn and completing the proof.

Part (5) is what every simplification below uses: a clutching function may be modified on the left by a section of $\text{Aut}(\pi_+^* E)$ over $X \times D_+$ and on the right by a section of $\text{Aut}(\pi_-^* E)$ over $X \times D_-$ without changing the clutched bundle. The two sides are not interchangeable, and what is required of the factor is continuity over the relevant hemisphere, not holomorphy. A factor independent of z extends over both hemispheres and so may be absorbed on either side.

Fix from now on the identification $S^2 = \mathbb{C} \cup \{\infty\}$ with $D_+ = \{|z| \leq 1\}$ and $D_- = \{|z| \geq 1\} \cup \{\infty\}$, so that a factor polynomial in z extends over D_+ and a factor polynomial in z^{-1} extends over D_- . With this convention the canonical line bundle $H \rightarrow \mathbb{CP}^1 = S^2$ has clutching function $z \mapsto z^{-1}$ (example 1.2.7), so $H = [\mathbb{C}, z^{-1}]$ and its dual is $H^\vee = [\mathbb{C}, z]$. The dual is written $H^\vee$, not H^* , throughout: in this book H^* always means cohomology. By part (2), multiplying a clutching function by z^m tensors the clutched bundle with $(H^\vee)^{\otimes m}$:

$$[E, z^m f] \cong [E, f] \otimes \pi_{S^2}^*((H^\vee)^{\otimes m}) \quad (2.1)$$

In $K^0(S^2)$ the class $[H] = 1 + \beta$ is a unit, with inverse $[H^*] = 1 - \beta$ because $\beta^2 = 0$ (example 2.1.12), so twisting by a power of H is an invertible operation on $K^0(X \times S^2)$ and nothing is lost by multiplying a clutching function by a power of z .

The proof now proceeds in three normalizations. An arbitrary clutching function is first replaced by one whose entries are Laurent polynomials in z ; the Laurent polynomial is then cleared to a polynomial and the polynomial reduced to degree one at the cost of enlarging E by summands that clutch trivially; and finally a degree-one clutching function is shown to split the bundle into a part on which it is homotopic to z and a part on which it is homotopic to the identity. Only the first step is analytic.

Lemma 2.2.10: Laurent Approximation of Clutching Functions

Let $E \rightarrow X$ be a complex vector bundle over a compact Hausdorff space and let f be a clutching function for E . Then f is homotopic, through clutching functions for E , to one of the form

$$\ell(x, z) = \sum_{k=-N}^N a_k(x) z^k$$

for some $N \geq 0$, where each a_k is a continuous section of $\text{End}(E)$ over X . Moreover, if two Laurent polynomial clutching functions for E are homotopic through clutching functions, then they are homotopic through Laurent polynomial clutching functions.

Proof:

Step 0: A norm. Choose a Hermitian metric on E (proposition 1.1.14; the construction there is stated for a real bundle, and the Hermitian case is the same partition-of-unity patching applied to local Hermitian forms, which are convex) and let $\|\cdot\|_x$ be the associated operator norm on $\text{End}(E_x)$. Fix a finite cover of X by open sets over which E is trivial, together with compact sets $X_1, \dots, X_r$ covering X with X_i inside the i -th open set. This is possible because X is compact Hausdorff, hence normal. Over each X_i a trivialization identifies $\text{End}(E)$ with $X_i \times M_n(\mathbb{C})$, and on the compact X_i the transported norm and any fixed norm on $M_n(\mathbb{C})$ are comparable with constants independent of the point. Statements below of the form “uniformly on $X \times S^1$ ” are meant with respect to $\|\cdot\|$, and are checked on each $X_i \times S^1$ in a trivialization, where ordinary uniform continuity of a matrix-valued function on a compact metric space applies.

Step 1: Fourier coefficients. For $k \in \mathbb{Z}$ define a section a_k of $\text{End}(E)$ over X by the fibrewise integral

$$a_k(x) = \frac{1}{2\pi} \int_0^{2\pi} f(x, e^{i\theta}) e^{-ik\theta} d\theta$$

For fixed x the integrand takes values in the single finite-dimensional vector space $\text{End}(E_x)$, so the integral is the coordinatewise Riemann integral in any basis. This is independent of the choice of trivialization for the following reason, which is the whole content of the step: π^*E has transition functions pulled back from X , so they do not depend on z , and a change of trivialization conjugates the integrand by a matrix $g(x)$ constant in θ . Conjugation is linear, so it passes through the integral:

$$\int_0^{2\pi} g(x) f(x, e^{i\theta}) g(x)^{-1} e^{-ik\theta} d\theta = g(x) \left(\int_0^{2\pi} f(x, e^{i\theta}) e^{-ik\theta} d\theta \right) g(x)^{-1}$$

Continuity of a_k in x is checked on each X_i , where f is a matrix-valued function, uniformly continuous on the compact $X_i \times S^1$, so the integrals converge uniformly in x .

Step 2: Cesàro means converge uniformly. Let

$$\sigma_N f(x, z) = \sum_{|k| \leq N} \left(1 - \frac{|k|}{N+1}\right) a_k(x) z^k$$

be the N -th Cesàro mean of the Fourier series of f . Substituting the definition of a_k and summing the resulting geometric sum gives $\sigma_N f(x, e^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} f(x, e^{i(\theta-\varphi)}) K_N(\varphi) d\varphi$, where

K_N is the Fejér kernel. Fejér's theorem ([16, Fejér's Theorem]) states $K_N \geq 0$ and that K_N has total mass 1, and its closed form

$$K_N(\varphi) = \frac{1}{N+1} \left(\frac{\sin((N+1)\varphi/2)}{\sin(\varphi/2)} \right)^2$$

gives the third property the estimate needs: for $\delta \leq |\varphi| \leq \pi$ the denominator is bounded below by $\sin(\delta/2) > 0$ and the numerator by 1, so the mass of K_N outside $|\varphi| < \delta$ is at most $((N+1)\sin^2(\delta/2))^{-1}$, which tends to 0. (That source normalizes on $\mathbb{R}/\mathbb{Z}$ and defines σ_N as the average of the partial sums, and the displayed weights $1 - |k|/(N+1)$ are what that average produces, and the change of variable to $[0, 2\pi]$ rescales both the kernel and the measure, leaving the total mass at 1.)

Now for fixed x and θ , both $\sigma_N f(x, e^{i\theta})$ and $f(x, e^{i\theta})$ lie in the same space $\text{End}(E_x)$, so their difference is

$$\sigma_N f(x, e^{i\theta}) - f(x, e^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} \left(f(x, e^{i(\theta-\varphi)}) - f(x, e^{i\theta}) \right) K_N(\varphi) d\varphi$$

using that K_N has total mass 1. Taking norms and using $\| \int g \| \leq \int \|g\|$, valid for a continuous integrand in a finite-dimensional normed space because it holds for each Riemann sum and passes to the limit, together with $K_N \geq 0$, bounds the left side by $\frac{1}{2\pi} \int_0^{2\pi} \|f(x, e^{i(\theta-\varphi)}) - f(x, e^{i\theta})\| K_N(\varphi) d\varphi$. Given $\varepsilon > 0$, Step 0 supplies a δ with $\|f(x, e^{i(\theta-\varphi)}) - f(x, e^{i\theta})\| < \varepsilon/2$ whenever $|\varphi| < \delta$, uniformly in x and θ ; splitting the integral at $|\varphi| = \delta$ and using the mass bound above on the outer part gives $\sigma_N f \rightarrow f$ uniformly on $X \times S^1$.

Step 3: Nearby sections are automorphisms. The set of automorphisms is open in End : for $g \in \text{Aut}(E_x)$ and $u \in \text{End}(E_x)$ with $\|u\| < \|g^{-1}\|^{-1}$, the endomorphism $g + u = g(\text{id} + g^{-1}u)$ is invertible because $\|g^{-1}u\| < 1$ and the Neumann series converges. The function $(x, z) \mapsto \|f(x, z)^{-1}\|^{-1}$ is continuous and positive on the compact $X \times S^1$, so it has a positive minimum ε . Choose N with $\|\sigma_N f - f\| < \varepsilon$ uniformly. Then $\ell = \sigma_N f$ is a clutching function for E , and the straight-line homotopy $f_t = (1-t)f + t\ell$ satisfies $\|f_t - f\| \leq \|\ell - f\| < \varepsilon$ for every t , so each f_t is an automorphism. Thus $f \simeq \ell$ through clutching functions, and ℓ has the stated Laurent form.

Step 4: Homotopies may be taken Laurent. Let ℓ_0, ℓ_1 be Laurent polynomial clutching functions for E and let F be a homotopy of clutching functions between them, that is a section of $\text{Aut}(\pi^*E)$ over $X \times S^1 \times [0, 1]$. Regard F as a clutching function for the bundle $E \times [0, 1]$ over the compact Hausdorff space $X \times [0, 1]$. Steps 0 to 3 apply verbatim with X replaced by $X \times [0, 1]$, producing a Laurent polynomial clutching function $L(x, t, z) = \sum_{|k| \leq M} c_k(x, t) z^k$ for $E \times [0, 1]$, with $\|L - F\| < \varepsilon$ uniformly, where ε is the minimum of $\|F^{-1}\|^{-1}$ over the compact $X \times S^1 \times [0, 1]$.

Then $t \mapsto L(\cdot, t, \cdot)$ is a homotopy through Laurent polynomial clutching functions for E , from L_0 to L_1 . Its endpoints satisfy $\|L_0 - \ell_0\| < \varepsilon$ and $\|L_1 - \ell_1\| < \varepsilon$, since $F_0 = \ell_0$ and $F_1 = \ell_1$, so the straight-line homotopies $(1-s)\ell_0 + sL_0$ and $(1-s)L_1 + s\ell_1$ consist of automorphisms by the estimate of Step 3. Each stage of a straight-line homotopy between Laurent polynomials of degree at most $\max(M, \deg \ell_0, \deg \ell_1)$ is again such a Laurent polynomial. Concatenating $\ell_0 \rightarrow L_0$, $L_0 \rightarrow L_1$, $L_1 \rightarrow \ell_1$ gives a homotopy from ℓ_0 to ℓ_1 through Laurent polynomial clutching functions, completing the proof.

Multiplying by z^N clears the negative powers, which by (2.1) only twists the clutched bundle by a power of H^* . So it is enough to normalize *polynomial* clutching functions, and the next step removes all but the linear term.

Lemma 2.2.11: Linearization of Polynomial Clutching Functions

Let $p(x, z) = \sum_{k=0}^n a_k(x)z^k$ be a clutching function for E whose coefficients are sections of $\text{End}(E)$. Then there is a clutching function Lp for $E^{\oplus(n+1)}$, of the form $Lp(x, z) = B(x)z + A(x)$ with A, B sections of $\text{End}(E^{\oplus(n+1)})$, such that

$$[E^{\oplus(n+1)}, Lp] \cong [E, p] \oplus \pi_X^*(E^{\oplus n})$$

Proof:

Define, in block form with respect to $E^{\oplus(n+1)} = E \oplus \cdots \oplus E$,

$$Lp(x, z) = \begin{pmatrix} a_0 & a_1 & a_2 & \cdots & a_n \\ -z & 1 & 0 & \cdots & 0 \\ 0 & -z & 1 & & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -z & 1 \end{pmatrix}$$

which is of the form $Bz + A$ with A the matrix of a_k 's and 1's and B the matrix with -1 in the subdiagonal blocks.

Step 1: Lp is invertible on the equator. Perform the column operations $C_1 \mapsto C_1 + zC_2$, then $C_1 \mapsto C_1 + z^2C_3$, and so on up to $C_1 \mapsto C_1 + z^nC_{n+1}$. The first replaces the column-1 entries $(a_0, -z, 0, \dots)^T$ by $(a_0 + za_1, 0, -z^2, 0, \dots)^T$; the second replaces these by $(a_0 + za_1 + z^2a_2, 0, 0, -z^3, \dots)^T$; after n such operations column 1 is $(p(x, z), 0, \dots, 0)^T$. Next clear the first row from the bottom up: for $j = n+1, n, \dots, 2$ in that order, the row operation $R_1 \mapsto R_1 - (\text{entry in position } (1, j)) \cdot R_j$ removes the $(1, j)$ entry, and since R_j has zeros in columns 1 and $j+1, \dots, n+1$ it disturbs only the $(1, j-1)$ entry, which is cleared at the next step. The result is

$$p(x, z) \oplus T(z), \quad T(z) = \begin{pmatrix} 1 & & & & \\ -z & 1 & & & \\ & \ddots & \ddots & & \\ & & & \ddots & \\ & & & -z & 1 \end{pmatrix}$$

an $n \times n$ lower triangular block matrix with 1 on the diagonal and $-z$ on the subdiagonal. The row and column operations clear the first row and column only. The subdiagonal entries of T are never touched, and clearing them by further row operations would reintroduce a $-z^2$ one column to the left at each stage.

Every operation used is an elementary block operation whose multiplier is a polynomial in z with coefficients in $\text{End}(E)$, so each is realized by a block-elementary matrix, which is invertible with inverse of the same shape. Since $T(z)$ is lower triangular with 1 on the diagonal it is invertible for every z . Hence Lp is invertible at (x, z) precisely when $p(x, z)$ is, and p is a clutching function, so Lp is invertible on all of $X \times S^1$ and is a clutching function for $E^{\oplus(n+1)}$.

Step 2: The elementary operations are homotopies. A block-elementary matrix $I + \lambda e_{ij}$ with $i \neq j$ is connected to the identity through invertible matrices by $t \mapsto I + t\lambda e_{ij}$, $t \in [0, 1]$, each of which is invertible since $e_{ij}^2 = 0$ for $i \neq j$. When λ depends on z this is a homotopy through clutching functions, because invertibility holds at every (x, z, t) . Applying proposition 2.2.9(1) to the composite of these homotopies,

$$[E^{\oplus(n+1)}, Lp] \cong [E^{\oplus(n+1)}, p \oplus T]$$

Step 3: Contracting the triangular block. For $s \in [0, 1]$ let $T_s(z)$ be $T(z)$ with the subdiagonal entries $-z$ replaced by $-sz$. Each $T_s(z)$ is lower triangular with 1 on the diagonal, hence invertible for every s, x and z , so $s \mapsto p \oplus T_s$ is a homotopy of clutching functions from $p \oplus T$ to $p \oplus T_0 = \text{diag}(p, 1, \dots, 1)$. By proposition 2.2.9(1) again,

$$[E^{\oplus(n+1)}, Lp] \cong [E^{\oplus(n+1)}, \text{diag}(p, 1, \dots, 1)]$$

Step 4: Splitting the block-diagonal form. By proposition 2.2.9(2) the right-hand side is $[E, p] \oplus [E^{\oplus n}, \text{id}]$, and by (3) the second summand is $\pi_X^*(E^{\oplus n})$, which is the claim, completing the proof.

Two recursions relate the linearizations at consecutive degrees. They are what makes the invariant constructed at the end of this section independent of the degree chosen to linearize at, and they are proved by the same column operations as lemma 2.2.11.

Lemma 2.2.12: Recursions for the Linearization

Let p be a polynomial clutching function for E of degree at most n , regarded also as one of degree at most $n + 1$ by setting $a_{n+1} = 0$. Then

1. $[E^{\oplus(n+2)}, L^{n+1}p] \cong [E^{\oplus(n+1)}, L^n p] \oplus [E, \text{id}]$;
2. $[E^{\oplus(n+2)}, L^{n+1}(zp)] \cong [E^{\oplus(n+1)}, L^n p] \oplus [E, z]$.

Proof:

Throughout, an elementary block operation with multiplier $\lambda(z)$ is realized by the homotopy $t \mapsto I + t\lambda(z)e_{ij}$, $i \neq j$, which is invertible at every (x, z, t) because $e_{ij}^2 = 0$, and proposition 2.2.9(1) then gives an isomorphism of clutched bundles. No appeal to part (5) is needed, so no hemisphere condition on λ arises.

Each homotopy below stays inside the *linear* clutching functions, which is what proposition 2.2.14(1) requires in order to compare the two splittings. Writing $L^{n+1}p = A + zB$ as in lemma 2.2.11, the two products used are

$$(A + zB)(I + tz e_{n+2, n+1}) = A + z(B + tA e_{n+2, n+1}), \quad (A + zB)(I + tz^{-1} e_{12}) = (A - t e_{22}) + zB$$

The first holds because $Be_{n+2, n+1} = 0$, the subdiagonal of B having no entry in the last column; the second because $Ae_{12} = 0$, the first column of A vanishing when the constant coefficient b_0 of zp is zero. In particular the z^{-1} multiplier produces no z^{-1} term, and both products are again linear in z at every t .

(1). With $a_{n+1} = 0$ the last column of $L^{n+1}p$ is $(0, \dots, 0, 1)^T$ and the last row is $(0, \dots, 0, -z, 1)$. Apply $C_{n+1} \mapsto C_{n+1} + zC_{n+2}$. Column $n + 1$ has $-z$ in row $n + 2$, and adding z times the last column replaces that entry by $-z + z = 0$, leaving every other entry unchanged. The

last row and column now meet the rest of the matrix nowhere, so the matrix is $L^n p \oplus 1$, and proposition 2.2.9(2),(3) give the claim. For $n = 0$ this reads $\begin{pmatrix} a_0 & 0 \\ -z & 1 \end{pmatrix} \mapsto \begin{pmatrix} a_0 & 0 \\ 0 & 1 \end{pmatrix}$, and for $n = 1$ it reads $\begin{pmatrix} a_0 & a_1 & 0 \\ -z & 1 & 0 \\ 0 & -z & 1 \end{pmatrix} \mapsto \begin{pmatrix} a_0 & a_1 & 0 \\ -z & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

(2). The coefficients of zp are $b_0 = 0$ and $b_k = a_{k-1}$ for $1 \leq k \leq n+1$, so the first row of $L^{n+1}(zp)$ is $(0, a_0, a_1, \dots, a_n)$ and the first column is $(0, -z, 0, \dots, 0)^T$. Apply $C_2 \mapsto C_2 + z^{-1}C_1$, legitimate on $X \times S^1$ where z^{-1} is defined. Column 1 scaled by z^{-1} is $(0, -1, 0, \dots, 0)^T$, and column 2 is $(a_0, 1, -z, 0, \dots)^T$, so the entry in row 2 becomes $1 - 1 = 0$. Row 2 is now $(-z, 0, \dots, 0)$ and column 1 is $(0, -z, 0, \dots, 0)^T$: they meet the rest of the matrix nowhere and isolate the single entry $-z$ at position $(2, 1)$. Deleting that row and column leaves, on rows $1, 3, 4, \dots, n+2$ and columns $2, \dots, n+2$, the matrix with first row $(a_0, \dots, a_n)$ and $-z, 1$ on the two diagonals below it, which is $L^n p$.

Bringing the isolated entry onto the diagonal is a transposition of two indices, realized by a permutation matrix P . Since $\mathrm{GL}_{n+2}(\mathbb{C})$ is path-connected, P is joined to the identity by a path of constant invertible matrices, and $t \mapsto P_t L^{n+1}(zp)$ is therefore a homotopy through linear clutching functions; each P_t is also constant in z , so it extends over both hemispheres and proposition 2.2.9(5) applies as well. Therefore $[E^{\oplus(n+2)}, L^{n+1}(zp)] \cong [E^{\oplus(n+1)}, L^n p] \oplus [E, -z]$. Finally $[E, -z] \cong [E, z]$, since $-\mathrm{id}$ is a z -independent automorphism of E and part (5) absorbs it. For $n = 0$ the computation reads $\begin{pmatrix} 0 & a_0 \\ -z & 1 \end{pmatrix} \mapsto \begin{pmatrix} 0 & a_0 \\ -z & 0 \end{pmatrix}$, which is $\mathrm{diag}(a_0, -z)$ after the transposition, completing the proof.

What remains is the case of a clutching function linear in z . Such a function $az + b$ is invertible on the unit circle and degenerates at the roots of $\det(a(x)z + b(x))$, a polynomial in z of degree at most the rank of E , so for each x there are finitely many bad points and they may lie anywhere off the circle. The treatment has two stages: first normalize so that the coefficient of z is the identity, which is a homotopy argument; then split the remaining endomorphism according to whether its eigenvalues lie inside or outside the unit circle, which is the spectral projection of a contour integral. The projection is taken of the *normalized* endomorphism and not of the pencil $az + b$ itself, which in general commutes with neither a nor b .

Lemma 2.2.13: Eigenvalue Splitting of an Endomorphism

Let $E \rightarrow X$ be a complex vector bundle over a compact Hausdorff space and let b be a section of $\mathrm{End}(E)$ such that $b(x)$ has no eigenvalue on the unit circle, for every $x \in X$. Then there are subbundles $E_+, E_- \subseteq E$ with

1. $E = E_+ \oplus E_-$;
2. $b(E_{\pm}) \subseteq E_{\pm}$;
3. every eigenvalue of $b|_{E_+}$ lies outside S^1 , and every eigenvalue of $b|_{E_-}$ lies inside S^1 .

The splitting is unique, and it respects direct sums: applied to $b_1 \oplus b_2$ on $E_1 \oplus E_2$ it gives $(E_1 \oplus E_2)_{\pm} = (E_1)_{\pm} \oplus (E_2)_{\pm}$.

Proof:

Step 1: The projection. For $x \in X$ and z on the unit circle, $z - b(x)$ is invertible, since otherwise

z would be an eigenvalue of $b(x)$ on S^1 . Define a section Π of $\text{End}(E)$ by the contour integral

$$\Pi(x) = \frac{1}{2\pi i} \oint_{|z|=1} (z - b(x))^{-1} dz$$

taken fibrewise in the finite-dimensional space $\text{End}(E_x)$. The resolvent $(z - b(x))^{-1}$ is jointly continuous in (x, z) on the compact set $X \times S^1$, being the inverse of a continuous family of invertible endomorphisms, so Π is a continuous section of $\text{End}(E)$.

Step 2: Π is idempotent. Write $R(z) = (z - b(x))^{-1}$ for fixed x . Since R is holomorphic on the annulus where $z - b(x)$ stays invertible, which by compactness contains $1 - \varepsilon < |z| < 1 + \varepsilon$ for some $\varepsilon > 0$ uniformly in x , Cauchy's theorem ([14, Cauchy's (Integral) Theorem]) lets the contour be taken on any circle in that annulus without changing Π . Choose radii $r_1 < r_2$ in the annulus, and use the resolvent identity

$$R(z)R(w) = \frac{R(w) - R(z)}{z - w} \quad (z \neq w)$$

which follows from $R(z)R(w)[(w - b) - (z - b)] = R(z) - R(w)$. Then

$$\Pi^2 = \frac{1}{(2\pi i)^2} \oint_{|z|=r_1} \oint_{|w|=r_2} \frac{R(w) - R(z)}{z - w} dw dz$$

In the term containing $R(w)$, integrating first in z over the smaller circle gives $\oint_{|z|=r_1} (z - w)^{-1} dz = 0$, because w lies outside that circle. In the term containing $R(z)$, integrating first in w over the larger circle gives $-\oint_{|w|=r_2} (z - w)^{-1} dw = \oint_{|w|=r_2} (w - z)^{-1} dw = 2\pi i$, because z lies inside it. Both evaluations are Cauchy's integral formula applied to the constant function 1. What survives is $\Pi^2 = \frac{1}{2\pi i} \oint_{|z|=r_1} R(z) dz = \Pi$.

Step 3: The subbundles. Set $E_- = \text{im } \Pi$ and $E_+ = \ker \Pi$. A continuous idempotent has locally constant rank, since $\text{rank } \Pi(x) = \text{tr } \Pi(x)$ is both continuous and integer-valued, so E_- and E_+ are subbundles and $E = E_+ \oplus E_-$, which is (1). Each is b -invariant because b commutes with $(z - b)^{-1}$, hence with Π , which is (2).

Step 4: Locating the eigenvalues. Fix x and decompose E_x into generalized eigenspaces of $b(x)$. On the generalized eigenspace for an eigenvalue λ , the resolvent $(z - b)^{-1}$ has a pole at $z = \lambda$ and no other singularity, so by the residue theorem ([14, Residue Theorem]) the integral Π restricts to the identity when $|\lambda| < 1$ and to zero when $|\lambda| > 1$. Hence E_- is the sum of the generalized eigenspaces for eigenvalues inside S^1 and E_+ the sum of those outside, which is (3), and this description also gives uniqueness: any splitting satisfying (1)-(3) must assign each generalized eigenspace to the side its eigenvalue dictates. Compatibility with direct sums is immediate from the same description, since the generalized eigenspaces of $b_1 \oplus b_2$ are the sums of those of b_1 and b_2 , which completes the proof.

Proposition 2.2.14: Linear Clutching Functions Split

Let $\ell(x, z) = a(x)z + b(x)$ be a clutching function for E , with a, b sections of $\text{End}(E)$. Then there is a splitting $E \cong E_+ \oplus E_-$ into subbundles with

$$[E, \ell] \cong [E_+, \text{id}] \oplus [E_-, z]$$

and the splitting respects direct sums of linear clutching functions. Moreover:

1. the isomorphism classes of E_+ and E_- are unchanged along a homotopy through *linear* clutching functions, and in particular do not depend on the choice of t_0 made in the proof;
2. for $\ell = \text{id}$ one has $E_- = 0$ and $E_+ = E$; for $\ell = z \cdot \text{id}$ one has $E_+ = 0$ and $E_- = E$.

Proof:

Step 1: Normalizing the coefficient of z . For $t \in [0, 1)$ consider the Möbius substitution $z \mapsto (z + t)/(1 + tz)$ with t real, which carries S^1 to itself: if $|z| = 1$ then $|1 + tz| = |z| |\bar{z} + t| = |\bar{z} + t| = |\overline{z + t}| = |z + t|$, using $|z| = 1$, $z\bar{z} = 1$ and $\bar{\bar{t}} = t$, so the quotient has modulus 1. Multiplying by the scalar $1 + tz$, which is nonzero for $|z| = 1$ and $t < 1$, gives

$$(1 + tz) \left[a(x) \frac{z + t}{1 + tz} + b(x) \right] = (a(x) + tb(x))z + (ta(x) + b(x)) \quad (2.2)$$

At $t = 0$ the right-hand side is $a(x)z + b(x)$. For each $t \in [0, 1)$ the left-hand side is invertible on $X \times S^1$, being the composite of the substitution, which preserves S^1 , with ℓ and a nonzero scalar. So (2.2) is a homotopy of clutching functions as t runs from 0 to any $t_0 < 1$.

At $t = 1$ the coefficient $a(x) + b(x)$ equals $\ell(x, 1)$, which is invertible. The function $t \mapsto \inf_{x \in X} |\det(a(x) + tb(x))|$ is continuous, and X compact makes the infimum attained, so it is nonzero at $t = 1$ and hence nonzero for $t = t_0$ close enough to 1. Fix such a t_0 . Then $a + t_0b$ is a section of $\text{Aut}(E)$, independent of z , so it extends over both hemispheres and proposition 2.2.9(5) permits multiplying the clutching function by its inverse on the right. Doing so replaces (2.2) at $t = t_0$ by

$$z + c(x), \quad c = (t_0a + b)(a + t_0b)^{-1}$$

Hence $[E, \ell] \cong [E, z + c]$ with c a section of $\text{End}(E)$.

Step 2: Splitting. Since $z + c(x)$ is invertible for $|z| = 1$, the endomorphism $-c(x)$, and therefore $c(x)$, has no eigenvalue on the unit circle. Apply lemma 2.2.13 to c to get $E = E_+ \oplus E_-$ preserved by c , with the eigenvalues of $c_+ = c|_{E_+}$ outside S^1 and those of $c_- = c|_{E_-}$ inside. Because the splitting is c -invariant, the clutching function decomposes, and proposition 2.2.9(2) gives

$$[E, z + c] \cong [E_+, z + c_+] \oplus [E_-, z + c_-]$$

Step 3: Contracting each summand. On E_+ every eigenvalue of c_+ lies outside S^1 , so for $s \in [0, 1]$ and $|z| = 1$ the endomorphism $sz + c_+$ is invertible: it fails to be so only if $-sz$ is an eigenvalue of c_+ , and $|-sz| = s \leq 1$ while every eigenvalue has modulus greater than 1. Thus $s \mapsto sz + c_+$ is a homotopy of clutching functions from $z + c_+$ to c_+ , and c_+ is a z -independent automorphism, so a further application of proposition 2.2.9(5) and (1) gives $[E_+, z + c_+] \cong [E_+, \text{id}]$.

On E_- every eigenvalue of c_- lies inside S^1 , so for $s \in [0, 1]$ and $|z| = 1$ the endomorphism $z + s c_-$ is invertible: it fails only if $-z/s$ is an eigenvalue of c_- , and $|-z/s| = 1/s \geq 1$ while every eigenvalue has modulus less than 1 (for $s = 0$ the endomorphism is $z \cdot \text{id}$, invertible). Thus $s \mapsto z + s c_-$ is a homotopy of clutching functions from $z + c_-$ to $z \cdot \text{id}$, giving $[E_-, z + c_-] \cong [E_-, z]$.

Combining the three steps yields $[E, \ell] \cong [E_+, \text{id}] \oplus [E_-, z]$. Compatibility with direct sums holds because Step 1 is applied coordinatewise to a direct sum of linear clutching functions and the splitting of lemma 2.2.13 respects direct sums.

Step 4: Homotopy invariance and the two special cases. Let $\ell_s = a_s z + b_s$, $s \in [0, 1]$, be a homotopy through linear clutching functions for E . Running Step 1 with the parameter space $X \times [0, 1]$ in place of X produces a single endomorphism c of $E \times [0, 1]$ over $X \times [0, 1]$, and lemma 2.2.13 applied there gives subbundles $(E \times [0, 1])_{\pm}$ restricting to the splittings at each s . A subbundle of $E \times [0, 1]$ restricts to isomorphic bundles at $s = 0$ and $s = 1$ by theorem 1.2.5, so the isomorphism classes of $E_{\pm}$ are constant along the homotopy. The choice of t_0 is covered by this. Step 1 produces an interval $(\tau, 1)$ on which $\inf_x |\det(a + tb)|$ stays positive, and any two values in that interval are joined inside it, along which (2.2) varies continuously through linear clutching functions.

For $\ell = \text{id}$, that is $a = 0$ and $b = \text{id}$, Step 1 gives $c = (t_0 \cdot 0 + \text{id})(0 + t_0 \text{id})^{-1} = t_0^{-1} \text{id}$, whose only eigenvalue t_0^{-1} exceeds 1; hence $E_+ = E$ and $E_- = 0$. For $\ell = z \cdot \text{id}$, that is $a = \text{id}$ and $b = 0$, Step 1 gives $c = (t_0 \text{id})(\text{id})^{-1} = t_0 \text{id}$, whose only eigenvalue $t_0 < 1$ lies inside the circle; hence $E_- = E$ and $E_+ = 0$, completing the proof.

Theorem 2.2.15: Bott Periodicity Isomorphism

Let X be a compact Hausdorff space. The external product

$$\mu: K^0(X) \otimes_{\mathbb{Z}} K^0(S^2) \longrightarrow K^0(X \times S^2)$$

is an isomorphism of rings. Equivalently, $K^0(X \times S^2)$ is a free $K^0(X)$ -module of rank 2 with basis $\{1, [H]\}$, and as a $K^0(X)$ -algebra it is $K^0(X)[H]/([H] - 1)^2$.

Consequently, for every finite pointed CW complex X (the hypothesis of proposition 2.2.5, which the next clause uses through the injectivity of q^*) the Bott map

$$\beta_X: \tilde{K}^0(X) \longrightarrow \tilde{K}^0(\Sigma^2 X)$$

of definition 2.2.7 is an isomorphism of abelian groups, natural in X .

Proof:

Write $\beta = [H] - 1 \in \tilde{K}^0(S^2)$, so that $K^0(S^2) = \mathbb{Z}\{1, \beta\}$ with $\beta^2 = 0$ and $[H] = 1 + \beta$ is a unit with inverse $[H^*] = 1 - \beta$ (example 2.1.12). For a bundle $F \rightarrow X$ and $k \in \mathbb{Z}$ abbreviate $F * [H]^k = \mu([F] \otimes [H]^k) = \pi_X^*[F] \cdot \pi_{S^2}^*[H]^k$.

Step 1: Surjectivity. Let F be a bundle over $X \times S^2$. By proposition 2.2.9(4) write $F \cong [E, f]$, and by lemma 2.2.10 and proposition 2.2.9(1) replace f by a Laurent polynomial clutching function. Multiplying by a power of z , which by (2.1) twists the clutched bundle by a power of H , write that function as $z^{-m}q$ with q polynomial of degree at most n , so that $[E, z^{-m}q] \cong [E, q] \otimes \pi_{S^2}^* H^{\otimes m}$.

By lemma 2.2.11, in $K^0(X \times S^2)$,

$$[E, q] = [E^{\oplus(n+1)}, L^n q] - \pi_X^*[E^{\oplus n}]$$

and by proposition 2.2.14 the first term splits as $[G_+, \text{id}] + [G_-, z]$, where $G = E^{\oplus(n+1)}$ and $G_{\pm}$ are the two summands produced there. Finally $[G_+, \text{id}] = \pi_X^*[G_+]$ by proposition 2.2.9(3), and $[G_-, z] = \pi_X^*[G_-] \cdot \pi_{S^2}^*[H^*]$ by (2.1). Assembling,

$$[F] = (G_+ * 1 + G_- * [H^*] - E^{\oplus n} * 1) \cdot [H]^m \quad (2.3)$$

every term of which lies in the image of μ . Since $K^0(X \times S^2)$ is generated by classes of bundles, μ is surjective.

Step 2: An inverse on the level of clutching data. Define, for a bundle presented as $[E, z^{-m}q]$ with $\deg q \leq n$,

$$\nu([E, z^{-m}q]) = -[G_-] \otimes \beta + [E] \otimes [H]^m \in K^0(X) \otimes K^0(S^2)$$

where $G_- = (E^{\oplus(n+1)})_-$ is the minus-summand of the splitting of $L^n q$. Four checks make this well defined.

Independence of n . Regarding q as having degree at most $n+1$, lemma 2.2.12(1) gives $[E^{\oplus(n+2)}, L^{n+1}q] \cong [E^{\oplus(n+1)}, L^n q] \oplus [E, \text{id}]$, an isomorphism realized by a homotopy of linear clutching functions. By proposition 2.2.14 the splitting respects direct sums and is homotopy invariant, and the minus-summand of $[E, \text{id}]$ is 0 by clause (2) there. Hence $(E^{\oplus(n+2)})_- \cong (E^{\oplus(n+1)})_-$, so G_- does not depend on n .

Independence of m . Replacing $z^{-m}q$ by $z^{-m-1}(zq)$ presents the same bundle. By lemma 2.2.12(2), $[E^{\oplus(n+2)}, L^{n+1}(zq)] \cong [E^{\oplus(n+1)}, L^n q] \oplus [E, z]$, and the minus-summand of $[E, z]$ is E by proposition 2.2.14(2). So the minus-summand for zq is $G_- \oplus E$, and

$$\nu([E, z^{-m-1}(zq)]) = -([G_-] + [E]) \otimes \beta + [E] \otimes [H]^{m+1}$$

Since $[H]^k = 1 + k\beta$ for every k , one has $[H]^{m+1} - \beta = 1 + (m+1)\beta - \beta = [H]^m$, so the last two terms combine to $[E] \otimes [H]^m$ and the whole expression equals $\nu([E, z^{-m}q])$.

Independence of the normalization constant. The constant t_0 in the proof of proposition 2.2.14 may be any value sufficiently close to 1, and proposition 2.2.14(1) records that the splitting is unchanged along a homotopy through linear clutching functions, so G_- does not depend on it.

Independence of the presenting clutching function. By proposition 2.2.9(6) the presentation may be taken normalized, and any two normalized clutching functions for isomorphic bundles are homotopic through clutching functions. By lemma 2.2.10 their Laurent approximations are then homotopic through Laurent polynomial clutching functions. Running lemma 2.2.11 and proposition 2.2.14 over $X \times [0, 1]$ with that homotopy as clutching function produces a bundle over $X \times [0, 1]$ whose minus-summand restricts to the two candidate G_- at the ends, so they are isomorphic by theorem 1.2.5.

Finally ν is additive: $L^n(q_1 \oplus q_2) = L^n q_1 \oplus L^n q_2$ directly from the matrix, and the splitting respects direct sums, so ν carries a direct sum of clutching data to the sum of its values. Since every bundle over $X \times S^2$ has such a presentation, ν extends to a group homomorphism $K^0(X \times S^2) \rightarrow K^0(X) \otimes K^0(S^2)$.

Step 3: $\nu\mu = \text{id}$. The group $K^0(S^2) = \mathbb{Z}\{1, \beta\}$ is generated by $[H]^0 = 1$ and $[H]^1 = 1 + \beta$, so it is generated by the classes $[H]^m$ with $m \geq 0$, and it suffices to evaluate $\nu\mu$ on elements $[E] \otimes [H]^m$. Now $\mu([E] \otimes [H]^m) = [E, z^{-m}]$, whose polynomial part is $q = \text{id}$ of degree 0. Taking $n = 0$, the linearization $L^0 \text{id}$ is id on E , and by proposition 2.2.14(2) its minus-summand is 0. Hence

$$\nu\mu([E] \otimes [H]^m) = -0 \otimes \beta + [E] \otimes [H]^m = [E] \otimes [H]^m$$

So $\nu\mu = \text{id}$, μ is injective, and with Step 1 it is an isomorphism. It is a ring map because the external product is multiplicative (definition 2.2.6), and the module description follows since $K^0(S^2) = \mathbb{Z}\{1, [H]\}$ with $([H] - 1)^2 = \beta^2 = 0$.

Step 4: The reduced form. Let X be pointed. Decompose $K^0(X) = \tilde{K}^0(X) \oplus \mathbb{Z}$ and $K^0(S^2) = \tilde{K}^0(S^2) \oplus \mathbb{Z}$. This uses only the basepoint, not connectedness: for the basepoint inclusion $i: \{*\} \rightarrow X$ and the collapse $c: X \rightarrow \{*\}$ one has $c \circ i = \text{id}_{\{*\}}$, so $i^*c^* = \text{id}$ on $K^0(\{*\}) = \mathbb{Z}$. Hence restriction to the basepoint is a split surjection, its kernel is $\tilde{K}^0(X)$ by definition, and c^* splits it. Under μ these give

$$K^0(X \times S^2) \cong (\tilde{K}^0(X) \otimes \tilde{K}^0(S^2)) \oplus \tilde{K}^0(X) \oplus \tilde{K}^0(S^2) \oplus \mathbb{Z}$$

Restriction to $X \vee S^2$ kills exactly the first summand: a class $\pi_X^*a \cdot \pi_{S^2}^*b$ with a, b both reduced restricts on $X \times \{s_0\}$ to π_X^*a times the pullback of $b|_{s_0} = 0$, and symmetrically on $\{x_0\} \times S^2$, while on the other three summands restriction is injective, being the identity on each factor. Hence

$$\ker(\tilde{K}^0(X \times S^2) \rightarrow \tilde{K}^0(X \vee S^2)) = \tilde{K}^0(X) \cdot \beta$$

and μ restricts to an isomorphism $\tilde{K}^0(X) \otimes \tilde{K}^0(S^2) \rightarrow$ that kernel. Since $\tilde{K}^0(S^2) = \mathbb{Z}\beta$, the map $a \mapsto \mu(a \otimes \beta)$ is an isomorphism of $\tilde{K}^0(X)$ onto the kernel. By proposition 2.2.5 the kernel is q^* of $\tilde{K}^0((X \times S^2)/(X \vee S^2)) = \tilde{K}^0(X \wedge S^2) = \tilde{K}^0(\Sigma^2 X)$, using the homeomorphism $X \wedge S^2 \cong S^2 \wedge X$ that interchanges the smash factors, and q^* is injective, so composing with $(q^*)^{-1}$ gives the Bott map β_X of definition 2.2.7. Therefore β_X is an isomorphism, natural in X because every construction used is, completing the proof.

The homotopy-theoretic form of periodicity is a consequence of the theorem rather than an input to it, obtained by evaluating the theorem on spheres and reading the answer through the representability of K^0 .

Theorem 2.2.16: Bott Periodicity

The stable homotopy groups of the unitary group are 2-periodic: $\pi_k(U) \cong \pi_{k+2}(U)$ for all $k \geq 0$, and

$$\pi_k(U) = \begin{cases} 0 & k \text{ even} \\ \mathbb{Z} & k \text{ odd} \end{cases}$$

The orthogonal group is 8-periodic, $\pi_k(O) \cong \pi_{k+8}(O)$, with

$$(\pi_k(O))_{k \bmod 8} = (\mathbb{Z}/2, \mathbb{Z}/2, 0, \mathbb{Z}, 0, 0, 0, \mathbb{Z})$$

The complex statement is proved here. The real statement is *not* proved in this section; it is recorded for comparison, with the argument in Atiyah's *K-Theory* [1], and the period 8 is derived from the mod-8 periodicity of the real Clifford algebras when real K-theory is developed.

Proof:

Only the complex statement is proved here. Take $X = S^k$ in theorem 2.2.15, in its reduced form: $\tilde{K}^0(S^k) \cong \tilde{K}^0(\Sigma^2 S^k) = \tilde{K}^0(S^{k+2})$, since $\Sigma S^k = S^{k+1}$. So $\tilde{K}^0(S^k)$ depends only on the parity of k . Its two values are computed directly: $\tilde{K}^0(S^1) = 0$, because every complex bundle over S^1 is trivial by the clutching correspondence (proposition 1.2.8) and the path-connectedness of $\mathrm{GL}_n(\mathbb{C})$, and $\tilde{K}^0(S^2) \cong \mathbb{Z}$ generated by β (example 2.1.12). Hence $\tilde{K}^0(S^k)$ is $\mathbb{Z}$ for k even and 0 for k odd.

By representability (theorem 2.1.9), $\tilde{K}^0(S^k) \cong [S^k, \mathbb{Z} \times BU]_* \cong \pi_k(\mathbb{Z} \times BU) \cong \pi_k(BU)$ for $k \geq 1$, the last step because $\mathbb{Z}$ is discrete. The fibration $U \rightarrow EU \rightarrow BU$ with EU contractible gives $\pi_k(BU) \cong \pi_{k-1}(U)$ ([13, Long Exact Sequence of a Fibration]). Combining,

$$\pi_{k-1}(U) \cong \pi_k(BU) \cong \tilde{K}^0(S^k)$$

which is $\mathbb{Z}$ for k even and 0 for k odd, that is $\pi_j(U)$ is $\mathbb{Z}$ for j odd and 0 for j even. Periodicity $\pi_j(U) \cong \pi_{j+2}(U)$ follows, completing the proof of the complex statement.

2.2.17 Note: The Loop-Space Form Periodicity is often stated as a homotopy equivalence $\mathbb{Z} \times BU \simeq \Omega^2(\mathbb{Z} \times BU)$, and in the real case $\mathbb{Z} \times BO \simeq \Omega^8(\mathbb{Z} \times BO)$. That form says more than the table above: it asserts an equivalence of spaces, not merely an isomorphism of homotopy groups, and producing it requires representing the Bott map by an actual map of spaces and then invoking Whitehead's theorem. The representability theorem available here (theorem 2.1.9) is stated on compact spaces, which are not the objects at which such an argument must be evaluated, so the equivalence is recorded rather than derived; see Atiyah's *K-Theory* [1]. Nothing later in this book uses the loop-space form, the table and the isomorphism β_X being what the computations consume.

2.2.18 Historical Note: Bott's Original Route Bott's proof was not this one. He studied the loop space of a compact symmetric space by Morse theory for the energy functional, whose critical points are the geodesics joining two fixed points. An index computation identifies the space of minimal geodesics well enough to give $\pi_k(M^\nu) \cong \pi_{k+1}(M)$, and iterating along the chain $U, O, U/O, Sp/U, Sp, \dots$ produces both periodicities at once. That argument is in Milnor's *Morse Theory* [11]. It needs infinite-dimensional Morse theory and the Morse index theorem, neither of which this book develops, and it proves the homotopy-group statement first, so the multiplicative form used throughout K-theory has to be recovered from it afterwards. The route taken above reverses that order and stays inside the material of chapter 1.

Periodicity collapses the doubly-infinite sequence of groups $\tilde{K}^{-n}$ to two, $\tilde{K}^0$ and $\tilde{K}^{-1}$, the latter conventionally written $\tilde{K}^1$. This is what is meant by saying complex K-theory is $\mathbb{Z}/2$ -graded, and the next section makes the whole family into a cohomology theory.

Exercise 2.2.1

1. Let f be a clutching function for E that extends to a section α of $\text{Aut}(\pi_+^* E)$ over $X \times D_+$, meaning $f = \alpha|_{X \times S^1}$. Show that $[E, f] \cong \pi_X^* E$. Deduce that a clutching function whose entries are polynomials in z with invertible value at every point of D_+ presents a pullback.
2. Using proposition 2.2.9(2), show that $[\mathbb{C}, z] \otimes [\mathbb{C}, z^{-1}]$ is the trivial line bundle over $X \times S^2$, and conclude that $[H]$ is a unit in $K^0(S^2)$ with $[H]^{-1} = [H^*]$.
3. Show that $[\mathbb{C}, z^k] \cong (H^\vee)^{\otimes k}$ for every $k \in \mathbb{Z}$, and deduce that the class of $[\mathbb{C}, z^k]$ in $K^0(S^2)$ is $1 - k\beta$. Which values of k give a bundle with vanishing reduced class?
4. Let $S_n f = \sum_{|k| \leq n} a_k z^k$ be the partial sums of the Fourier series of a clutching function. Verify that the Cesàro mean $\frac{1}{N+1} \sum_{n=0}^N S_n f$ equals $\sum_{|k| \leq N} (1 - \frac{|k|}{N+1}) a_k z^k$, the expression used in the proof of lemma 2.2.10.
5. Write out the matrix Lp of lemma 2.2.11 for a polynomial clutching function $p = a_0 + a_1 z + a_2 z^2$ of degree 2, and carry out the column and row operations of Step 1 explicitly. Exhibit the intermediate matrix $p \oplus T$ and check that T is not the identity, so that the homotopy of Step 3 is needed.
6. Let $E = X \times \mathbb{C}^2$ be trivial and let b be the constant endomorphism $\text{diag}(\lambda, \mu)$ with $|\lambda| < 1 < |\mu|$. Compute the projection Π of lemma 2.2.13 directly from the contour integral and identify E_+ and E_- .
7. Take $a = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $b = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}$, so that $\ell(z) = az + b$ is invertible on S^1 . Compute $P = \frac{1}{2\pi i} \oint_{|z|=1} \ell(z)^{-1} a dz$ and verify that $P^2 = P$ but $Pa \neq aP$. Explain why this does not contradict proposition 2.2.14, and identify which step of that proof the normalization to $z + c$ is doing.
8. Use theorem 2.2.15 with $X = S^2$ to compute $K^0(S^2 \times S^2)$ as a ring, and give its rank as an abelian group. Compare with $H^{\text{even}}(S^2 \times S^2; \mathbb{Z})$.
9. Path-connectedness of $\text{GL}_n(\mathbb{C})$ is used twice in this section: once in the proof of lemma 2.2.12, and once in the proof of theorem 2.2.16. Identify both, and identify where the argument uses that every endomorphism of a complex vector space has an eigenvalue. Then explain why the

section does not establish a 2-periodicity for real vector bundles, and state the period that replaces it.

2.3 K-Theory as a Generalized Cohomology Theory

A cohomology theory is more than a single functor: it is a graded family $\{h^n\}_{n \in \mathbb{Z}}$ tied together by suspension isomorphisms and by the long exact sequence of a pair. This section assembles topological K-theory into such a theory. The negative groups $K^{-n}(X)$ are defined by iterated reduced suspension, the relative groups $K^0(X, A)$ by collapsing the subspace A , and periodicity (theorem 2.2.15) is what collapses the doubly-infinite family down to two groups.

The passage from a single functor to a graded family rests on one construction: suspension. For ordinary cohomology the suspension isomorphism $\tilde{H}^n(X) \cong \tilde{H}^{n+1}(\Sigma X)$ shifts degree by smashing with a circle. K-theory has no a priori notion of degree, so the strategy is reversed: one *defines* the negative groups by suspending the space. The reduced suspension is forced, rather than the unreduced cone-based one, because $\tilde{K}^0$ is a functor on pointed spaces.

Definition 2.3.1: Negative K-Groups

Let X be a compact Hausdorff space and let $n \geq 0$ be an integer. The n -th *negative reduced K-group* of a pointed space X is

$$\tilde{K}^{-n}(X) = \tilde{K}^0(\Sigma^n X)$$

where $\Sigma^n X$ is the n -fold reduced suspension and $\Sigma^0 X = X$. For an unpointed compact space X , the n -th *negative (unreduced) K-group* is

$$K^{-n}(X) = \tilde{K}^0(\Sigma^n(X_+)) = \tilde{K}^{-n}(X_+)$$

With $n = 0$ this recovers $K^0(X) = \tilde{K}^0(X_+)$ and $\tilde{K}^0(X)$ for the unreduced and reduced groups respectively.

The two are reconciled by the splitting $X_+ = X \sqcup \{*\}$. Adding a disjoint basepoint converts an unpointed space into a pointed one without losing information, and the reduced theory of X_+ is the unreduced theory of X . Concretely, the inclusion of the added basepoint and the collapse $X_+ \rightarrow S^0$ give a natural splitting $K^{-n}(X) \cong \tilde{K}^{-n}(X) \oplus \tilde{K}^{-n}(S^0)$ when X itself is pointed; since $\tilde{K}^{-n}(S^0) = \tilde{K}^0(\Sigma^n S^0) = \tilde{K}^0(S^n)$, the discrepancy between reduced and unreduced is the K-theory of a sphere. This is the K-theoretic analogue of the $\tilde{H}^*$ versus H^* bookkeeping in ordinary cohomology, and it is the reason every formula below comes in a reduced and an unreduced version.

Example 2.3.2: Negative Groups of a Point

Take $X = \{*\}$ the one-point space, regarded as pointed. Then $\Sigma^n \{*\} = \{*\}$ for every n , since suspending a point gives a point: $\Sigma \{*\} = S^1 \wedge \{*\} = \{*\}$. Hence

$$\tilde{K}^{-n}(\{*\}) = \tilde{K}^0(\{*\}) = 0$$

for all $n \geq 0$, the reduced group of a point being trivial by definition 2.1.4. For the unreduced

groups, $\{*\}_+ = S^0$, so

$$K^{-n}(\{*\}) = \tilde{K}^0(\Sigma^n S^0) = \tilde{K}^0(S^n)$$

The right-hand side is $\mathbb{Z}$ when n is even and 0 when n is odd, as computed in the proof of theorem 2.2.16: every complex bundle over an odd sphere is stably trivial by the clutching correspondence (proposition 1.2.8), while $\tilde{K}^0(S^2) \cong \mathbb{Z}$ is generated by the reduced class of the canonical line bundle. Thus the coefficient groups $K^{-n}(\{*\})$ exhibit the 2-periodic pattern $\mathbb{Z}, 0, \mathbb{Z}, 0, \dots$ that periodicity imposes on every space.

Relative K-theory measures the K-theory of X *rel* a subspace A . The model is the quotient X/A , which collapses A to a single basepoint; classes in the relative group are classes on X that are trivialized over A . This is the geometric content of the cofiber: X/A is the cofiber of the inclusion $A \hookrightarrow X$, and reduced K-theory of the cofiber is the receptacle for relative classes.

Definition 2.3.3: Relative K-Theory

Let (X, A) be a compact pair, meaning X is compact Hausdorff and $A \subseteq X$ is a closed (hence compact) subspace. For $n \geq 0$ the *relative K-groups* are

$$K^{-n}(X, A) = \tilde{K}^{-n}(X/A) = \tilde{K}^0(\Sigma^n(X/A))$$

where X/A is basepointed at the collapsed image of A . When $A = \emptyset$ one collapses the empty set, which by convention adjoins a disjoint basepoint, so $X/\emptyset = X_+$ and $K^{-n}(X, \emptyset) = K^{-n}(X)$ recovers the absolute groups.

The convention $X/\emptyset = X_+$ is what makes the relative theory a generalization of the absolute one. It encodes the principle that “no subspace collapsed” should leave the absolute group unchanged, and adding a disjoint basepoint is the operation that turns the absolute unreduced group into a reduced group on which the machinery operates. With this convention the long exact sequence of the pair $(X, \emptyset)$ degenerates correctly and excision reduces to a tautology.

Example 2.3.4: Relative K-Theory of a Disk Rel its Boundary

Let $X = D^2$ be the closed 2-disk and $A = \partial D^2 = S^1$ its boundary circle. The quotient collapses the boundary to a point, and

$$D^2/S^1 \cong S^2$$

since a disk with its boundary identified to a single point is a 2-sphere. Therefore

$$K^0(D^2, S^1) = \tilde{K}^0(D^2/S^1) = \tilde{K}^0(S^2) \cong \mathbb{Z}$$

the generator being the reduced class $[H] - 1$ of the canonical line bundle $H \rightarrow \mathbb{CP}^1 = S^2$. This is the relative group housing the Bott generator of definition 2.2.7. More generally $D^{2k}/\partial D^{2k} \cong S^{2k}$, so $K^0(D^{2k}, \partial D^{2k}) \cong \tilde{K}^0(S^{2k}) \cong \mathbb{Z}$, while collapsing the boundary of an odd-dimensional disk gives an odd sphere and the relative group vanishes.

2.3.5 Convention: Periodic Grading Let X be a finite pointed CW complex (box 2.2.1). By theorem 2.2.15 the groups $\tilde{K}^{-n}(X)$ depend only on the parity of n . One therefore extends the indexing to all $n \in \mathbb{Z}$ by declaring $\tilde{K}^n(X) = \tilde{K}^{n-2}(X) = \tilde{K}^{n+2}(X)$, so that $\tilde{K}^n = \tilde{K}^0$ for n even and $\tilde{K}^n = \tilde{K}^1$ for n odd, where $\tilde{K}^1(X) = \tilde{K}^0(\Sigma X) = \tilde{K}^{-1}(X)$. The same convention applies to the unreduced and relative groups. Henceforth $\tilde{K}^*$ denotes this $\mathbb{Z}/2$ -graded family.

A generalized cohomology theory on compact pairs is, by the axioms developed in the generalized-cohomology chapter of [13], a sequence of contravariant functors h^n together with natural coboundary maps $\delta: h^n(A) \rightarrow h^{n+1}(X, A)$ satisfying homotopy invariance, exactness for pairs, excision, and additivity. The K-groups satisfy all of these, with the suspension isomorphism supplied by periodicity. The next theorem records this; its proof verifies each axiom, drawing the exactness from the cofiber sequence and excision from the homeomorphism type of the relevant quotients.

Theorem 2.3.6: K-Theory is a Generalized Cohomology Theory

The family $\{K^{-n}, \tilde{K}^{-n}\}_{n \geq 0}$ of definition 2.3.1, together with the relative groups of definition 2.3.3, satisfies Axioms 1 to 3 and Axiom 5 below on compact Hausdorff pairs. On finite CW pairs, periodicity folds that family into the $\mathbb{Z}/2$ -graded family $\{K^n, \tilde{K}^n\}_{n \in \mathbb{Z}}$ of box 2.3.5, Axiom 4 holds, and together with the long exact sequence of theorem 2.3.7, whose proof uses only Axiom 2, the result is a multiplicative generalized cohomology theory in the sense of [13]. The division of hypotheses is box 2.2.1. Explicitly:

1. (*Homotopy invariance.*) If $f_0 \simeq f_1: (X, A) \rightarrow (Y, B)$ are homotopic maps of pairs, then $f_0^* = f_1^*: K^n(Y, B) \rightarrow K^n(X, A)$.
2. (*Exactness.*) For a compact pair (X, A) with inclusions $i: A \hookrightarrow X$ and $j: (X, \emptyset) \hookrightarrow (X, A)$, the sequence

$$K^n(X, A) \xrightarrow{j^*} K^n(X) \xrightarrow{i^*} K^n(A)$$
 is exact at $K^n(X)$.
3. (*Excision.*) If (X, A) is a compact pair and $U \subseteq A$ is open with $\bar{U} \subseteq \text{int} A$, then the inclusion $(X \setminus U, A \setminus U) \hookrightarrow (X, A)$ induces an isomorphism $K^n(X, A) \cong K^n(X \setminus U, A \setminus U)$.
4. (*Suspension.*) There are natural isomorphisms $\tilde{K}^n(X) \cong \tilde{K}^{n+1}(\Sigma X)$, and the periodicity $\tilde{K}^n \cong \tilde{K}^{n+2}$ holds.
5. (*Additivity.*) For a finite disjoint union, restriction to the pieces is an isomorphism $K^n(\coprod_{j=1}^m X_j) \cong \bigoplus_{j=1}^m K^n(X_j)$.

Moreover the external product of definition 2.2.6 makes K^* a multiplicative theory.

Proof:

Axiom 1: Homotopy invariance. For absolute groups this is theorem 1.2.5: homotopic maps induce the same map on isomorphism classes of bundles, hence the same map on Grothendieck groups K^0 (definition 2.1.1), and the same on $\tilde{K}^0$. For negative groups, a homotopy f_t induces a homotopy $\Sigma^n f_t$ of the suspended maps, so $f_0^* = f_1^*$ on $\tilde{K}^{-n}(Y) = \tilde{K}^0(\Sigma^n Y)$. For relative groups, a homotopy of pairs $(X, A) \rightarrow (Y, B)$ induces a homotopy of the quotient maps $X/A \rightarrow Y/B$,

since the homotopy carries A into B at every time and hence descends to the quotients; homotopy invariance of $\tilde{K}^0$ on the quotients gives the claim.

Axiom 2: Exactness. The inclusion $A \xrightarrow{i} X$ has cofiber X/A , and there is a cofiber sequence

$$A \xrightarrow{i} X \xrightarrow{q} X/A$$

where q is the collapse. The content is that the composite $A \xrightarrow{i} X \xrightarrow{q} X/A$ is the constant map to the basepoint, so $i^*q^* = 0$, and conversely any class on X restricting to zero on A extends over the quotient. Concretely, work with the reduced groups via $K^n(X) = \tilde{K}^n(X_+)$ and $K^n(X, A) = \tilde{K}^n(X/A)$. The collapse $q: X_+ \rightarrow X/A$ (sending A and the added basepoint to the basepoint) and the inclusion $i: A_+ \rightarrow X_+$ give

$$\tilde{K}^n(X/A) \xrightarrow{q^*=j^*} \tilde{K}^n(X_+) \xrightarrow{i^*} \tilde{K}^n(A_+)$$

Exactness at the middle term is lemma 2.2.3, proved in section 2.2 because the smash-product identification there needed it. The same argument applied to Σ^n yields exactness in every degree.

Axiom 3: Excision. For a compact pair the excision isomorphism is a tautology in this model, because the relative group is defined through the quotient and the quotient is unchanged by excising an open set buried inside A . If $U \subseteq A$ is open with $\bar{U} \subseteq \text{int} A$, then collapsing A in X and collapsing $A \setminus U$ in $X \setminus U$ produce homeomorphic quotients:

$$(X \setminus U)/(A \setminus U) \cong X/A$$

The map is induced by the inclusion $X \setminus U \hookrightarrow X$; it is a continuous bijection of compact Hausdorff spaces (every point of A maps to the single basepoint on both sides, and points outside A are untouched), hence a homeomorphism. Applying $\tilde{K}^n$ to this homeomorphism gives the excision isomorphism $K^n(X, A) \cong K^n(X \setminus U, A \setminus U)$.

Axiom 5: Additivity. A vector bundle on a disjoint union is exactly a tuple of vector bundles on the pieces, since the pieces are open and closed, and the same holds for isomorphisms; so $\text{Vect}(\coprod_j X_j) \cong \prod_j \text{Vect}(X_j)$ as monoids, and applying the Grothendieck completion of definition 2.1.1, which preserves finite products, gives the claim in degree 0. For the negative degrees the substitution $X_j \mapsto \Sigma^n((X_j)_+)$ is not available, because $(\coprod_j X_j)_+$ is the wedge $\bigvee_j (X_j)_+$ rather than a disjoint union. Instead

$$K^{-n}\left(\coprod_j X_j\right) = \tilde{K}^0\left(\Sigma^n \bigvee_j (X_j)_+\right) = \tilde{K}^0\left(\bigvee_j \Sigma^n((X_j)_+)\right) \cong \bigoplus_j \tilde{K}^0(\Sigma^n((X_j)_+)) = \bigoplus_j K^{-n}(X_j)$$

using that suspension carries a wedge to a wedge and, for the middle isomorphism, lemma 2.2.2 applied inductively to the finitely many summands. On finite CW pairs periodicity carries the statement to the $\mathbb{Z}/2$ -graded family. Only finite unions occur in this book, all spaces being compact.

Axiom 4: Suspension and periodicity. For a finite pointed CW complex X and any integer $m \geq 0$ the definition of the negative groups gives, by the very meaning of suspension,

$$\tilde{K}^{-(m+1)}(X) = \tilde{K}^0(\Sigma^{m+1} X) = \tilde{K}^0(\Sigma^m(\Sigma X)) = \tilde{K}^{-m}(\Sigma X)$$

which is the suspension isomorphism $\tilde{K}^n(X) \cong \tilde{K}^{n+1}(\Sigma X)$ at $n = -(m+1)$, holding tautologically for all negative degrees. Periodicity (theorem 2.2.15) extends this identity to every integer

n by shifting degrees in steps of 2 until they become negative, so the suspension isomorphism holds in the $\mathbb{Z}/2$ -graded family of box 2.3.5. The substantive content beyond the tautology is the periodicity $\tilde{K}^n \cong \tilde{K}^{n+2}$ itself, which is exactly theorem 2.2.15. Together these identify the abstract suspension isomorphism required by the generalized-cohomology axioms of [13] with the maps in hand.

Multiplicativity. The external product μ of definition 2.2.6 is associative and commutative up to natural isomorphism, since the bundle tensor product is, and the diagonal $X \rightarrow X \times X$ pulls μ back to the internal ring product of proposition 2.1.10. The Bott class β is the unit for the periodicity action, so the graded group $K^*(X)$ is a module over the coefficient ring $K^*(\{*\}) = \mathbb{Z}[\beta, \beta^{-1}]$, a Laurent polynomial ring on the invertible degree -2 generator β . This is the multiplicative structure, completing the verification of all axioms.

The theorem places K-theory on the same axiomatic footing as ordinary cohomology, but with one decisive difference: the coefficient ring $K^*(\{*\}) = \mathbb{Z}[\beta, \beta^{-1}]$ is not concentrated in degree 0. The dimension axiom of Eilenberg and Steenrod, which would force $h^n(\{*\}) = 0$ for $n \neq 0$, fails, and it is this failure that makes K-theory a genuinely *generalized* cohomology theory rather than ordinary cohomology in disguise. Every consequence of the axioms valid for an arbitrary generalized theory, the long exact sequence of a pair, Mayer-Vietoris, and the Atiyah-Hirzebruch spectral sequence of [13] computing K^* from ordinary cohomology H^* , now applies to K-theory.

The exactness axiom is local in the sense that it relates three terms; iterating it along the suspensions of a pair produces a long exact sequence running infinitely in both directions. Because of periodicity this long sequence wraps around into a six-term cyclic sequence, the topological precursor of the six-term sequence for C^* -algebras developed in [5]. The next theorem assembles it.

Theorem 2.3.7: The Long Exact Sequence of a Pair

Let (X, A) be a finite CW pair (box 2.2.1) with inclusions $i: A \hookrightarrow X$ and $j: (X, \emptyset) \hookrightarrow (X, A)$. There is a long exact sequence

$$\cdots \rightarrow K^n(X, A) \xrightarrow{j^*} K^n(X) \xrightarrow{i^*} K^n(A) \xrightarrow{\delta} K^{n+1}(X, A) \xrightarrow{j^*} K^{n+1}(X) \rightarrow \cdots$$

natural in the pair, where δ is the coboundary map. By Bott periodicity (theorem 2.2.15) this sequence is 2-periodic and collapses to the six-term exact sequence

$$\begin{array}{ccccc} K^0(X, A) & \xrightarrow{j^*} & K^0(X) & \xrightarrow{i^*} & K^0(A) \\ \delta \uparrow & & & & \downarrow \delta \\ K^1(A) & \xleftarrow{i^*} & K^1(X) & \xleftarrow{j^*} & K^1(X, A) \end{array}$$

Proof:

Step 1: The Puppe sequence of the inclusion. The inclusion $i: A \hookrightarrow X$ of a subcomplex extends to its Puppe (cofiber) sequence of pointed spaces ([13, The Puppe Sequence and its Coexactness], applicable because (X, A) is a CW pair)

$$A_+ \xrightarrow{i} X_+ \xrightarrow{q} X/A \xrightarrow{\partial} \Sigma A_+ \xrightarrow{\Sigma i} \Sigma X_+ \xrightarrow{\Sigma q} \Sigma(X/A) \rightarrow \cdots$$

where each space is the cofiber of the preceding map and $\partial: X/A \rightarrow \Sigma A_+$ is the connecting map,

identifying the cofiber of q with the suspension ΣA_+ , as constructed in [13]. The construction in [13] establishes that applying any reduced cohomology functor to a cofiber sequence produces an exact sequence: for the reduced functor $\tilde{K}^0$ each consecutive pair of maps in the Puppe sequence gives an exact triple, the exactness being the case $n = 0$ of Axiom 2 in theorem 2.3.6.

Step 2: Applying $\tilde{K}^0$. Apply $\tilde{K}^0$ to the Puppe sequence. Each three-term stretch $W' \rightarrow W \rightarrow W''$ of cofiber and cofiber-of-cofiber yields exactness of $\tilde{K}^0(W'') \rightarrow \tilde{K}^0(W) \rightarrow \tilde{K}^0(W')$, and assembling all of them gives the exact sequence

$$\cdots \rightarrow \tilde{K}^0(\Sigma(X/A)) \rightarrow \tilde{K}^0(\Sigma X_+) \rightarrow \tilde{K}^0(\Sigma A_+) \xrightarrow{\partial^*} \tilde{K}^0(X/A) \rightarrow \tilde{K}^0(X_+) \rightarrow \tilde{K}^0(A_+)$$

Translating with the definitions $\tilde{K}^0(X_+) = K^0(X)$, $\tilde{K}^0(X/A) = K^0(X, A)$, and $\tilde{K}^0(\Sigma X_+) = \tilde{K}^{-1}(X_+) = K^{-1}(X)$, this reads

$$\cdots \rightarrow K^{-1}(X, A) \xrightarrow{j^*} K^{-1}(X) \xrightarrow{i^*} K^{-1}(A) \xrightarrow{\delta} K^0(X, A) \xrightarrow{j^*} K^0(X) \xrightarrow{i^*} K^0(A)$$

where the coboundary δ is induced by the Puppe connecting map ∂ . Iterating the Puppe sequence (each further suspension adds three more terms) extends this to a sequence infinite in both directions, exactly the displayed long exact sequence with K^n written for $\tilde{K}^0 \circ \Sigma^{-n}$ in the periodic sense.

Step 3: Naturality. A map of pairs $(X, A) \rightarrow (Y, B)$ induces a map of Puppe sequences $A \rightarrow X \rightarrow X/A \rightarrow B \rightarrow Y \rightarrow Y/B$, hence a map of the associated Puppe sequences, hence a commuting ladder after applying $\tilde{K}^0$. Thus the long exact sequence is natural in the pair, and in particular δ is a natural transformation.

Step 4: Collapsing by periodicity. By theorem 2.2.15 the groups satisfy $K^n \cong K^{n-2}$, so the doubly-infinite sequence has only two distinct rows of groups, those in even and odd degree. Identifying $K^{-1} = K^1$ via periodicity folds the long sequence into the cyclic six-term sequence

$$K^0(X, A) \rightarrow K^0(X) \rightarrow K^0(A) \xrightarrow{\delta} K^1(X, A) \rightarrow K^1(X) \rightarrow K^1(A) \xrightarrow{\delta} K^0(X, A)$$

which is the hexagon displayed in the statement. Exactness at each of the six vertices is inherited from the long exact sequence, completing the proof.

The six-term sequence is the computational engine of K-theory. Where ordinary cohomology gives a long exact sequence with infinitely many distinct groups, periodicity compresses the K-theory sequence into six terms arranged in a hexagon, and a single coboundary δ in each parity carries all the connecting information. Two standard specializations deserve to be recorded. The first relates the reduced theory of X , A , and X/A ; the second is the Mayer-Vietoris sequence for a cover.

Corollary 2.3.8: Reduced Cofiber Sequence

For a finite CW pair (X, A) with A nonempty there is an exact sequence of reduced groups

$$\tilde{K}^0(X/A) \xrightarrow{q^*} \tilde{K}^0(X) \xrightarrow{i^*} \tilde{K}^0(A) \xrightarrow{\delta} \tilde{K}^1(X/A) \xrightarrow{q^*} \tilde{K}^1(X) \xrightarrow{i^*} \tilde{K}^1(A) \xrightarrow{\delta} \tilde{K}^0(X/A)$$

Proof:

Choose the basepoint of X in A . Then $\tilde{K}^n(X) = K^n(X, \text{pt})$ and the splitting $K^n(X) = \tilde{K}^n(X) \oplus K^n(\text{pt})$ is natural in the based maps $A \rightarrow X \rightarrow X/A$, all of which preserve the basepoint. Subtracting the constant-point summand from the six-term sequence of theorem 2.3.7 for (X, A) leaves exactly the displayed sequence of reduced groups, since $K^n(X, A) = \tilde{K}^n(X/A)$ by definition 2.3.3 and the relative term is already reduced, completing the proof.

Corollary 2.3.9: Mayer-Vietoris in K-Theory

Let $X = X_1 \cup X_2$ be a finite CW complex written as the union of two subcomplexes, and set $X_0 = X_1 \cap X_2$, assumed nonempty. There is a six-term exact Mayer-Vietoris sequence

$$\begin{array}{ccccc} K^0(X) & \longrightarrow & K^0(X_1) \oplus K^0(X_2) & \longrightarrow & K^0(X_0) \\ \delta \uparrow & & & & \downarrow \delta \\ K^1(X_0) & \longleftarrow & K^1(X_1) \oplus K^1(X_2) & \longleftarrow & K^1(X) \end{array}$$

with the horizontal maps the differences of restrictions.

Proof:

Apply the long exact sequence of theorem 2.3.7 to the pairs (X, X_2) and (X_1, X_0) . The two are compared by a homeomorphism $X_1/X_0 \cong X/X_2$, proved directly rather than by excision: the inclusion $X_1 \hookrightarrow X$ carries $X_0 = X_1 \cap X_2$ into X_2 , so it descends to a continuous map $X_1/X_0 \rightarrow X/X_2$. This map is a bijection, because $X = X_1 \cup X_2$ makes every point of $X \setminus X_2$ lie in $X_1 \setminus X_0$ and both quotients collapse their second subspace to a single point; and it is a continuous bijection of compact Hausdorff spaces, hence a homeomorphism. Note that Axiom 3 of theorem 2.3.6 does *not* apply here: it requires an open U with $\overline{U} \subseteq \text{int}A$, and for $U = X \setminus X_1$, $A = X_2$ this fails already for $X = S^1$ covered by two closed arcs. Applying $\tilde{K}^n$ to the homeomorphism gives an isomorphism of relative groups

$$K^n(X, X_2) \cong K^n(X_1, X_0)$$

The Mayer-Vietoris sequence is the standard consequence of this excision isomorphism together with the two long exact sequences of pairs: splicing the pair sequences for (X, X_2) and (X_1, X_0) along the common relative term, and using periodicity to fold the result into six terms, produces the displayed hexagon, completing the proof.

These two corollaries are the tools used in section 2.4 to compute the K-theory of spheres, projective spaces, and other spaces built by gluing. The reduced cofiber sequence handles spaces presented as a quotient, and Mayer-Vietoris handles spaces presented as a union; periodicity guarantees that in either case only two groups, $\tilde{K}^0$ and $\tilde{K}^1$, ever appear.

Exercise 2.3.1

1. Using definition 2.3.1 and the computation of $\tilde{K}^0(S^n)$ in example 2.3.2, determine $\tilde{K}^{-n}(S^1)$ for all $n \geq 0$. Then identify $K^1(\{*\})$ and explain why it vanishes.

2. Let $\beta \in \tilde{K}^0(S^2)$ be the Bott generator and let β_X be the Bott map of definition 2.2.7. Show directly from the external product that $\beta_{S^0}: \tilde{K}^0(S^0) \rightarrow \tilde{K}^0(S^2)$ carries the generator $\gamma = [\mathbb{C}_-]$ of $\tilde{K}^0(S^0)$ to β , and deduce that β_{S^0} is an isomorphism without invoking theorem 2.2.15. (Note that neither the unit $1 = [\mathbb{C}]$ nor $[\mathbb{C}_-] - [\mathbb{C}_+]$ lies in the reduced group: both have nonzero rank at the basepoint, and $\tilde{K}^0(S^0)$ is the kernel of that rank.)
3. Use the reduced cofiber sequence (corollary 2.3.8) for the pair (D^2, S^1) to recompute $\tilde{K}^0(S^2)$ and $\tilde{K}^1(S^2)$, given that $\tilde{K}^*(D^2) = 0$ (contractibility) and $\tilde{K}^0(S^1) = 0$, $\tilde{K}^1(S^1) \cong \mathbb{Z}$.
4. Verify the excision homeomorphism $(X \setminus U)/(A \setminus U) \cong X/A$ used in the proof of theorem 2.3.6 in the explicit case $X = [0, 2]$, $A = [0, 1]$, $U = [0, \frac{1}{2}]$, by exhibiting the homeomorphism of quotients.
5. Let $X = S^2 \vee S^2$. Using Mayer-Vietoris (corollary 2.3.9) or the reduced cofiber sequence, compute $\tilde{K}^0(S^2 \vee S^2)$ and $\tilde{K}^1(S^2 \vee S^2)$.
6. Show that the coefficient ring $K^*(\{*\})$ of complex K-theory is the Laurent polynomial ring $\mathbb{Z}[\beta, \beta^{-1}]$ with β in degree -2 , and contrast this with the coefficient ring of ordinary cohomology. Explain in one sentence why this difference is exactly the failure of the dimension axiom.

2.4 Computations in K-Theory

The machinery assembled in the previous two sections (the ring $K^0(X)$ of section 2.1, the Bott periodicity isomorphism, and the long exact sequence that turns K^* into a cohomology theory in section 2.3) is worth very little until it produces numbers. This section computes K^* for the spaces a topologist meets first: spheres, complex projective spaces, and products. Every input is exhibited explicitly. The reduced groups $\tilde{K}^0(S^n)$ fall out of the suspension and clutching descriptions together with 2-periodicity; the ring $K^0(\mathbb{C}P^n)$ is computed from the projective-bundle structure and the Leray-Hirsch theorem; and the external product yields a Künneth statement for $K^*(X \times Y)$. The recurring theme is that K-theory, unlike ordinary cohomology, sees the multiplicative geometry of line bundles directly: the generator of $\tilde{K}^0(\mathbb{C}P^n)$ is literally $[H] - 1$ for the tautological line bundle H , and the single relation $([H] - 1)^{n+1} = 0$ encodes the whole ring.

Throughout, $K^0(X) = KU(X)$ denotes complex topological K-theory unless stated otherwise, and $\tilde{K}^0(X)$ is the reduced group, the kernel of the rank map $K^0(X) \rightarrow \mathbb{Z}$ at a chosen basepoint (definition 2.1.4). For a finite CW pair the negative groups $K^{-n}(X)$ and the relative groups $K^0(X, A)$ are those of definitions 2.3.1 and 2.3.3, and Bott periodicity supplies the isomorphism $\tilde{K}^0(X) \cong \tilde{K}^0(\Sigma^2 X)$ used at every turn (theorem 2.2.15).

The spheres are the natural starting point, since their bundle theory is already in hand. The clutching description of section 1.2 already computed the unreduced bundle sets $\text{Vect}_{\mathbb{C}}^n(S^k)$; what remains is to package them into the K-groups and to read off the ring structure. The cleanest route is through reduced K-theory and the identity $\tilde{K}^0(S^k) \cong \tilde{K}^0(\Sigma^2 S^k) = \tilde{K}^0(S^{k+2})$ supplied by Bott periodicity, which reduces every sphere to one of two base cases. The base cases are S^0 and S^1 , both of which are immediate from the bundle classifications already in hand.

Example 2.4.1: The K-Theory of Spheres

For complex K-theory the reduced groups of the spheres are

$$\tilde{K}^0(S^n) \cong \begin{cases} \mathbb{Z} & n \text{ even} \\ 0 & n \text{ odd} \end{cases} \quad \tilde{K}^{-1}(S^n) \cong \begin{cases} 0 & n \text{ even} \\ \mathbb{Z} & n \text{ odd} \end{cases}$$

and for $k \geq 1$ the full unreduced ring of the even sphere is $K^0(S^{2k}) \cong \mathbb{Z}[\eta]/(\eta^2)$, where $\eta = \beta^k \in \tilde{K}^0(S^{2k})$ is the k -fold Bott class and the generator squares to zero.

Step 1: The two base cases. The space S^1 is the suspension ΣS^0 , and every complex vector bundle over it is trivial: by proposition 1.2.8 the isomorphism classes of n -dimensional complex bundles over S^1 are in bijection with $[S^0, \mathrm{GL}_n(\mathbb{C})] = \pi_0(\mathrm{GL}_n(\mathbb{C}))$, which is a single point because $\mathrm{GL}_n(\mathbb{C})$ is path-connected (this is the content of example 1.2.11, item (1), on the complex side). Hence $\mathrm{Vect}_{\mathbb{C}}^n(S^1)$ is a single class for each n , every element of $K^0(S^1)$ is a difference of trivial bundles, and $\tilde{K}^0(S^1) = 0$. For $S^0 = \{+, -\}$ a two-point space, a complex bundle is just a pair of vector spaces, so $K^0(S^0) \cong \mathbb{Z} \oplus \mathbb{Z}$ by rank at each point, and the reduced group (kernel of rank at $+$) is $\tilde{K}^0(S^0) \cong \mathbb{Z}$, generated by the class $[\mathbb{C}_-]$ of the line bundle supported at the non-basepoint, which has rank 0 at the basepoint and so lies in the kernel of restriction there.

Step 2: Suspension and clutching for S^2 . For $k \geq 2$ the correspondence proposition 1.2.8 identifies $\mathrm{Vect}_{\mathbb{C}}^n(S^k)$ with $[S^{k-1}, \mathrm{GL}_n(\mathbb{C})] \cong \pi_{k-1}(\mathrm{GL}_n(\mathbb{C}))$ (based homotopy classes, using path-connectedness to drop the basepoint). In the stable range, and after deformation-retracting $\mathrm{GL}_n(\mathbb{C})$ onto $U(n)$, the reduced group $\tilde{K}^0(S^k)$ is the colimit $\varinjlim_n \pi_{k-1}(U(n)) = \pi_{k-1}(U)$, exactly the stabilization theorem 1.3.1. For $k = 2$ this is $\pi_1(U) \cong \mathbb{Z}$ by Bott periodicity (theorem 2.2.16), so

$$\tilde{K}^0(S^2) \cong \pi_1(U) \cong \mathbb{Z}$$

A generator is the class $\beta = [H] - 1$, where $H \rightarrow \mathbb{CP}^1 = S^2$ is the tautological line bundle and 1 is the trivial line bundle; this β is the *Bott class*, the same element implementing the periodicity isomorphism theorem 2.2.15.

Step 3: Periodicity propagates the answer. Each sphere is a double suspension of one two lower, $S^n = \Sigma^2 S^{n-2}$, so Bott periodicity theorem 2.2.15 gives

$$\tilde{K}^0(S^n) \cong \tilde{K}^0(S^{n-2})$$

Starting from $\tilde{K}^0(S^0) \cong \mathbb{Z}$ and $\tilde{K}^0(S^1) = 0$, induction yields $\tilde{K}^0(S^{2k}) \cong \mathbb{Z}$ and $\tilde{K}^0(S^{2k+1}) = 0$. The statement for $\tilde{K}^{-1}$ follows by the same periodicity from the suspension identity $\tilde{K}^{-1}(S^n) = \tilde{K}^0(\Sigma S^n) = \tilde{K}^0(S^{n+1})$.

Step 4: The ring structure. It remains to see that the generator of $\tilde{K}^0(S^{2k})$ squares to zero. Write $\eta = \beta^k$ for the k -fold external Bott class, a generator of $\tilde{K}^0(S^{2k}) \cong \mathbb{Z}$. The product η^2 lies in $\tilde{K}^0(S^{2k}) \cdot \tilde{K}^0(S^{2k})$, and the cup product on a reduced group factors through the smash product: multiplication $\tilde{K}^0(S^{2k}) \otimes \tilde{K}^0(S^{2k}) \rightarrow \tilde{K}^0(S^{2k})$ is induced by the diagonal $\Delta : S^{2k} \rightarrow S^{2k} \times S^{2k}$ followed by the projection to the smash $S^{2k} \wedge S^{2k} = S^{4k}$. The composite $S^{2k} \xrightarrow{\Delta} S^{2k} \times S^{2k} \rightarrow S^{2k} \wedge S^{2k}$ is null-homotopic, because the diagonal of a sphere lands in the wedge $S^{2k} \vee S^{2k}$ up to homotopy and the wedge maps to a point in the smash. Hence $\eta^2 = 0$, giving

$$K^0(S^{2k}) \cong \mathbb{Z}[\eta]/(\eta^2)$$

with $\eta = \beta^k$ the generator of the reduced group, completing the computation.

The reduction to S^0 and S^1 is the whole story: K-theory of spheres is $\mathbb{Z}$ in even degrees and 0 in odd degrees, with all higher structure governed by the single Bott class. The vanishing $\eta^2 = 0$ is forced by dimension once one knows that products of reduced classes live on smash products: there is simply no room for β^{2k} in $\tilde{K}^0(S^{2k})$ when $4k > 2k$. This is the K-theoretic form of the fact that $H^*(S^n)$ has trivial cup products in positive degrees, and it is the reason the even spheres are “square-zero” extensions of $\mathbb{Z}$. The contrast with $\mathbb{CP}^n$ below is instructive: there the generator does *not* square to zero until the $(n+1)$ -st power, and the difference is exactly that $\mathbb{CP}^n$ has cohomology in many even degrees while S^{2k} has it in only one.

A remark on the real theory rounds out the picture. The same suspension-and-clutching argument applied to KO uses $\pi_{k-1}(O)$ in place of $\pi_{k-1}(U)$, and the 8-fold periodicity of theorem 2.2.16 produces the well-known pattern $\tilde{KO}^0(S^n) \cong \mathbb{Z}, \mathbb{Z}/2, \mathbb{Z}/2, 0, \mathbb{Z}, 0, 0, 0$ for $n \equiv 0, 1, \dots, 7 \pmod{8}$. The two $\mathbb{Z}/2$'s in dimensions 1 and 2 are precisely the Möbius-type phenomena recorded in example 1.2.11; they have no analogue in the complex theory, where $\pi_{2j}(U) = 0$ kills the odd spheres outright.

2.4.2 Remark: The Index-Theoretic Meaning of β The generator $\beta \in \tilde{K}^0(S^2)$ is not an arbitrary choice. Under the identification $S^2 = \mathbb{CP}^1$ it is $[H] - 1$, and under the isomorphism $\tilde{K}^0(S^2) \cong K^{-2}(\{*\})$ it is the periodicity element whose inverse multiplication *is* the Bott map. When one later builds the analytic index of [13], β becomes the K-theory class of the symbol of the Dolbeault operator on $\mathbb{CP}^1$, and the equation $\beta^2 = 0$ on the sphere is the statement that this symbol class has no room to interact with itself. The reader should keep $\beta = [H] - 1$ in mind: it is the single generator from which the K-theory of all the spaces in this section is built.

The spheres gave a ring that is essentially trivial ($\mathbb{Z}$ with a square-zero generator). Complex projective space is the first genuinely multiplicative example, and it is the model computation of the subject. The target is the statement that $K^0(\mathbb{CP}^n)$ is a truncated polynomial ring on one generator, with the truncation degree equal to $n+1$. Two routes are given. The first is an induction over the cell structure of $\mathbb{CP}^n$, using only the long exact sequence of theorem 2.3.7 and the K-theory of spheres; it is elementary and self-contained, but it needs one input from the second route to identify the generator. The second is the Atiyah-Hirzebruch spectral sequence, which explains *why* the answer has the same shape as $H^*(\mathbb{CP}^n)$.

Theorem 2.4.3: K-Theory of Complex Projective Space

Let $H \rightarrow \mathbb{CP}^n$ be the tautological (canonical) complex line bundle and set $t = [H] - 1 \in K^0(\mathbb{CP}^n)$. Then $t^{n+1} = 0$ and

$$K^0(\mathbb{CP}^n) \cong \mathbb{Z}[t]/(t^{n+1})$$

as a ring, a free abelian group of rank $n+1$ on the basis $1, t, t^2, \dots, t^n$. The odd group vanishes: $K^{-1}(\mathbb{CP}^n) = 0$. The reduced group $\tilde{K}^0(\mathbb{CP}^n)$ is the ideal (t) , free of rank n on $t, \dots, t^n$.

The interpretive content is that the entire ring is generated by the single line bundle H , and that the only relation is the vanishing of $t^{n+1} = ([H] - 1)^{n+1}$. Geometrically, $([H] - 1)^{n+1} = 0$ is a K-theoretic Cayley-Hamilton identity: the line bundle H satisfies a degree- $(n+1)$ “polynomial” because $\mathbb{CP}^n$ is $2n$ -dimensional and there is no room for a nonzero class in $\tilde{K}^0$ beyond the n -th power. The ring is abstractly isomorphic to the cohomology ring $H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1})$ with $|x| = 2$, but the

isomorphism is not the identity on generators: since $x = c_1(H^\vee)$ and hence $c_1(H) = -x$, the class t corresponds under the Chern character to $e^{-x} - 1 = -x + x^2/2 - \dots$, not to x itself. The two proofs below make the ring structure and the vanishing relation completely explicit.

Proof:

The computation is carried out by the projective-bundle method; an alternative via the spectral sequence is given afterward.

Step 1: $\mathbb{CP}^n$ as a projective bundle and the basic class. Recall that $\mathbb{CP}^n = P(\mathbb{C}^{n+1})$ is the projectivization of the trivial bundle $\mathbb{C}^{n+1} \rightarrow \{*\}$. The tautological line bundle H has fiber over a line $\ell \in \mathbb{CP}^n$ equal to ℓ itself. Its first Chern class $x = c_1(H^\vee) \in H^2(\mathbb{CP}^n; \mathbb{Z})$ (with $H^\vee$ the dual of the tautological bundle, so that x is the standard positive generator) generates the cohomology ring,

$$H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1})$$

a free abelian group with basis $1, x, \dots, x^n$, concentrated in even degrees ([13]). This is the cohomological input.

Step 2: The candidate basis. The class $t = [H] - 1 \in \tilde{K}^0(\mathbb{CP}^n)$ is reduced, and the powers $1, t, \dots, t^n$ are the candidate basis. That they are one is proved by the induction of Step 3 over the cell structure, which is self-contained; theorem 3.2.2 is a statement about singular cohomology and is not available here, and the projective-bundle theorem in K-theory, which would give the result directly, is not developed in this book.

Step 3: Induction via the subcomplex $\mathbb{CP}^{n-1} \hookrightarrow \mathbb{CP}^n$. The pair $(\mathbb{CP}^n, \mathbb{CP}^{n-1})$ has quotient $\mathbb{CP}^n/\mathbb{CP}^{n-1} = S^{2n}$, the top cell. This yields the cofiber sequence and hence the long exact sequence of theorem 2.3.7 (with the relative group definition 2.3.3)

$$\dots \rightarrow \tilde{K}^{-1}(\mathbb{CP}^{n-1}) \rightarrow \tilde{K}^0(S^{2n}) \rightarrow \tilde{K}^0(\mathbb{CP}^n) \rightarrow \tilde{K}^0(\mathbb{CP}^{n-1}) \rightarrow \tilde{K}^1(S^{2n}) \rightarrow \dots$$

By example 2.4.1, $\tilde{K}^0(S^{2n}) \cong \mathbb{Z}$ and $\tilde{K}^{-1}(S^{2n}) = \tilde{K}^1(S^{2n}) = 0$, and by the inductive hypothesis $\tilde{K}^{-1}(\mathbb{CP}^{n-1}) = 0$ and $\tilde{K}^0(\mathbb{CP}^{n-1})$ is free of rank $n-1$. The sequence therefore collapses to the short exact sequence

$$0 \rightarrow \tilde{K}^0(S^{2n}) \rightarrow \tilde{K}^0(\mathbb{CP}^n) \rightarrow \tilde{K}^0(\mathbb{CP}^{n-1}) \rightarrow 0$$

with the two outer groups free of ranks 1 and $n-1$. Since the right-hand group is free, the sequence splits and $\tilde{K}^0(\mathbb{CP}^n)$ is free of rank n ; together with the rank-1 summand $K^0(\{*\}) = \mathbb{Z}$ this gives $K^0(\mathbb{CP}^n)$ free of rank $n+1$. The base case $n=1$ is $\tilde{K}^0(\mathbb{CP}^1) = \tilde{K}^0(S^2) \cong \mathbb{Z}$ from example 2.4.1. The vanishing of the odd groups propagates the same way, giving $K^{-1}(\mathbb{CP}^n) = 0$ at every stage.

Step 4: Identifying the basis with powers of t . The class t^n generates the top summand $\tilde{K}^0(S^{2n})$. First, t^n lies in that summand: under the splitting of Step 3 the image of t^n in $\tilde{K}^0(\mathbb{CP}^{n-1})$ is 0 (because t^n restricts to a class on $\mathbb{CP}^{n-1}$ which is the n -th power of t there, and $t^n = 0$ on $\mathbb{CP}^{n-1}$ by induction), so t^n comes from $\tilde{K}^0(S^{2n}) \cong \mathbb{Z}$. That it is a *generator* of that group, and not a proper multiple, is the one point the cell induction does not settle on its own. It follows from the multiplicative structure of the Atiyah-Hirzebruch spectral sequence ([13, The Atiyah-Hirzebruch Spectral Sequence]), worked out in the alternative proof below: t has filtration at least 2 with image the generator $x \in H^2(\mathbb{CP}^n; \mathbb{Z})$, so t^n has filtration at least $2n$ with leading term x^n , which generates $H^{2n}(\mathbb{CP}^n; \mathbb{Z})$. Hence $1, t, \dots, t^n$ is a $\mathbb{Z}$ -basis: each t^k generates the rank-1 subquotient

supported on the $2k$ -cell. Consequently t^{n+1} , restricting to $t^{n+1} = 0$ on $\mathbb{CP}^n$ (it is supported beyond the top cell), vanishes. This is the single relation, and

$$K^0(\mathbb{CP}^n) \cong \mathbb{Z}[t]/(t^{n+1})$$

completing the proof.

Proof Alternative proof:

The Atiyah-Hirzebruch spectral sequence explains the answer structurally. For any finite CW complex X the spectral sequence (the Atiyah-Hirzebruch spectral sequence of [13], built from the skeletal filtration and the fact that K^* is a generalized cohomology theory in the sense of the generalized-cohomology axioms of [13]) has

$$E_2^{p,q} = H^p(X; K^q(\{*\})) \implies K^{p+q}(X)$$

where $K^q(\{*\}) = \mathbb{Z}$ for q even and 0 for q odd, by Bott periodicity theorem 2.2.15. For $X = \mathbb{CP}^n$ the cohomology $H^p(\mathbb{CP}^n; \mathbb{Z})$ is $\mathbb{Z}$ for p even with $0 \leq p \leq 2n$ and 0 otherwise. Thus on the E_2 page every nonzero entry $E_2^{p,q}$ has both p and q even, so $p+q$ is always even. Every differential $d_r : E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$ changes the parity of $p+q$ when r is even and, for r odd, maps between a $(p+q)$ -even entry and a $(p+q)$ -odd entry, which is zero. More precisely, d_r raises total degree by 1, hence sends an even-total-degree class to an odd-total-degree group, and all odd-total-degree groups vanish. Therefore every differential is zero and the spectral sequence collapses at E_2 :

$$E_\infty^{p,q} = E_2^{p,q} = H^p(\mathbb{CP}^n; K^q(\{*\}))$$

The associated graded of $K^0(\mathbb{CP}^n)$ is $\bigoplus_p E_\infty^{p,-p} = \bigoplus_{k=0}^n H^{2k}(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}^{n+1}$, and that of $K^{-1}(\mathbb{CP}^n)$ is 0 since all odd cohomology vanishes. As each filtration quotient is free, the filtration of $K^0(\mathbb{CP}^n)$ splits and $K^0(\mathbb{CP}^n) \cong \mathbb{Z}^{n+1}$, with $K^{-1}(\mathbb{CP}^n) = 0$. The multiplicative structure is read off from the filtration: $t = [H] - 1$ has filtration ≥ 2 (it is reduced) and its image in $E_\infty^{2,-2} = H^2(\mathbb{CP}^n; \mathbb{Z})$ is the generator x , so t^k has filtration $\geq 2k$ with leading term x^k . Since $x^{n+1} = 0$ in $H^*(\mathbb{CP}^n)$ and there is nothing in filtration $> 2n$, the products $1, t, \dots, t^n$ are a basis and $t^{n+1} = 0$, recovering $K^0(\mathbb{CP}^n) \cong \mathbb{Z}[t]/(t^{n+1})$, as we sought to show.

The two proofs illuminate different features. The induction over cells is hands-on: it builds the answer one cell at a time and pins the generator t^k to the $2k$ -cell. The spectral sequence is conceptual: the collapse is forced purely by parity ($K^*(\{*\})$ is concentrated in even degrees and $\mathbb{CP}^n$ has cohomology only in even degrees, so no differential can connect two nonzero groups), and it shows that the abstract shape of $K^0(\mathbb{CP}^n)$ is dictated by $H^*(\mathbb{CP}^n)$. What the spectral sequence does *not* give for free is the precise generator; that $t = [H] - 1$ rather than some other lift of x is what makes the ring genuinely computable, and it is why the multiplicative statement is sharper than “ $K^0(\mathbb{CP}^n)$ is a free group of rank $n+1$ ”. The relation $([H] - 1)^{n+1} = 0$ should be compared with the Chern-class relation $c_1(H)^{n+1} = 0$: the latter holds in H^* , the former in K^0 , and the Chern character intertwines them. The comparison of the two rings is carried out in example 3.3.10, once the Chern character has been constructed in section 3.3.

The relation $t^{n+1} = 0$ also lets one compute the K-theory of $\mathbb{CP}^\infty$ as a limit, which is the generating case behind the splitting principle and the theory of Chern classes.

2.4.4 Remark: $K^0(\mathbb{CP}^\infty)$ as a Power Series Ring Passing to the colimit $\mathbb{CP}^\infty = \varinjlim_n \mathbb{CP}^n$ requires care, because theorem 2.4.3 computes K^0 of the finite stages only. The bridge is the Milnor exact sequence [13, The Milnor Exact Sequence], which for a based CW complex written as an increasing union of subcomplexes fits $\tilde{h}^n(X)$ into a short exact sequence between $\varprojlim_m^1 \tilde{h}^{n-1}(X_m)$ and $\varprojlim_m \tilde{h}^n(X_m)$; here $\tilde{h}^* = \tilde{K}^*$, which is a reduced cohomology theory by theorem 2.3.6, and $X_m = \mathbb{CP}^m$. The transition maps $K^0(\mathbb{CP}^n) \rightarrow K^0(\mathbb{CP}^{n-1})$ are the restrictions $\mathbb{Z}[t]/(t^{n+1}) \rightarrow \mathbb{Z}[t]/(t^n)$ sending $t \mapsto t$, which are surjective, so the tower satisfies the Mittag-Leffler condition and the $\varprojlim^1$ term vanishes by [13, Mittag-Leffler Collapse of the Milnor Sequence]. Hence

$$K^0(\mathbb{CP}^\infty) \cong \varprojlim_n \mathbb{Z}[t]/(t^{n+1}) \cong \mathbb{Z}[[t]]$$

the formal power series ring in $t = [H] - 1$. This is the home of the universal Chern class: a complex line bundle on any space is classified by a map to $\mathbb{CP}^\infty = BU(1)$, and its K-theory invariant is a power series in t . The same computation in cohomology gives $H^*(\mathbb{CP}^\infty; \mathbb{Z}) \cong \mathbb{Z}[[x]]$ truncated to a polynomial ring in each finite degree; the appearance of genuine power series in K-theory (rather than polynomials) reflects that K-theory is not connective in the way ordinary cohomology is. The formal-group-law structure on $\mathbb{Z}[[t]]$ coming from the tensor product of line bundles, $t(L_1 \otimes L_2) = t(L_1) + t(L_2) + t(L_1)t(L_2)$, is the multiplicative formal group, in contrast to the additive formal group $x(L_1 \otimes L_2) = x(L_1) + x(L_2)$ of ordinary cohomology.

The third computation concerns products. Computing K^* of a product reduces, when it works, to the tensor product of the factors' K-theory. The mechanism is the external product, a pairing $K^0(X) \otimes K^0(Y) \rightarrow K^0(X \times Y)$ built from the external tensor product of bundles. After recording it and a worked instance, the section closes with the Künneth statement, proved by comparing the two sides as cohomology theories in one variable.

Example 2.4.5: The External Product on $S^2 \times S^2$

For vector bundles $E \rightarrow X$ and $F \rightarrow Y$, the *external tensor product* is $E \boxtimes F = \pi_X^*(E) \otimes \pi_Y^*(F) \rightarrow X \times Y$, where π_X, π_Y are the two projections. On classes this defines the external product

$$\times : K^0(X) \otimes K^0(Y) \longrightarrow K^0(X \times Y), \quad a \times b = \pi_X^*(a) \cdot \pi_Y^*(b)$$

which is a ring homomorphism out of the tensor-product ring. Take $X = Y = S^2$ with $\tilde{K}^0(S^2) = \mathbb{Z}\beta$, $\beta = [H] - 1$ and $\beta^2 = 0$ (example 2.4.1). Write $\beta_1 = \pi_1^*(\beta)$ and $\beta_2 = \pi_2^*(\beta)$ in $K^0(S^2 \times S^2)$. Since pullback is a ring map and $\beta^2 = 0$ on each factor, $\beta_1^2 = \beta_2^2 = 0$, while the cross term $\beta_1\beta_2 = \beta \times \beta$ is the external square. The external product gives a ring isomorphism

$$K^0(S^2 \times S^2) \cong K^0(S^2) \otimes_{\mathbb{Z}} K^0(S^2) \cong \frac{\mathbb{Z}[\beta_1, \beta_2]}{(\beta_1^2, \beta_2^2)}$$

a free abelian group of rank 4 on the basis $1, \beta_1, \beta_2, \beta_1\beta_2$. The reduced group $\tilde{K}^0(S^2 \times S^2)$ is the rank-3 ideal (β_1, β_2) , and one checks directly that $\beta_1\beta_2$ generates the top class supported on the 4-cell $S^2 \wedge S^2 = S^4$: indeed under the collapse $S^2 \times S^2 \rightarrow S^2 \wedge S^2 = S^4$ the class $\beta_1\beta_2$ pulls back from a generator of $\tilde{K}^0(S^4) \cong \mathbb{Z}$, consistent with $\tilde{K}^0(S^4) \cong \tilde{K}^0(S^2)$ under Bott periodicity. The element $\beta_1\beta_2$ is the external Bott class $\beta \times \beta = \beta^2 \in \tilde{K}^0(S^4)$, the same $\eta = \beta^2$ that appeared as the generator of $\tilde{K}^0(S^4)$ in example 2.4.1; the apparent clash with $\beta^2 = 0$ is resolved by noting that the *internal* square on a single S^2 vanishes while the *external* square on $S^2 \times S^2$ does not, because the diagonal $S^2 \rightarrow S^2 \times S^2$ kills it but the product space retains it.

The example exhibits the general pattern: the K-theory of a product is the tensor product of the K-theories, provided no torsion or extension obstructions intervene. The precise statement is the Künneth theorem in K-theory, which holds cleanly when one factor has torsion-free, finitely generated K-theory.

Theorem 2.4.6: Künneth Theorem in K-Theory

Let X and Y be finite CW complexes, and suppose at least one of $K^*(X)$, $K^*(Y)$ is a finitely generated free abelian group (for instance whenever that factor is S^n or $\mathbb{C}P^n$). Then the external product is a $\mathbb{Z}/2$ -graded ring isomorphism

$$K^*(X) \otimes_{\mathbb{Z}} K^*(Y) \xrightarrow{\sim} K^*(X \times Y)$$

In particular, if both $K^*(X)$ and $K^*(Y)$ are finitely generated and free, then so is $K^*(X \times Y)$, with rank equal to the product of the ranks.

In words: when the K-theory of one factor is free, computing K^* of a product is pure algebra, the tensor product of graded rings. The freeness hypothesis is what avoids a Tor correction term; in full generality there is a short exact Künneth sequence with a $\text{Tor}_1^{K^*(\{*\})}$ summand, exactly as in ordinary cohomology, but the spaces of this chapter (spheres, projective spaces, and their products) all have free K-theory, so the clean statement applies to every example here.

Proof:

Step 1: The external product is well-defined and natural. For $E \rightarrow X$ and $F \rightarrow Y$ the bundle $E \boxtimes F = \pi_X^* E \otimes \pi_Y^* F$ depends only on the isomorphism classes of E and F by functoriality of pullback (proposition 1.2.3) and bilinearity of the tensor product over $\oplus$ (definition 1.1.17). Hence $a \times b = \pi_X^*(a) \cdot \pi_Y^*(b)$ extends to a homomorphism $K^0(X) \otimes K^0(Y) \rightarrow K^0(X \times Y)$, and using the negative groups (definition 2.3.1) to a graded homomorphism $K^*(X) \otimes K^*(Y) \rightarrow K^*(X \times Y)$. It is a ring map because pullback is a ring map and the cup product on $K^*(X \times Y)$ is graded-commutative.

Step 2: Both sides are cohomology theories in the inducting variable. By symmetry of the external product (the two factors enter on equal footing, since $X \times Y \cong Y \times X$) it costs nothing to label the factors so that Y is the one whose K-theory is finitely generated and free; this is exactly the hypothesis. Fix this free factor Y and regard both sides as cohomology theories in the remaining variable X , over whose cells the induction will run. The functor $X \mapsto K^*(X \times Y)$ is a generalized cohomology theory (it is K^* precomposed with $- \times Y$, and $- \times Y$ preserves cofiber sequences), and $X \mapsto K^*(X) \otimes_{\mathbb{Z}} K^*(Y)$ is a cohomology theory because the fixed factor $K^*(Y)$ is a *free* $\mathbb{Z}$ -module, so tensoring with it is exact and preserves the long exact sequences (theorem 2.3.7). The external product is a natural transformation between these two theories.

Step 3: The transformation is an isomorphism on a point, hence everywhere. Regard K^* as $\mathbb{Z}/2$ -graded via Bott periodicity, so that the coefficient ring is $K^*(\{*\}) = \mathbb{Z}$ concentrated in even degree. On $X = \{*\}$ the external product is

$$K^*(\{*\}) \otimes_{\mathbb{Z}} K^*(Y) = \mathbb{Z} \otimes_{\mathbb{Z}} K^*(Y) \xrightarrow{\sim} K^*(Y)$$

the unit isomorphism of the tensor product, since $K^*(\{*\}) = \mathbb{Z}$. A natural transformation of cohomology theories on finite CW complexes that is an isomorphism on the point (together with the additivity axiom, which handles disjoint unions and hence S^0) is an isomorphism on all finite CW complexes: this is the standard comparison via the five lemma and induction over

cells, run on the skeletal filtration. Concretely, suppose the transformation is an isomorphism for all complexes with fewer cells than X . Attaching a cell to a subcomplex X' gives a cofiber sequence $X' \hookrightarrow X \rightarrow X/X' = S^m$, and naturality produces a map of long exact sequences (theorem 2.3.7) in which the maps for X' and for S^m are isomorphisms (the latter because $K^*(S^m) \otimes K^*(Y) \rightarrow K^*(S^m \times Y)$ is an iterated suspension of the point case, using Bott periodicity theorem 2.2.15). The five lemma then forces the map for X to be an isomorphism.

Step 4: Conclusion. The induction starts from $X = \{*\}$ (Step 3) and covers every finite CW complex X , so for the fixed free factor Y the external product

$$K^*(X) \otimes_{\mathbb{Z}} K^*(Y) \xrightarrow{\sim} K^*(X \times Y)$$

is an isomorphism for *every* finite CW complex X , with no freeness assumed on $K^*(X)$. The freeness that was used is that of the fixed factor $K^*(Y)$, which is what made $X \mapsto K^*(X) \otimes K^*(Y)$ a cohomology theory in Step 2; by the relabelling at the start of that step this covers the stated hypothesis that at least one of $K^*(X)$, $K^*(Y)$ is finitely generated and free. When *both* factors are finitely generated and free, the rank and freeness statements follow since a tensor product of finitely generated free abelian groups is finitely generated and free of the product rank, completing the proof.

With the Künneth theorem the K-theory of every product of spheres and projective spaces is now mechanical. For instance $K^0((\mathbb{CP}^n)^{\times m}) \cong \mathbb{Z}[t_1, \dots, t_m]/(t_1^{n+1}, \dots, t_m^{n+1})$, the tensor power of $\mathbb{Z}[t]/(t^{n+1})$, with each $t_i = \pi_i^*([H] - 1)$ the pulled-back generator from the i -th factor. This is the K-theoretic analogue of the cohomology of a product of projective spaces, and it is the computation that underlies the splitting principle: a sum of line bundles over a base B is classified by a map to $(\mathbb{CP}^\infty)^{\times m} = BU(1)^m$, and its K-theory class is a polynomial in the pulled-back generators.

Exercise 2.4.1

1. Using only the suspension identity $\tilde{K}^{-1}(X) = \tilde{K}^0(\Sigma X)$ and the values $\tilde{K}^0(S^0) \cong \mathbb{Z}$, $\tilde{K}^0(S^1) = 0$, together with Bott periodicity, compute $\tilde{K}^0(S^n)$ and $\tilde{K}^{-1}(S^n)$ for $n = 3, 4, 5, 6$. State the unreduced groups $K^0(S^n)$ and $K^{-1}(S^n)$ in each case.
2. Let $t = [H] - 1 \in K^0(\mathbb{CP}^2)$. Compute, in the basis $1, t, t^2$ of $\mathbb{Z}[t]/(t^3)$, the products $t \cdot t$, $t \cdot t^2$, and $(1 + t)^2$. Deduce the K-theory class of $H^{\otimes 2} = H^2$ as an element of $\mathbb{Z}[t]/(t^3)$.
3. Use the Atiyah-Hirzebruch spectral sequence to compute $K^*(S^2 \times S^4)$. State the E_2 page explicitly, explain why all differentials vanish, and give the ranks of K^0 and K^{-1} . Confirm the answer agrees with the Künneth theorem applied to $K^*(S^2) \otimes K^*(S^4)$.
4. Let $H \rightarrow \mathbb{CP}^1 = S^2$ be the tautological line bundle and $\beta = [H] - 1$. Show directly, without invoking the smash-product argument, that $\beta^2 = 0$ in $K^0(\mathbb{CP}^1)$ by using the relation $t^{n+1} = 0$ of theorem 2.4.3 for $n = 1$. Then explain why the external square $\beta \times \beta$ is nonzero in $K^0(S^2 \times S^2)$.
5. Compute the rank of $\tilde{K}^0(\mathbb{CP}^n \times \mathbb{CP}^m)$ and write down a $\mathbb{Z}$ -basis in terms of the pulled-back generators $t_1 = \pi_1^*([H] - 1)$ and $t_2 = \pi_2^*([H] - 1)$. Identify the single class supported on the top cell.
6. The reduced suspension formula gives $\tilde{K}^0(\Sigma X) \cong \tilde{K}^{-1}(X)$. Using $\mathbb{CP}^2/\mathbb{CP}^1 = S^4$ and the cofiber long exact sequence, recompute $\tilde{K}^0(\mathbb{CP}^2)$ and verify it is free of rank 2, matching

theorem 2.4.3.

The Chern Character and Characteristic Classes

The previous chapter turned the monoid of vector bundles into the ring $K^0(X)$ and showed it is periodic. Ordinary cohomology is the other invariant attached to a bundle, and this chapter builds the two bridges between them. In one direction a bundle determines cohomology classes, the *characteristic classes*, which measure how far it is from trivial: the Stiefel-Whitney classes $w_i(E) \in H^i(B; \mathbb{Z}/2\mathbb{Z})$ of a real bundle, the Chern classes $c_i(E) \in H^{2i}(B; \mathbb{Z})$ of a complex one, the Pontryagin classes $p_i(E) \in H^{4i}(B; \mathbb{Z})$ of a real bundle, and the Euler class $e(E) \in H^n(B; \mathbb{Z})$ of an oriented bundle of rank n . Their construction belongs to algebraic topology and is cited to [13]; what is developed here is the bundle-theoretic reading of each class as an obstruction, and the computations.

In the other direction the whole ring $K^0(X)$ maps to cohomology. That map is the Chern character, and theorem 3.3.8 shows it is an isomorphism after tensoring with $\mathbb{Q}$: rationally, K-theory carries exactly the information of even-dimensional cohomology. Integrally it does not, and the gap is where the arithmetic of the subject lives. Section 3.1 sets up the four classes and reads them off the skeleta of the base; section 3.2 constructs them and identifies each as an obstruction; section 3.3 builds ch and proves the rational isomorphism; section 3.4 measures how far that isomorphism is from integral; and section 3.5 evaluates characteristic classes against a fundamental class, producing the numerical invariants that decide when a manifold bounds.

3.1 Definition and Intuitions

A characteristic class assigns cohomology to a bundle, naturally in the base. Two pieces of data go into the definition and must be kept apart: the *structure group* G , which says what kind of bundle is being classified, and the *coefficient group* A , which says where the class lives. Conflating them is

easy and produces nonsense, since the four classes below use three different structure groups and two different coefficient groups.

Definition 3.1.1: Characteristic Classes

Let G be a topological group and let A be an abelian group. Write $b_G(-)$ for the contravariant functor sending a space B to the set of isomorphism classes of principal G -bundles over B , and a map f to the pullback f^* . A *characteristic class for G -bundles with coefficients in A* is a natural transformation

$$c: b_G \Rightarrow H^*(-; A)$$

Concretely, c assigns to each principal G -bundle $P \rightarrow X$ a class $c(P) \in H^*(X; A)$ subject to

$$c(f^*P) = f^*c(P)$$

the f^* on the left being pullback of bundles and the one on the right the induced map in cohomology. The component landing in $H^i(-; A)$ is written c_i .

A rank- n real vector bundle has structure group $O(n)$ and is the bundle associated to its frame bundle, so $b_{O(n)}(B)$ is identified with $\text{Vect}_{\mathbb{R}}^n(B)$; the complex case replaces $O(n)$ by $U(n)$ and the oriented real case by $SO(n)$. Throughout this section $\text{Vect}^n(B)$ means $\text{Vect}_{\mathbb{R}}^n(B)$, and $\text{Vect}^n(B) \cong [B, \text{Grass}_n]$ by theorem 1.2.13. In particular, if $f: B \rightarrow \text{Grass}_n$ is a classifying map and $E = f^*(E_n)$ for the universal bundle E_n , then

$$c_i(E) = c_i(f^*(E_n)) = f^*(c_i(E_n))$$

so if f induces the trivial map on cohomology with the relevant coefficients, every $c_i(E)$ vanishes.

The definitions below name which (G, A) each class belongs to. Naming the pair does *not* single out one class: there are many natural transformations $\text{Vect}^n(-) \Rightarrow H^*(-; \mathbb{Z}/2\mathbb{Z})$, so each definition names a type and then says which member of that type it means. In each case the member is named by its construction upstream, as the pullback along a classifying map of a fixed universal class: w_i and c_i from the polynomial generators of $H^*(BO(n); \mathbb{Z}/2\mathbb{Z})$ and $H^*(BU(n); \mathbb{Z})$, the Euler class from the transgression class on $BSO(n)$. The axiom lists of theorem 3.2.4 and theorem 3.2.5 record what those families satisfy, and are what the computations below actually use; they are not offered here as characterizations, since neither theorem is stated with a uniqueness clause. The Pontryagin classes are of type $(O(n), \mathbb{Z})$ like the rest, but definition 3.1.3 names its member by an explicit formula in the Chern classes of the complexification rather than by a universal class.

The four classes are the following instances.

Definition 3.1.2: Stiefel-Whitney Classes

Take $G = O(n)$, so that $b_G(B) = \text{Vect}^n(B)$, and coefficients $A = \mathbb{Z}/2\mathbb{Z}$. A characteristic class of this type is a *Stiefel-Whitney class*, written $w_i(E) \in H^i(B; \mathbb{Z}/2\mathbb{Z})$, with total class $w(E) = 1 + w_1(E) + w_2(E) + \dots$. The particular family so named is the one constructed in [13, Stiefel-Whitney Classes and the Total Class] as $w_i(E) = f_E^*w_i$, the pullback along a classifying map $f_E: B \rightarrow BO(n)$ of the degree- i polynomial generator of $H^*(BO(n); \mathbb{Z}/2\mathbb{Z})$.

These classes shall be of interest to us as they shall generate the cohomology ring $H^*(\text{Grass}_n; \mathbb{Z}/2\mathbb{Z})$, in case:

$$H^*(\text{Grass}_n; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}[w_1(E_n), \dots, w_n(E_n)]$$

This is proved in [13, The Cohomology of $BO(n)$]. It fails with $\mathbb{Z}$ coefficients, and modulo torsion the Pontryagin classes are what characterize the integral case:

Definition 3.1.3: Pontryagin Classes

For a real vector bundle $E \rightarrow B$, the *Pontryagin classes* are the integral characteristic classes

$$p_i(E) = (-1)^i c_{2i}(E \otimes_{\mathbb{R}} \mathbb{C}) \in H^{4i}(B; \mathbb{Z})$$

obtained from the Chern classes of the complexification, with $p_0(E) = 1$. The total Pontryagin class is $p(E) = 1 + p_1(E) + p_2(E) + \cdots$. Only the even Chern classes of $E \otimes_{\mathbb{R}} \mathbb{C}$ are used: the odd ones are 2-torsion, because $E \otimes_{\mathbb{R}} \mathbb{C}$ is isomorphic to its own conjugate.

The classes of interest sit in H^{4i} because the odd Chern classes of $E \otimes_{\mathbb{R}} \mathbb{C}$ are 2-torsion, so they are the only ones of infinite order; they need not vanish, and over a base with 2-torsion in its integral cohomology they generally do not. For an oriented bundle the classifying map may be taken as $f: B \rightarrow \widetilde{\text{Grass}}_n$ into the oriented Grassmannian, which yields the following.

Definition 3.1.4: Euler Class

Let B be paracompact. Take $G = SO(n)$, so that $b_G(B) = \widetilde{\text{Vect}}^n(B)$ is the set of isomorphism classes of *oriented* real rank- n bundles, classified by maps into the oriented Grassmannian $\widetilde{\text{Grass}}_n = BSO(n)$, and coefficients $A = \mathbb{Z}$. The *Euler class* is the characteristic class $e(E) \in H^n(B; \mathbb{Z})$ of this type constructed in [13, Euler Class of an Oriented Bundle], namely the pullback

$$e(E) = f_E^* e, \quad e \in H^n(BSO(n); \mathbb{Z})$$

of the universal class e along a classifying map $f_E: B \rightarrow BSO(n)$ for E , where e is the transgression of the oriented fiber generator in the Gysin sequence of the universal sphere bundle $S^{n-1} \rightarrow BSO(n-1) \rightarrow BSO(n)$. The orientation is part of the data and not merely a hypothesis: reversing it replaces the classifying map by its composite with the deck involution of $BSO(n) \rightarrow BO(n)$, which negates the universal class and hence $e(E)$, so e is not a function on orientable bundles.

For an oriented real vector bundle, f induces the trivial map exactly when the Stiefel-Whitney, Pontryagin, and Euler classes all vanish.

The complex case is separated out because removing 0 from $\mathbb{C}$ leaves a path-connected space, so complex bundles carry no orientability obstruction:

Definition 3.1.5: Chern Class

Take $G = U(n)$, so that $b_G(B) = \text{Vect}_{\mathbb{C}}^n(B)$, and coefficients $A = \mathbb{Z}$. A characteristic class of this type is a *Chern class*, written $c_i(E) \in H^{2i}(B; \mathbb{Z})$, with total class $c(E) = 1 + c_1(E) + c_2(E) + \cdots$. The family so named is the one constructed in [13, Chern Classes and the Total Chern Class] as $c_i(E) = f_E^* c_i$, the pullback along a classifying map $f_E: B \rightarrow BU(n)$ of the degree- $2i$ polynomial generator of $H^*(BU(n); \mathbb{Z})$.

The complex Grassmannian satisfies

$$H^*(\text{Grass}_n(\mathbb{C}); \mathbb{Z}) \cong \mathbb{Z}[c_1, \dots, c_n]$$

where $c_i = c_i(E_n)$ for the universal complex bundle $E_n \rightarrow \text{Grass}_n(\mathbb{C})$

Several classes were needed above to detect triviality, because vanishing characteristic classes are necessary but not sufficient for it. The simplest nontrivial example is the tangent bundle of S^5 , which is nontrivial (S^5 is not parallelizable) yet has every characteristic class zero. Two separate reasons combine. The Stiefel-Whitney, Chern and Pontryagin classes are *stably invariant*, meaning unchanged by direct sum with a trivial bundle, and TS^5 is stably trivial, so all of them vanish. The Euler class is *not* stably invariant (indeed $e(E \oplus \mathbb{R}) = 0$ for every oriented E over a finite-type CW base, by proposition 3.2.12(1) and (2): the trivial line bundle is pulled back from a bundle over a point, so $e(\mathbb{R}) = 0$, and the Whitney formula multiplies by it), and it vanishes here for the different reason that $\langle e(TS^5), [S^5] \rangle = \chi(S^5) = 0$. Every tangent bundle of a sphere is stably trivial, since TS^n plus the trivial normal line is $\mathbb{R}^{n+1}$, so the stably invariant classes cannot distinguish odd-dimensional spheres.

Let us now think about for a moment what characteristic classes are detecting. The first “obstruction” it detects is orientability. For a vector bundle $E \rightarrow B$ with B path-connected there is a map

$$\pi_1(B) \rightarrow \mathbb{Z}/2\mathbb{Z}$$

where it assigns 0 or 1 to each loop according to whether orientation of fibers is preserved or reversed (as one goes around the loop). As $\mathbb{Z}/2\mathbb{Z}$ is abelian, the homomorphism factors through the abelianization $H_1(B)$ of $\pi_1(B)$, and the homomorphisms $H_1(B) \rightarrow \mathbb{Z}/2\mathbb{Z}$ are elements of $H^1(B; \mathbb{Z}/2\mathbb{Z})$. From this one obtains an element of $H^1(B; \mathbb{Z}/2\mathbb{Z})$ associated to E , and it vanishes exactly when E is orientable. This shall be the interpretation of the first Stiefel-Whitney class $w_1(E)$.

To generalize to higher Stiefel-Whitney classes, let us re-interpret this result when B is a CW complex. Then E is orientable if and only if the restriction of the 1-skeleton B^1 of B is orientable, as all loops in B can be deformed to lie in B^1 . Then a vector bundle over a 1-dimensional CW complex is trivial if and only if it is orientable, so $w_1(E)$ measures whether E is trivial over the 1-skeleton B^1 . The fibers may be higher dimensional: over each loop the object in question is a vector bundle on S^1 . The next question is whether triviality over the 1-skeleton extends to the 2-skeleton. Assuming E is trivial over B^1 and its fibers are n -dimensional, choose n orthonormal sections over B^1 and ask whether they extend to orthonormal sections over each 2-cell. Each 2-cell is given by a characteristic map $D^2 \rightarrow B$, so E pulls back over the disk D^2 , where the sections over B^1 pullback to sections over ∂D^2 . As D^2 is contractible, this bundle is trivial. Choosing a trivialization, the sections over ∂D^2 determine a map:

$$\partial D^2 \rightarrow O(n)$$

Choosing a trivialization with the same orientation as the trivialization of E over B^1 puts the image in $SO(n)$. Such a map is an element of $\pi_1(SO(n))$, so each 2-cell is assigned an element of $\pi_1(SO(n))$. Then B^1 extends to orthonormal sections over B^2 if and only if this element of $\pi_1(SO(n))$ is zero for each 2-cell.

This groups is also relatively easy to compute : when $n = 1$ it is 0, when $n = 2$ it is $\mathbb{Z}$ (as $SO(2)$ is homeomorphic to S^1). For $n \geq 3$, it is $\mathbb{Z}/2\mathbb{Z}$ (exercise). Thus, the inclusion $SO(2) \hookrightarrow SO(n)$ induces a surjection $\pi_1(SO(2)) \twoheadrightarrow \pi_1(SO(n))$, so the generator of $\pi_1(SO(n))$ can be represented by the loop given by rotating the plane through a full turn, fixing the orthogonal complement of the plane. This defines a function assigning to each 2-cell (regarded as a 2-dimensional cellular cochain) with

coefficients in $\pi_1(SO(n))$. This shall be a cellular cocycle, and so an element of $H^2(B; \pi_1(SO(n)))$. This element shall be called $\omega_2(E)$, and will not depend on the choice of sections over B^1 . So E is trivial over B^2 if and only if $\omega_2(E)$ is zero in $H^2(B; \pi_1(SO(n)))$. The name is deliberately not w_2 : the obstruction takes its coefficients in $\pi_1(SO(n))$, which varies with n , while the Stiefel-Whitney class of definition 3.1.2 always lives in $H^2(B; \mathbb{Z}/2\mathbb{Z})$. If $n = 1$ then $\omega_2(E) = 0$. If $n = 2$ then $\omega_2(E) \in H^2(B; \mathbb{Z})$ and is the Euler class $e(E)$ covered below, whose mod-2 reduction is $w_2(E)$ by proposition 3.2.8. Only for $n \geq 3$ is $\omega_2(E)$ itself the class $w_2(E) \in H^2(B; \mathbb{Z}/2\mathbb{Z})$.

The natural next question to ask is if E is trivial over B^2 , can it be extended to be trivial over B^3 which again is given by looking at elements of $\pi_2(SO(n))$. Fortunately for us, $\pi_2(SO(n)) = 0$ for all $n \in \mathbb{N}_{>0}$ and so the sections automatically extend over B^3 . For extension over 4-cells, $\pi_3(SO(n))$ is relatively simple to compute (see [10, p.75]). For higher dimensional extension B^k for $k \geq 5$, things become a good deal more complicated as the groups $\pi_{k-1}(SO(n))$ become increasingly difficult to compute. They are finitely generated abelian groups whose free part is fairly easy to compute, but whose torsion part is complicated¹. In the range of $n > k$, then Bott periodicity gives the value of $\pi_{k-1}(SO(n))$. For $k = 4i$,

$$\pi_{4i-1}(SO(n)) \cong \mathbb{Z}$$

which is why the Pontryagin class lies in $H^{4i}(B; \mathbb{Z})$, and it detects the non-triviality of bundles through the obstruction

$$\omega_{4i}(E) \in H^{4i}(B; \pi_{4i-1}(SO(n)))$$

which is the Pontryagin class $p_i(E)$ of definition 3.1.3 rather than a Stiefel-Whitney class, the coefficient group being $\mathbb{Z}$.

In the stable range $n > k$ the other nontrivial groups are

$$\pi_{k-1}(SO(n)) \cong \mathbb{Z}/2\mathbb{Z} \quad k = 8i + 1 \ (i \geq 1), \quad k = 8i + 2 \ (i \geq 0)$$

The restriction on the first family matters, and it is the only one. At $i = 0$, $k = 1$ the group is $\pi_0(SO(n)) = 0$, since $SO(n)$ is connected, whereas Bott's $\mathbb{Z}/2$ in degree 0 is $\pi_0(O) = \mathbb{Z}/2\mathbb{Z}$, which belongs to the full orthogonal group; that discrepancy is confined to degree 0, and in the positive degrees $8i$ with $i \geq 1$ the group $\pi_{8i}(SO(n)) \cong \mathbb{Z}/2\mathbb{Z}$ is present for $SO(n)$ as well. The second family runs from $i = 0$: there $k = 2$ and the group is $\pi_1(SO(n)) \cong \mathbb{Z}/2\mathbb{Z}$ for $n \geq 3$, the case computed above.

In degree $k = 2$ the obstruction is the class named $\omega_2(E)$ above, which for $n \geq 3$ is the second Stiefel-Whitney class. In every other degree of the two families, that is for $k = 8i + 1$ and $k = 8i + 2$ with $i \geq 1$, the obstruction $\omega_k(E)$ is not detected by the Stiefel-Whitney or Pontryagin classes; ?? takes up that claim and identifies these obstructions through KO -theory instead.

A second way of measuring triviality leads to the Euler class. Consider an n -dimensional vector bundle $E \rightarrow B$ and the obstruction to the existence of n orthonormal sections. Take first the obstruction to a single orthonormal section, that is, to a section of unit vectors. The induction again runs over the skeleta of B . Given a section over B^{k-1} , extending over a k -cell pulls E back to the trivial bundle over D^k via the characteristic map, leaving a unit-vector section on ∂D^k . This time the relevant homotopy group is that of the fiber sphere rather than of the structure group. Hence, if $k < n$, such an extension always is possible, so the first (possibly) nontrivial obstruction occurs when $k = n$ where

$$\pi_{k-1}(S^{n-1}) \cong \mathbb{Z}$$

¹Interestingly, this contrasts with elliptic curves over rationals $E(\mathbb{Q})$ for which it is relatively easy to compute the torsion but a million dollar problem (literally) to compute the free part

Note that to identify $\mathbb{Z}$ to $\pi_{n-1}(S^{n-1})$ in a consistent way, it is *necessary* be that E is orientable. Thus class is the Euler class and shall be denoted $e(E)$. Dropping the orientability condition, the coefficients must be $\mathbb{Z}/2\mathbb{Z}$, giving the Stiefel-Whitney class $w_n(E) \in H^n(B; \mathbb{Z}/2\mathbb{Z})$.

3.2 The Classes

The skeleton-by-skeleton picture of section 3.1 says what the classes mean but computes almost nothing: it needs the homotopy groups $\pi_{k-1}(SO(n))$, which are not available in general. This section takes the algebraic route instead, recording the axioms the classes satisfy and the projective-bundle machinery that produces them, and returns at the end to show that the two descriptions agree. The base space is assumed paracompact throughout, which by box 1.0.1 includes Hausdorff, and the first order of business is that the classes are defined there at all.

Proposition 3.2.1: The Classes over a Paracompact Base

Let B be paracompact and let $E \rightarrow B$ be a real (respectively complex) vector bundle of rank n . Then for $1 \leq i \leq n$ the formulas

$$w_i(E) = f_E^* w_i, \quad c_i(E) = f_E^* c_i$$

of [13, Stiefel-Whitney Classes and the Total Class] and [13, Chern Classes and the Total Chern Class] define classes $w_i(E) \in H^i(B; \mathbb{Z}/2\mathbb{Z})$ and $c_i(E) \in H^{2i}(B; \mathbb{Z})$, agreeing with those definitions whenever B is a CW complex, and the convention $w_i(E) = 0$ for $i > n$ and $c_i(E) = 0$ for $i > n$ is carried over unchanged. They are natural: for a map $g: B' \rightarrow B$ of paracompact spaces, $w_i(g^*E) = g^*w_i(E)$ and $c_i(g^*E) = g^*c_i(E)$.

Proof:

The universal classes are the polynomial generators $w_i \in H^i(BO(n); \mathbb{Z}/2\mathbb{Z})$ and $c_i \in H^{2i}(BU(n); \mathbb{Z})$ of [13, The Cohomology of $BO(n)$] and [13, The Cohomology of $BU(n)$]; they are classes on the classifying spaces and involve no hypothesis on B . What the definitions additionally require is a classifying map, and theorem 1.2.13 supplies one over any paracompact B , determined by E up to homotopy. Homotopic maps induce the same map on cohomology, so $f_E^* w_i$ and $f_E^* c_i$ depend only on the isomorphism class of E , and the two displayed formulas are well defined. Above the rank there is no universal class to pull back, and the value 0 is taken by convention exactly as upstream. When B is a CW complex the two definitions agree: a CW complex is paracompact, so theorem 1.2.13 applies to it as well, and its uniqueness clause makes any classifying map used upstream homotopic to f_E , whence the two pullbacks of w_i agree.

For naturality, g^*E is classified by $f_E \circ g$: pulling the universal bundle back along $f_E \circ g$ gives $g^*(f_E^* E_n) \cong g^*E$, and theorem 1.2.13 makes the classifying map of g^*E unique up to homotopy, so $f_{g^*E} \simeq f_E \circ g$. Hence

$$w_i(g^*E) = f_{g^*E}^* w_i = (f_E \circ g)^* w_i = g^*(f_E^* w_i) = g^*w_i(E)$$

and identically for c_i , completing the proof.

The proposition is what lets this chapter work over paracompact bases rather than CW complexes,

and the distinction is not idle: the flag bundle $F(E)$ of proposition 3.2.3, which need not be a CW complex and over which the splitting argument computes, while the splitting argument computes, carries no CW structure, and *Algebraic Topology* [13] defines the classes only over CW complexes. Naturality is stated here as well, for the same reason. The two results below supply the construction:

Theorem 3.2.2: Leray-Hirsch Theorem

Let R be a commutative ring and let $p: E \rightarrow B$ be a fiber bundle with fiber F over a paracompact base. In [13] it was shown $H^*(E; R)$ is a $H^*(B; R)$ -module where

$$\alpha\beta = p^*(\alpha) \smile \beta$$

with $\alpha \in H^*(B)$ and $\beta \in H^*(E)$. If

1. for *every* fiber F the inclusion $F \hookrightarrow E$ induces a surjection on $H^*(-; R)$
2. $H^n(F; R)$ is a free R -module of finite rank for each n , of finite total rank.

Then $H^*(E; R)$ is a free $H^*(B; R)$ -module, with basis any set of elements of $H^*(E; R)$ restricting to a basis of $H^*(F; R)$ in *every* fiber F .

Proof:

Over a *path-connected* base this is [13, The Leray-Hirsch Theorem], proved there via the collapse of the Serre spectral sequence; a fiber bundle over a paracompact base is a Hurewicz fibration by [13, Fiber Bundles over Paracompact are Hurewicz Fibrations], which supplies the fibration hypothesis, and clause (2) supplies the finitely many classes that statement asks for. Over a base that need not be path-connected, the version stated above, with the surjectivity of clause (1) imposed at *every* fiber, is the one recorded in [10, §3.1]; the series does not prove it, and it is cited rather than reproved. The quantifier is what does the work: over a path-connected base a surjection at one fiber gives one at all, and over a disconnected base it does not.

Both quantifiers are needed over a disconnected base, and neither can be weakened to a single fiber. The one in the conclusion fails the more cheaply: over $B = \text{pt} \sqcup \text{pt}$ take the product bundle $E = S^1 \sqcup S^1$ with $R = \mathbb{Z}$, where both clauses hold, and let $c_0 = (1, 1)$ and $c_1 = (u, 0)$. These restrict at the first fiber to a basis of $H^*(S^1; \mathbb{Z})$, but $H^*(B; \mathbb{Z})$ is concentrated in degree 0, so their span misses $(0, u)$ and they are no basis. The quantifier in clause (1) is what the following example is about, and it cannot be weakened either. Take $R = \mathbb{Z}$, fiber $F = S^1$, base $B = \text{pt} \sqcup S^1$, and $E = S^1 \sqcup K$ with K the Klein bottle as the circle bundle over S^1 given by the mapping torus of a reflection. The inclusion of the fiber over the point component is the inclusion of a summand, hence surjective on cohomology, and clause (2) holds. Yet $H^2(K; \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$, while a free graded module over $H^*(B; \mathbb{Z})$ is torsion-free as an abelian group, so $H^*(E; \mathbb{Z})$ is not free. What fails is clause (1) at the other component: $H^1(K; \mathbb{Z}) = \mathbb{Z}$ restricts to zero on a fiber circle, since that circle is 2-torsion in $H_1(K) = \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$. Imposing surjectivity at every fiber excludes exactly this.

Proposition 3.2.3: Splitting Principle

Let $\pi: E \rightarrow B$ be a vector bundle over a paracompact base. There exist a space $F(E)$ and a map $p: F(E) \rightarrow B$ such that p^*E splits as a direct sum of line bundles and p^* is injective on cohomology, in each of the two settings the book uses:

1. for a real bundle E , $p^*: H^*(B; \mathbb{Z}/2\mathbb{Z}) \hookrightarrow H^*(F(E); \mathbb{Z}/2\mathbb{Z})$;
2. for a complex bundle E , $p^*: H^*(B; \mathbb{Z}) \hookrightarrow H^*(F(E); \mathbb{Z})$.

In (2) the summands are complex line bundles, and their first Chern classes are the *Chern roots* of E . Moreover $F(E)$ is again paracompact, so the classes of proposition 3.2.1 are defined on it and the construction may be iterated.

Proof:

This is the splitting principle, proved in full as [13, Characteristic Classes, Splitting Principle], whose statement carries both the real and the complex clause over a paracompact base, with no connectivity hypothesis: the flag bundle $F(E) \rightarrow B$ is built as a tower of projectivizations, and the Leray-Hirsch theorem makes each stage injective on cohomology, so p^* is injective and p^*E splits into a sum of line bundles. The property of $F(E)$ itself is new here. Each stage of the tower is a fiber bundle whose fiber is a projective space, hence compact Hausdorff, so [8, Locally Trivial Maps with Compact Fiber Preserve Paracompactness] carries paracompactness and the Hausdorff property from the base to the total space at every stage. Both clauses of that theorem are used, since *Topology* takes paracompactness to be the covering condition alone while box 1.0.1 takes it to include Hausdorff. Inducting up the finitely many stages, $F(E)$ is paracompact in the sense of box 1.0.1, completing the proof.

Theorem 3.2.4: Existence of Stiefel-Whitney Classes

Let $E \rightarrow B$ be a real vector bundle over a paracompact base. Then there exists a sequence of functions $w_1, w_2, \dots$ where $w_i(E) \in H^i(B; \mathbb{Z}/2\mathbb{Z})$ that depends only on the isomorphism type of E such that

1. $w_i(f^*(E)) = f^*(w_i(E))$ for a pullback $f^*(E)$
2. $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$ where $w = 1 + w_1 + w_2 + \dots \in H^*(B; \mathbb{Z}/2\mathbb{Z})$
3. $w_i(E) = 0$ if $i > \dim E$
4. Given $E \rightarrow \mathbb{R}P^\infty$ is the canonical line bundle, $w_1(E)$ is a generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}/2\mathbb{Z})$.

The Stiefel-Whitney class w is called the *total Stiefel-Whitney class*.

Note that by (3), w only has finitely many nonzero terms, showing it is well-defined in $H^*(B; \mathbb{Z}/2\mathbb{Z})$. The $\oplus$ to $\smile$ is shorthand for the Whitney sum formula:

$$w_n(E_1 \oplus E_2) = \sum_{i+j=n} w_i(E_1) \smile w_j(E_2)$$

where $w_0 = 1$. Property (4) is a non-triviality condition: without it, setting $w_i(E) = 0$ for all $i > 0$ would satisfy (1)-(3).

The proof for Chern classes runs the same way, so both are stated together.

Theorem 3.2.5: Existence of Chern Classes

Let $E \rightarrow B$ be a complex vector bundle over a paracompact base. Then there exists a sequence of functions $c_1, c_2, \dots$ where $c_i(E) \in H^{2i}(B; \mathbb{Z})$ that depends only on the isomorphism type of E such that

1. $c_i(f^*(E)) = f^*(c_i(E))$ for a pullback $f^*(E)$
2. $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$ where $c = 1 + c_1 + c_2 + \dots \in H^*(B; \mathbb{Z})$
3. $c_i(E) = 0$ if i exceeds the complex rank of E
4. Given $E \rightarrow \mathbb{CP}^\infty$ is the canonical line bundle, $c_1(E)$ is a generator of $H^2(\mathbb{CP}^\infty; \mathbb{Z})$ specified in advance.

The extra condition for (4) is a non triviality *and* a normalization condition, as each function c_i can be replaced by $k^i c_i$ for a fixed integer k which gives new functions satisfying (1)-(3).

Proof:

Both families are constructed in [13] as pullbacks of universal classes, and the four properties are consequences of that construction rather than a characterization of it. By [13, Stiefel-Whitney Classes and the Total Class] and [13, Chern Classes and the Total Chern Class], $w_i(E) = f_E^* w_i$ and $c_i(E) = f_E^* c_i$ for the polynomial generators of $H^*(BO(n); \mathbb{Z}/2\mathbb{Z})$ and $H^*(BU(n); \mathbb{Z})$ supplied by [13, The Cohomology of $BO(n)$] and [13, The Cohomology of $BU(n)$], and proposition 3.2.1 extends the definition to the paracompact bases used here.

Property (1) is the naturality clause of proposition 3.2.1. Property (2) is [13, Whitney Sum Formula], which is stated there over a paracompact base and covers the real and complex cases together. Property (3) holds because $H^*(BO(n); \mathbb{Z}/2\mathbb{Z})$ and $H^*(BU(n); \mathbb{Z})$ are polynomial on generators of degrees $1, \dots, n$ and $2, \dots, 2n$ respectively, so there is no universal class w_i or c_i to pull back once i exceeds n ; the upstream definitions set the value to 0 there. Property (4) is the normalization computed in [13, Stiefel-Whitney Class of the Tautological Bundle over $\mathbb{RP}^\infty$], where the tautological line bundle over $BO(1) = \mathbb{RP}^\infty$ has w_1 the generator of $H^1(\mathbb{RP}^\infty; \mathbb{Z}/2\mathbb{Z})$, which is the unique nonzero element and so needs no further choice. For Chern classes the generator has to be named, since $H^2(\mathbb{CP}^\infty; \mathbb{Z}) \cong \mathbb{Z}$ has two: the normalization taken here is the polynomial generator c_1 of [13, The Cohomology of $BU(n)$] at $n = 1$, fixed by [13, The First Chern Class of a Line Bundle] so that the *tautological* line bundle over $\mathbb{CP}^1$ has $c_1 = -x$, where $x = c_1(H^\vee)$ is the positive generator in this book's convention. That is the choice "specified in advance" in property (4), completing the proof.

Note that for property (4), the canonical line bundle over $\mathbb{RP}^1$ could have been used instead of $\mathbb{RP}^\infty$ as $\mathbb{RP}^1 \hookrightarrow \mathbb{RP}^\infty$ induces an isomorphism

$$H^1(\mathbb{RP}^\infty; \mathbb{Z}/2\mathbb{Z}) \cong H^1(\mathbb{RP}^1; \mathbb{Z}/2\mathbb{Z})$$

and similarly for Chern classes.

Example 3.2.6: Stiefel-Whitney and Chern Classes

1. By property (1), the trivial bundle $B \times \mathbb{R}^n$ has

$$w_i(E) = 0 \quad \forall i > 0$$

as the product is the pullback of the bundle over a point.

2. By property (2), the direct sum of a bundle with a product bundle does not change its Stiefel-Whitney classes; this property is called *stable*. The tangent bundle TS^n is stably trivial, and hence

$$w_i(TS^n) = 0 \quad \forall i > 0$$

More generally, as :

$$(1 + w_1 + \cdots)(1 + w'_1 + \cdots) = 1 + (w_1 + w'_1) + (w_2 + w_1w'_1 + w'_2) + \cdots$$

given $w(E_1)$ and $w(E_1 \oplus E_2)$, the class $w(E_2)$ is determined by

$$\begin{aligned} w_1 + w'_1 &= a_1 \\ w_2 + w_1w'_1 + w'_2 &= a_2 \\ &\vdots \\ \sum_i w_{n-i}w'_i &= a_n \end{aligned}$$

If $E_1 \oplus E_2$ is the trivial bundle, $a_i = 0$ for all $i > 0$, so $w(E_1)$ determines $w(E_2)$ uniquely by the above formulas. The same applies to Chern classes.

3. There is no bundle $E \rightarrow \mathbb{R}P^\infty$ whose sum with the line bundle $E_1(\mathbb{R}^\infty)$ is trivial. For

$$w(E_1(\mathbb{R}^\infty)) = 1 + \omega$$

where ω is a generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}/2\mathbb{Z})$. Over $\mathbb{Z}/2\mathbb{Z}$ coefficients this gives

$$w(E) = (1 + \omega)^{-1} = 1 + \omega + \omega^2 + \cdots$$

So $w_i(E) = \omega^i$ is nonzero for all i , contradicting $w_i(E) = 0$ for all $i > \dim E$. This is why compactness is necessary in proposition 1.1.16.

4. [10, p.82] constructs an n -dimensional vector bundle $E \rightarrow B$ with $w_i(E)$ nonzero for each $i \leq \dim E$.

The cohomology rings of the Grassmannians, $H^*(\text{Grass}_n(\mathbb{C}); \mathbb{Z}) = \mathbb{Z}[c_1, \dots, c_n]$ and $H^*(\text{Grass}_n(\mathbb{R}); \mathbb{Z}/2\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}[w_1, \dots, w_n]$, are computed in [13] (there as $H^*(BU(n))$ and $H^*(BO(n); \mathbb{Z}/2\mathbb{Z})$), realizing the Chern and Stiefel-Whitney classes as the polynomial generators.

The Stiefel-Whitney and Chern classes of the previous section live in $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}$ cohomology and are built uniformly, as pullbacks of the polynomial generators of $H^*(BO(n); \mathbb{Z}/2\mathbb{Z})$ and $H^*(BU(n); \mathbb{Z})$. The two remaining classes from definition 3.1.4 and definition 3.1.3 are different in character. The Euler class $e(E)$ is an integral class attached to an *oriented* real bundle of rank exactly equal to its

top degree; it is the integral refinement of the top Stiefel-Whitney class and the precise obstruction to a nowhere-zero section. The Pontryagin classes $p_i(E)$ are integral classes of an arbitrary real bundle obtained by complexifying E and reading off its Chern classes. This section develops both from the existence theorems already proved (theorem 3.2.5), relates the Euler class to the Thom class and to the top Chern class, establishes the Whitney-sum and naturality behaviour of each, and ends by attaching to a closed smooth manifold its Pontryagin and Stiefel-Whitney *numbers*, the characteristic numbers that govern cobordism.

Throughout, $E \rightarrow B$ is a vector bundle over a paracompact base, and cohomology is singular cohomology with the indicated coefficients. The orientation of a real bundle of rank n is a choice, for each $b \in B$, of generator of $H^n(E_b, (E_b)_0; \mathbb{Z}) \cong \mathbb{Z}$, where $(E_b)_0$ denotes the fiber with the origin removed, the choices fitting together continuously. This is the data carried by the oriented bundles that make up $\widehat{\text{Vect}}^n(B)$ in definition 3.1.4.

The Euler class has a second description, through the Thom isomorphism, and it is that description rather than the Gysin construction of definition 3.1.4 that yields the formal properties below. Let $\pi: E \rightarrow B$ be an oriented real vector bundle of rank n and let $E_0 \subseteq E$ be the complement of the zero section. A *Thom class* for E is a class

$$u(E) \in H^n(E, E_0; \mathbb{Z})$$

restricting on each fiber $(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ to the generator of $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; \mathbb{Z}) \cong \mathbb{Z}$ determined by the orientation of that fiber ([13, Thom Class]). Such a class exists, is unique, and the map $\alpha \mapsto \pi^*(\alpha) \smile u(E)$ is an isomorphism $H^k(B; \mathbb{Z}) \xrightarrow{\sim} H^{k+n}(E, E_0; \mathbb{Z})$ for every k , provided B admits a finite trivializing open cover (for instance B compact, or a finite CW complex) or is a CW complex of finite type. This is the Thom isomorphism of [13, The Thom Isomorphism], proved there by a cellular induction under exactly those hypotheses. The finite-type hypothesis belongs to that route rather than to the theorem: [12, Theorem 10.4] obtains the same isomorphism with no hypothesis on the base whatsoever. The stronger hypothesis is nonetheless what proposition 3.2.7 and proposition 3.2.12 state, because the identification of the two definitions of the Euler class is available here only along the finite-type route; The zero section $z: B \rightarrow E$ of definition 1.1.9 is a homotopy equivalence, so $H^*(E; \mathbb{Z}) \cong H^*(B; \mathbb{Z})$, and one may pull the Thom class back along the inclusion $(E, \emptyset) \hookrightarrow (E, E_0)$ to obtain a class in $H^n(E; \mathbb{Z}) \cong H^n(B; \mathbb{Z})$.

Proposition 3.2.7: Euler Class as Restriction of the Thom Class

Let $\pi: E \rightarrow B$ be an oriented real vector bundle of rank n over a CW complex B of finite type, with Thom class $u(E) \in H^n(E, E_0; \mathbb{Z})$. Write $j^*: H^n(E, E_0; \mathbb{Z}) \rightarrow H^n(E; \mathbb{Z})$ for the map induced by the inclusion of pairs $(E, \emptyset) \hookrightarrow (E, E_0)$, and let $z: B \rightarrow E$ be the zero section. Then the Euler class of definition 3.1.4 is

$$e(E) = z^*(j^*(u(E))) \in H^n(B; \mathbb{Z})$$

Equivalently, under the identification $\pi^*: H^n(B; \mathbb{Z}) \xrightarrow{\sim} H^n(E; \mathbb{Z})$, the class $j^*(u(E))$ corresponds to $e(E)$.

The two formulations agree because z and π are mutually inverse homotopy equivalences, so $z^* = (\pi^*)^{-1}$ on cohomology. The content of the proposition is that the Euler class is what remains of the Thom class once the relative information (the behaviour near the zero section) is forgotten and only the absolute class on B is retained.

Proof:

This is [13, The Euler Class is the Restriction of the Thom Class], stated there for an oriented real bundle of rank n over a CW complex of finite type, with the Thom class supplied by [13, The Thom Isomorphism]; those are the hypotheses imposed above. Note that the identification is not a formality: definition 3.1.4 defines $e(E)$ as the pullback of the Gysin transgression class on $BSO(n)$, whereas [12, §9] defines the Euler class to be the restriction of the Thom class, so this proposition is what makes the two agree. The identification is made by transporting the long exact sequence of the pair (E, E_0) through the Thom isomorphism and matching the result against the Gysin sequence that [13, Euler Class of an Oriented Bundle] uses to define $e(E)$, which is the definition recorded in definition 3.1.4, completing the proof.

This description makes two features immediate. First, the degree of $e(E)$ equals the rank of E , in contrast to the Chern and Pontryagin classes which live in twice or four times the index. Second, the Euler class is genuinely an orientation-dependent invariant: reversing the orientation of E negates the Thom class, hence negates $e(E)$. The reduction of $e(E)$ modulo 2 loses the orientation data and recovers the top Stiefel-Whitney class.

Proposition 3.2.8: Mod-2 Reduction of the Euler Class

Let $E \rightarrow B$ be an oriented real vector bundle of rank n . Then the image of $e(E)$ under the coefficient reduction $\rho : H^n(B; \mathbb{Z}) \rightarrow H^n(B; \mathbb{Z}/2\mathbb{Z})$ is the top Stiefel-Whitney class:

$$\rho(e(E)) = w_n(E)$$

Proof:

Proved in [13, Top Class is the Euler Class]: the mod-2 reduction of the integral Euler class is the top Stiefel-Whitney class.

The Euler class comes into its own for complex bundles, where it recovers a Chern class. A complex vector bundle of complex rank n has a canonical orientation as a real bundle of rank $2n$: a complex basis $v_1, iv_1, \dots, v_n, iv_n$ of a fiber is an oriented real basis, and any two complex bases are related by an element of $\mathrm{GL}_n(\mathbb{C})$, which is connected, so the induced real orientation is independent of the choice. Thus every complex bundle has a well-defined Euler class of its underlying oriented real bundle, and one expects it to coincide with the top Chern class, which is the only Chern class living in the top real degree $2n$. The next theorem confirms this.

Theorem 3.2.9: Euler Class Equals Top Chern Class

Let $E \rightarrow B$ be a complex vector bundle of complex rank n , and let $E_{\mathbb{R}}$ denote its underlying real vector bundle of rank $2n$, with the canonical orientation described above. Then

$$e(E_{\mathbb{R}}) = c_n(E)$$

in $H^{2n}(B; \mathbb{Z})$, where $c_n(E)$ is the top Chern class of definition 3.1.5.

Proof:

Proved in [13, Top Class is the Euler Class] by the splitting principle: for a complex bundle the top Chern class equals the Euler class of the underlying canonically-oriented real bundle.

Proposition 3.2.10: Relating Chern and Stiefel-Whitney Classes

Let $E \rightarrow B$ be a complex n -dimensional vector bundle over a paracompact base, and regard it as a $2n$ -dimensional real vector bundle $E_{\mathbb{R}}$. Then

$$w_{2i+1}(E_{\mathbb{R}}) = 0$$

and $w_{2i}(E_{\mathbb{R}})$ is the image of $c_i(E)$ under the coefficient homomorphism $\rho_2: H^{2i}(B; \mathbb{Z}) \rightarrow H^{2i}(B; \mathbb{Z}/2\mathbb{Z})$. Equivalently, $w(E_{\mathbb{R}}) = \rho_2(c(E))$.

Proof:

Step 1: Reduction to line bundles. Let $p: F(E) \rightarrow B$ be the complex flag bundle of proposition 3.2.3(2), so that p^*E splits as a sum of complex line bundles $L_1 \oplus \cdots \oplus L_n$. That proposition gives injectivity of p^* on $H^*(-; \mathbb{Z})$; injectivity on $H^*(-; \mathbb{Z}/2\mathbb{Z})$ holds for the same reason, since $F(E) \rightarrow B$ is built as a tower of projectivizations and theorem 3.2.2 applies over any commutative coefficient ring R , in particular $R = \mathbb{Z}/2\mathbb{Z}$, making $H^*(F(E); \mathbb{Z}/2\mathbb{Z})$ a free $H^*(B; \mathbb{Z}/2\mathbb{Z})$ -module on a basis containing 1. Both $E \mapsto w(E_{\mathbb{R}})$ and $E \mapsto \rho_2(c(E))$ are natural (theorem 3.2.4(1), theorem 3.2.5(1)) and multiplicative under direct sums (theorem 3.2.4(2), theorem 3.2.5(2), together with the fact that ρ_2 is a ring homomorphism and that $(E \oplus E')_{\mathbb{R}} = E_{\mathbb{R}} \oplus E'_{\mathbb{R}}$). So both sides pull back to $F(E)$ as products over the line summands, and it suffices to prove the identity for a complex line bundle.

Step 2: The line bundle case. Let $L \rightarrow B$ be a complex line bundle, so $L_{\mathbb{R}}$ is a real bundle of rank 2 carrying the canonical orientation of theorem 3.2.9. Its first Stiefel-Whitney class vanishes. To see this, pass to the universal case: by theorem 1.2.13 in its complex form with $n = 1$, there is a map $f: B \rightarrow \text{Grass}_1(\mathbb{C}^\infty) = \mathbb{CP}^\infty$ and an isomorphism $L \cong f^*\gamma$ with $\gamma \rightarrow \mathbb{CP}^\infty$ the universal complex line bundle. Realification commutes with pullback, so $L_{\mathbb{R}} \cong f^*(\gamma_{\mathbb{R}})$, and naturality (theorem 3.2.4(1)) gives $w_1(L_{\mathbb{R}}) = f^*w_1(\gamma_{\mathbb{R}})$. Now $\mathbb{CP}^\infty$ is a path-connected CW complex, with one cell in each even dimension and none in odd dimensions ([13, $\mathbb{CP}^\infty$ as $K(\mathbb{Z}, 2)$]), and $\gamma_{\mathbb{R}}$ is oriented by the canonical orientation of a complex line, hence orientable. So [13, First Stiefel-Whitney Class is the Obstruction to Orientability], whose hypothesis of a connected CW base is met by $\mathbb{CP}^\infty$, gives $w_1(\gamma_{\mathbb{R}}) = 0$, and therefore $w_1(L_{\mathbb{R}}) = f^*(0) = 0$. (Applying that criterion to $L_{\mathbb{R}}$ directly is not available here: B is only assumed paracompact, and in the application of Step 1 the base is the flag bundle $F(E)$, which need not be a connected CW complex. What the argument does ask of that base is paracompactness, so that theorem 1.2.13 supplies f and proposition 3.2.1 defines w_1 there, and that is supplied by the last clause of proposition 3.2.3.) Its top Stiefel-Whitney class is $w_2(L_{\mathbb{R}}) = \rho_2(e(L_{\mathbb{R}}))$ by proposition 3.2.8, and $e(L_{\mathbb{R}}) = c_1(L)$ by theorem 3.2.9 applied with $n = 1$. Hence $w_2(L_{\mathbb{R}}) = \rho_2(c_1(L))$, and $w_i(L_{\mathbb{R}}) = 0$ for $i > 2$ by theorem 3.2.4(3) since $\text{rk} L_{\mathbb{R}} = 2$. Therefore

$$w(L_{\mathbb{R}}) = 1 + \rho_2(c_1(L)) = \rho_2(1 + c_1(L)) = \rho_2(c(L))$$

Step 3: Conclusion. By Step 1 the two total classes agree after pulling back along p , and p^* is injective on $H^*(B; \mathbb{Z}/2\mathbb{Z})$, so $w(E_{\mathbb{R}}) = \rho_2(c(E))$ on B . Comparing degrees, the odd-degree

components of $\rho_2(c(E))$ vanish because $c_i(E)$ sits in degree $2i$, giving $w_{2i+1}(E_{\mathbb{R}}) = 0$ and $w_{2i}(E_{\mathbb{R}}) = \rho_2(c_i(E))$, as we sought to show.

The theorem identifies the top Chern class with a section-theoretic invariant: $c_n(E)$ is exactly the obstruction to a nowhere-zero section of the complex bundle E . This is the algebraic-topology incarnation of the fact that a generic section of a rank- n complex bundle over a $2n$ -manifold vanishes at a finite set of points whose signed count is $\langle c_n(E), [B] \rangle$. The identification also gives a uniform formula for the Euler class of any complex bundle, which is the route by which Euler classes are computed in practice.

Example 3.2.11: Euler Class of the Tangent Bundle of S^2

The tangent bundle TS^2 is an oriented real rank-2 bundle. Its Euler class lies in $H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}$, so it is determined by the Euler number $\langle e(TS^2), [S^2] \rangle$. Two routes compute it.

First, $S^2 = \mathbb{CP}^1$ and $T\mathbb{CP}^1$ is a complex line bundle. As recorded in section 1.2, TS^2 is the oriented plane bundle $E_2 \rightarrow S^2$ with clutching function $z \mapsto z^2$ of degree 2. The Euler number of $E_m \rightarrow S^2$ is m relative to the standard orientation, so

$$\langle e(TS^2), [S^2] \rangle = 2$$

By theorem 3.2.9 this also equals $\langle c_1(T\mathbb{CP}^1), [\mathbb{CP}^1] \rangle$.

Second, the Euler number of the tangent bundle of any closed oriented surface equals its Euler characteristic ([13, Euler Class of the Tangent Bundle Computes the Euler Characteristic]; the differential-geometric form is the Poincaré-Hopf theorem of [15, Poincaré-Hopf]). Since $\chi(S^2) = 2$, this agrees: $\langle e(TS^2), [S^2] \rangle = 2$. In particular $e(TS^2) \neq 0$, which is the cohomological statement of the Hairy Ball theorem: TS^2 has no nowhere-zero section.

The Euler class shares the formal structure of the other characteristic classes: naturality under pullback and multiplicativity under Whitney sum. The following collects those two, together with the torsion of the odd-rank case and the section-theoretic vanishing criterion that motivates the obstruction-theoretic viewpoint later in this section.

Proposition 3.2.12: Properties of the Euler Class

Let $E, E_1, E_2 \rightarrow B$ be oriented real vector bundles over a CW complex B of finite type, with E of rank n . Then:

1. (Naturality) For $f: B' \rightarrow B$ continuous with B' paracompact, $e(f^*E) = f^*(e(E))$, with f^*E given the pulled-back orientation.
2. (Whitney sum) $e(E_1 \oplus E_2) = e(E_1) \smile e(E_2)$, with $E_1 \oplus E_2$ given the direct-sum orientation.
3. (Odd rank torsion) If n is odd then $2e(E) = 0$.
4. (Vanishing) If B is connected and E admits a nowhere-zero section, then $e(E) = 0$.

Proof:

1. By [13, Euler Class of an Oriented Bundle], recorded in definition 3.1.4, the Euler class is $e(E) = f_E^* e$ for a classifying map $f_E: B \rightarrow BSO(n)$ of the oriented bundle E and the universal class $e \in H^n(BSO(n); \mathbb{Z})$. The pullback f^*E with its pulled-back orientation is classified by $f_E \circ f$, so

$$e(f^*E) = (f_E \circ f)^* e = f^*(f_E^* e) = f^*(e(E))$$

Paracompactness of B' is what supplies the classifying map for f^*E , unique up to homotopy, so that $e(f^*E)$ is defined at all; no finite-type hypothesis on B' enters. The same argument gives the consequence used in section 3.1, that a trivial bundle of rank $n > 0$ has $e = 0$, since it is pulled back from a bundle over a point, where H^n vanishes.

2. This is [13, Whitney Sum Formula for the Euler Class], whose hypotheses are the ones stated here: two oriented real bundles over a CW complex of finite type, with $E_1 \oplus E_2$ carrying the direct-sum orientation. There the Thom class of the sum is identified with $q_1^*(u_{E_1}) \smile q_2^*(u_{E_2})$ for the two projections q_i , and proposition 3.2.7 converts that identity into the stated one.
3. This is [13, The Euler Class of an Odd-Rank Bundle is 2-Torsion], again for an oriented bundle over a CW complex of finite type. The mechanism is that fiberwise multiplication by -1 acts on each fiber pair by the degree $(-1)^n$ of the antipodal map, so for odd n it reverses every fiber orientation and negates the Thom class; since it covers the identity of B and preserves the zero section, restricting along the zero section turns that sign into $e(E) = -e(E)$.
4. This is the first clause of [13, Top Class is the Obstruction to a Nowhere-Zero Section], stated for a real bundle of rank n over a connected CW complex: an oriented such bundle with a nowhere-zero section has $e(E) = 0$. That is where the connectedness hypothesis comes from, and B is a CW complex by assumption, completing the proof.

Property (4) is the bridge to the obstruction-theoretic reading taken up later in this section: a nonzero Euler class is an obstruction to the existence of a nowhere-zero section, and it is in fact the *primary* such obstruction, the first one encountered as one extends a section over the skeleta of B . Property (3) explains why Euler classes of odd-rank bundles carry only 2-torsion information and are therefore usually studied for even-rank (in particular complex) bundles. The Whitney formula (2) is consistent with theorem 3.2.9: for complex bundles F_1, F_2 of ranks n_1, n_2 the component in $H^{2(n_1+n_2)}$ of $c(F_1 \oplus F_2) = c(F_1) \smile c(F_2)$ (theorem 3.2.5(2)) is $c_{n_1}(F_1) \smile c_{n_2}(F_2)$, since $c_i(F_j) = 0$ for $i > n_j$ by theorem 3.2.5(3), so the top Chern class is multiplicative in exactly the way the Euler class is.

The Pontryagin classes of definition 3.1.3 are defined for every real bundle, with no orientation needed, and refine the obstruction information in degrees divisible by 4. The efficient definition realizes them through Chern classes of the *complexification*. Given a real vector bundle $E \rightarrow B$, its complexification is the complex bundle

$$E \otimes \mathbb{C} := E \otimes_{\mathbb{R}} \mathbb{C}$$

of complex rank equal to the real rank of E , where $\mathbb{C}$ is the trivial complex line bundle; fiberwise this tensors each real fiber with $\mathbb{C}$. The complexification carries a canonical conjugation, and the

central computational fact is that $E \otimes \mathbb{C}$ is isomorphic to its own conjugate.

Theorem 3.2.13: Pontryagin Classes via Chern Classes

Let $E \rightarrow B$ be a real vector bundle. Then the complexification $E \otimes \mathbb{C}$ is isomorphic to its conjugate bundle $\overline{E} \otimes \mathbb{C}$, so its odd Chern classes are 2-torsion:

$$2 c_{2i+1}(E \otimes \mathbb{C}) = 0$$

The Pontryagin class of definition 3.1.3 is given by

$$p_i(E) = (-1)^i c_{2i}(E \otimes \mathbb{C}) \in H^{4i}(B; \mathbb{Z})$$

and the total Pontryagin class $p(E) = 1 + p_1(E) + p_2(E) + \cdots$ satisfies, modulo elements of order 2,

$$p(E_1 \oplus E_2) \equiv p(E_1) \smile p(E_2) \pmod{2\text{-torsion}}$$

Proof:

Proved in [13, Complexification is Self-Conjugate] for the first clause and [13, Naturality and the Whitney Sum Formula for Pontryagin Classes] for the Whitney formula, with [13, Pontryagin Classes] supplying the definition: the complexification $E \otimes \mathbb{C}$ is self-conjugate, so its odd Chern classes are 2-torsion and the Pontryagin classes are its even Chern classes with the sign $(-1)^i$, giving $p_i(E) = (-1)^i c_{2i}(E \otimes \mathbb{C})$.

The theorem reduces every question about Pontryagin classes to a question about Chern classes of the complexification, where the full machinery of the splitting principle and the Whitney formula is available. The 2-torsion caveat in the Whitney formula is genuine and not an artifact: the failure of strict multiplicativity is detected precisely by the odd Chern classes of the complexifications, which are 2-torsion but need not vanish. Over a base with no 2-torsion in its integral cohomology, or after tensoring with $\mathbb{Z}[\frac{1}{2}]$ or $\mathbb{Q}$, the Pontryagin classes are strictly multiplicative.

A first consequence relates the Pontryagin classes of a complex bundle to its own Chern classes, which is the form in which Pontryagin classes of tangent bundles of complex manifolds are computed.

Corollary 3.2.14: Pontryagin Classes of a Complex Bundle

Let $F \rightarrow B$ be a complex vector bundle of complex rank n , regarded as an oriented real bundle $F_{\mathbb{R}}$. Then $F_{\mathbb{R}} \otimes \mathbb{C} \cong F \oplus \overline{F}$, and the Pontryagin classes of $F_{\mathbb{R}}$ are the degree- $(4i)$ components of the product $c(F) \smile c(\overline{F})$ up to sign: for each i ,

$$p_i(F_{\mathbb{R}}) = (-1)^i \sum_{r+s=2i} (-1)^s c_r(F) \smile c_s(F)$$

in $H^{4i}(B; \mathbb{Z})$. Equivalently, the equality of total classes

$$1 - p_1(F_{\mathbb{R}}) + p_2(F_{\mathbb{R}}) - \cdots = (1 + c_1(F) + \cdots + c_n(F)) \smile (1 - c_1(F) + c_2(F) - \cdots)$$

holds in the part of $H^*(B; \mathbb{Z})$ of degrees divisible by 4; the degree- $(4i+2)$ components of the right-hand side are 2-torsion and are discarded.

Proof:

Step 1: The splitting $F_{\mathbb{R}} \otimes \mathbb{C} \cong F \oplus \bar{F}$. Let $J: F_{\mathbb{R}} \rightarrow F_{\mathbb{R}}$ be the real bundle endomorphism given by multiplication by i in the complex structure of F , so $J^2 = -\text{id}$. Extend J complex-linearly to $F_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$. On that bundle J still satisfies $J^2 = -\text{id}$, so its minimal polynomial divides $t^2 + 1 = (t - i)(t + i)$, which has distinct roots in $\mathbb{C}$; hence $F_{\mathbb{R}} \otimes \mathbb{C}$ splits as the direct sum of the $(+i)$ - and $(-i)$ -eigenbundles of J ,

$$F_{\mathbb{R}} \otimes \mathbb{C} = F^{1,0} \oplus F^{0,1}$$

and the eigenbundle projections $\frac{1}{2}(\text{id} \mp iJ)$ are continuous, so both summands are subbundles. The map $v \mapsto \frac{1}{2}(v - iJv)$ is a complex-bundle isomorphism $F \xrightarrow{\sim} F^{1,0}$, and composing with conjugation identifies $F^{0,1} \cong \bar{F}$. Hence $F_{\mathbb{R}} \otimes \mathbb{C} \cong F \oplus \bar{F}$.

Step 2: Reading off the classes. By theorem 3.2.5(2), $c(F_{\mathbb{R}} \otimes \mathbb{C}) = c(F) \smile c(\bar{F})$, and $c_k(\bar{F}) = (-1)^k c_k(F)$ because conjugation acts on the Chern roots by negation, so $c(\bar{F}) = 1 - c_1(F) + c_2(F) - \cdots$. Substituting into theorem 3.2.13, $p_i(F_{\mathbb{R}}) = (-1)^i c_{2i}(F_{\mathbb{R}} \otimes \mathbb{C})$, and expanding the product in degree $4i$ gives

$$p_i(F_{\mathbb{R}}) = (-1)^i \sum_{r+s=2i} (-1)^s c_r(F) \smile c_s(F)$$

which is the displayed formula; the equality of total classes is the same identity read degree by degree, with the degree- $(4i+2)$ components 2-torsion by theorem 3.2.13 and therefore discarded, completing the proof.

Example 3.2.15: Pontryagin Class of the Tangent Bundle of $\mathbb{CP}^2$

Let $a \in H^2(\mathbb{CP}^2; \mathbb{Z})$ be the generator with $\langle a^2, [\mathbb{CP}^2] \rangle = 1$, normalized so that $a = c_1$ of the hyperplane bundle. The cohomology ring is $H^*(\mathbb{CP}^2; \mathbb{Z}) = \mathbb{Z}[a]/(a^3)$. The total Chern class of the tangent bundle is

$$c(T\mathbb{CP}^2) = (1 + a)^3 = 1 + 3a + 3a^2$$

using the Euler sequence $T\mathbb{CP}^2 \oplus \underline{\mathbb{C}} \cong (H^\vee)^{\oplus 3}$ and theorem 3.2.5(2), where $H^\vee$ is the dual of the tautological bundle with $c_1(H^\vee) = a$. Thus $c_1 = 3a$ and $c_2 = 3a^2$.

To compute p_1 , complexify the real tangent bundle. By corollary 3.2.14, $(T\mathbb{CP}^2)_{\mathbb{R}} \otimes \mathbb{C} \cong T\mathbb{CP}^2 \oplus \overline{T\mathbb{CP}^2}$, with

$$c(\overline{T\mathbb{CP}^2}) = 1 - 3a + 3a^2$$

since $c_k(\bar{F}) = (-1)^k c_k(F)$. Multiplying,

$$c((T\mathbb{CP}^2)_{\mathbb{R}} \otimes \mathbb{C}) = (1 + 3a + 3a^2)(1 - 3a + 3a^2) = 1 + (3 + 3 - 9)a^2 = 1 - 3a^2$$

where the degree-6 and higher terms vanish since $a^3 = 0$. Hence $c_2((T\mathbb{CP}^2)_{\mathbb{R}} \otimes \mathbb{C}) = -3a^2$, and by theorem 3.2.13,

$$p_1(T\mathbb{CP}^2) = (-1)^1 c_2((T\mathbb{CP}^2)_{\mathbb{R}} \otimes \mathbb{C}) = 3a^2$$

So the total Pontryagin class is $p(T\mathbb{CP}^2) = 1 + 3a^2$.

Everything so far has constructed the four classes and computed with them. What the construction leaves out is what they *measure*, and the answer is uniform: each class is the obstruction to a

structure the bundle would have if it were trivial enough. The Euler class obstructs a nowhere-zero section, w_1 obstructs an orientation, w_2 obstructs a spin structure, and the top Chern class obstructs a nowhere-zero section of a complex bundle. The method behind all four is the same. Each structure is a section of a fibration built from E : an orientation is a section of the orientation double cover, a spin structure is a lift of the classifying map along $B\text{Spin}(n) \rightarrow BSO(n)$, and a nowhere-zero section is a section of the sphere bundle. One then runs the obstruction theory of [13, Remark: Obstructions to lifting and to sections] on that fibration, and the primary obstruction, living in $H^{m+1}(B; \pi_m(F))$ for the fiber F , turns out to be the characteristic class. Each of the four identifications is proved in full upstream and is cited here at its statement; what this book adds is the torsor counts, which the upstream statements omit, and the worked examples.

The geometric problem “does E admit a nowhere-zero section” is, after normalizing lengths, the problem of finding a section of the unit sphere bundle. This reduction is recorded once for later use.

Definition 3.2.16: Sphere Bundle

Let $p: E \rightarrow B$ be a real vector bundle of rank n equipped with an inner product (which exists by proposition 1.1.14). The *sphere bundle* $S(E) \rightarrow B$ is the subspace of E consisting of unit vectors,

$$S(E) = \{v \in E \mid \langle v, v \rangle = 1\}$$

with the restricted projection. It is a fiber bundle with fiber the unit sphere $S^{n-1} \subset \mathbb{R}^n$.

A nowhere-zero section $s: B \rightarrow E$ of the vector bundle is the same datum as a section of $S(E)$, since dividing s by its (nowhere-zero, continuous) length $|s|$ lands in $S(E)$, and conversely a section of $S(E)$ is in particular a nowhere-zero section of E (see definition 1.1.8 for the notion of section). The fiber S^{n-1} is $(n-2)$ -connected with $\pi_{n-1}(S^{n-1}) \cong \mathbb{Z}$, so the obstruction machinery places the first obstruction to a section of $S(E)$ in $H^n(B; \pi_{n-1}(S^{n-1}))$. The remaining input is that this group is $H^n(B; \mathbb{Z})$ once an orientation has been fixed, since identifying $\pi_{n-1}(S^{n-1})$ with $\mathbb{Z}$ canonically is exactly a choice of orientation of the fiber. This is the geometric content of the orientability hypothesis in the definition of the Euler class.

The Euler class $e(E) \in H^n(B; \mathbb{Z})$ of an oriented rank- n bundle was introduced in definition 3.1.4 as a characteristic class, defined for the universal oriented bundle and pulled back. The obstruction-theoretic description identifies it with the primary obstruction to a section of the sphere bundle, and this is the statement that makes the name “Euler class” meaningful: when E is the tangent bundle, the obstruction to a nowhere-zero vector field is the Euler characteristic.

Theorem 3.2.17: Euler Class as Primary Obstruction

Let $p: E \rightarrow B$ be an oriented real vector bundle of rank $n \geq 2$ over a connected CW complex B . The primary obstruction to the existence of a nowhere-zero section of E is the Euler class:

$$[\sigma(s)] = e(E) \in H^n(B; \pi_{n-1}(S^{n-1})) \cong H^n(B; \mathbb{Z})$$

where the identification $\pi_{n-1}(S^{n-1}) \cong \mathbb{Z}$ is the one fixed by the orientation of E . In particular, if E admits a nowhere-zero section then $e(E) = 0$; and if in addition $\dim B \leq n$, the converse holds, so that E admits a nowhere-zero section if and only if $e(E) = 0$.

Without the orientation hypothesis the same argument gives the mod-2 statement: a nowhere-zero section forces $w_n(E) = 0$ in $H^n(B; \mathbb{Z}/2\mathbb{Z})$.

The theorem says two things at once: that the obstruction cocycle of the sphere bundle, an a priori sections-theoretic invariant, agrees with the Euler class, an a priori cohomological invariant; and that the orientation is exactly the bookkeeping needed to read the obstruction in integral cohomology rather than mod 2.

Proof:

Both clauses are proved in full in [13, Top Class is the Obstruction to a Nowhere-Zero Section], which is upstream of this book and whose statement carries exactly the two clauses above: the oriented case with the converse under $\dim B \leq n$, and the mod-2 case for the top Stiefel-Whitney class. The hypotheses match: B is a connected CW complex and E is a real bundle of rank n , oriented for the first clause. The identification of the sphere bundle as the relevant fibration is definition 3.2.16 together with the remark following it, which is what places the obstruction in $H^n(B; \pi_{n-1}(S^{n-1}))$ and identifies that group with $H^n(B; \mathbb{Z})$ once the orientation is fixed.

The vanishing of the Euler class is necessary but not sufficient for a nowhere-zero section, because $e(E) = 0$ only guarantees that the section extends over the n -skeleton; further obstructions, the secondary obstructions of [13], may block extension over higher skeleta. For $\text{rk} E = \dim B$ the n -skeleton is everything and the Euler class is a complete obstruction, which is the case relevant to vector fields on closed manifolds.

Corollary 3.2.18: Euler Class of the Tangent Bundle

Let M be a closed oriented smooth n -manifold and let TM be its tangent bundle, oriented by the orientation of M . Then evaluating the Euler class on the fundamental class gives the Euler characteristic,

$$\langle e(TM), [M] \rangle = \chi(M)$$

Consequently, if M admits a nowhere-zero vector field then $\chi(M) = 0$.

The corollary is the Poincaré-Hopf theorem in the language of characteristic classes: the obstruction to combing a manifold (finding a nowhere-zero tangent vector field) is a single integer, the Euler characteristic, and that integer is the Euler class evaluated on the fundamental class.

Proof:

Proved in [13, Euler Class of the Tangent Bundle Computes the Euler Characteristic]: the Euler class of the tangent bundle pairs with the fundamental class to give the Euler characteristic.

Example 3.2.19: The Tangent Bundle of S^{2n} Has No Nowhere-Zero Section

Let $M = S^{2n}$ with its standard orientation, $n \geq 1$. The Euler characteristic of an even-dimensional sphere is

$$\chi(S^{2n}) = 1 + (-1)^{2n} = 2$$

(its homology is $\mathbb{Z}$ in degrees 0 and $2n$ and 0 otherwise). By corollary 3.2.18,

$$\langle e(TS^{2n}), [S^{2n}] \rangle = \chi(S^{2n}) = 2 \neq 0$$

so $e(TS^{2n})$ is twice a generator of $H^{2n}(S^{2n}; \mathbb{Z}) \cong \mathbb{Z}$ and in particular is nonzero. By theorem 3.2.17 the tangent bundle TS^{2n} admits no nowhere-zero section, that is, S^{2n} carries no nowhere-vanishing vector field. For $n = 1$ this is the hairy-ball theorem recalled after definition 1.1.9, now obtained from the Euler class.

The contrast with odd spheres is sharp: $\chi(S^{2n+1}) = 1 + (-1)^{2n+1} = 0$, consistent with the standard nowhere-zero field $x \mapsto (-x_2, x_1, -x_4, x_3, \dots)$ on $S^{2n+1} \subset \mathbb{R}^{2n+2}$. Here the vanishing of χ , hence of the integral pairing, is necessary for the field to exist, and on spheres it is also sufficient because the only obstruction is the primary one.

The Euler class required an orientation even to be defined. The obstruction that decides whether an orientation exists at all is the first Stiefel-Whitney class. Orientability of a rank- n bundle is a reduction of the structure group from $O(n)$ to $SO(n)$, equivalently a reduction of the associated fiber datum from the two-point set $O(n)/SO(n) = \{\pm 1\}$ to a point; the obstruction to such a reduction is a class in $H^1(B; \mathbb{Z}/2\mathbb{Z})$.

Theorem 3.2.20: First Stiefel-Whitney Class as Orientability Obstruction

Let $p: E \rightarrow B$ be a real vector bundle of rank n over a path-connected CW complex B . Then E is orientable if and only if $w_1(E) = 0$ in $H^1(B; \mathbb{Z}/2\mathbb{Z})$. More precisely, $w_1(E)$ is the primary obstruction to reducing the structure group of E from $O(n)$ to $SO(n)$, and the orientations of E , when they exist, form a torsor under $H^0(B; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$.

The first Stiefel-Whitney class is thus the complete and only obstruction to orientability, with no higher obstructions, because the relevant fiber $O(n)/SO(n)$ is a discrete two-point set with a free $\mathbb{Z}/2\mathbb{Z}$ -action and has no higher homotopy. The class records, loop by loop, whether transporting an orientation of the fiber around the loop returns it to itself or reverses it.

Proof:

The criterion itself is proved in [13, First Stiefel-Whitney Class is the Obstruction to Orientability]: $w_1(E)$ is the class in $H^1(B; \mathbb{Z}/2\mathbb{Z})$ of the orientation double cover $\text{Or}(E) \rightarrow B$, which admits a section (equivalently, E is orientable) exactly when $w_1(E) = 0$. That statement carries no torsor clause, so the count is proved here.

Suppose E is orientable. An orientation is precisely a section of the double cover $\text{Or}(E) \rightarrow B$, whose fiber over b is the two-element set of orientations of E_b . Given two sections s, s' , the assignment $b \mapsto \varepsilon(b)$, where $\varepsilon(b) \in \mathbb{Z}/2\mathbb{Z}$ records whether $s(b)$ and $s'(b)$ agree, is locally constant: on any open set over which E is trivialized the two sections are constant in the induced local labelling, so ε is constant there. A locally constant $\mathbb{Z}/2\mathbb{Z}$ -valued function on B is exactly an element of $H^0(B; \mathbb{Z}/2\mathbb{Z})$. Conversely any such ε acts on a section by flipping it over the locus where $\varepsilon = 1$, and the result is again a section. The action is free (if ε fixes s then $\varepsilon = 0$) and transitive (the ε built from s, s' carries one to the other), so the orientations form a torsor under $H^0(B; \mathbb{Z}/2\mathbb{Z})$. For B path-connected $H^0(B; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, so an orientable bundle over a connected base has exactly two orientations, completing the proof.

Example 3.2.21: The Möbius Bundle Is Non-Orientable

Let $E \rightarrow S^1$ be the Möbius bundle (example 1.1.4), the rank-1 bundle $I \times \mathbb{R}/\sim$ with (s, t) for $s \in I = [0, 1]$ the base coordinate and $t \in \mathbb{R}$ the fiber coordinate, glued by $(0, t) \sim (1, -t)$. Transporting a chosen fiber orientation once around the base circle reverses it: an orientation is a choice of sign on $\mathbb{R}$, and traversing the base from $s = 0$ to $s = 1$ identifies the fiber at $s = 1$ with the fiber at $s = 0$ through $t \mapsto -t$, which carries the orientation $+$ to $-$. Hence the orientation-transport homomorphism $\pi_1(S^1) = \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ of section 3.1 sends the generator to 1,

so it is the nonzero homomorphism and

$$w_1(E) \neq 0 \quad \text{in} \quad H^1(S^1; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$$

Therefore E is non-orientable. Equivalently, the orientation double cover $\text{Or}(E) \rightarrow S^1$ is the connected double cover $S^1 \xrightarrow{z \mapsto z^2} S^1$, which has no section. This recovers, through the orientability obstruction, the fact recorded after definition 1.1.9 that the Möbius bundle is not isomorphic to the trivial bundle $S^1 \times \mathbb{R}$: a trivial bundle is orientable, while $w_1(E) \neq 0$.

By contrast, the trivial line bundle has $w_1 = 0$ and the cylinder $S^1 \times \mathbb{R}$ admits the constant orientation; the two bundles are distinguished by $w_1 \in H^1(S^1; \mathbb{Z}/2\mathbb{Z})$, which takes both of its values.

Once a bundle is orientable, so $w_1(E) = 0$, the next reduction of the structure group is from $SO(n)$ to its connected double cover $\text{Spin}(n)$. A spin structure is exactly a lift of the structure group along $\text{Spin}(n) \rightarrow SO(n)$, and the obstruction to such a lift is the second Stiefel-Whitney class. The relevant standard fact (proved in [13, Second Stiefel-Whitney Class is the Obstruction to a Spin Structure], whose Lie-theoretic input is [17]) is that for $n \geq 3$ the group $SO(n)$ has $\pi_1(SO(n)) \cong \mathbb{Z}/2\mathbb{Z}$, and $\text{Spin}(n)$ is its universal (double) cover.

Definition 3.2.22: Spin Structure

Let $E \rightarrow B$ be an oriented real vector bundle of rank $n \geq 3$, with principal $SO(n)$ -bundle of oriented orthonormal frames $P_{SO} \rightarrow B$. A *spin structure* on E is a principal $\text{Spin}(n)$ -bundle $P_{\text{Spin}} \rightarrow B$ together with a map $P_{\text{Spin}} \rightarrow P_{SO}$ over B that restricts on each fiber to the double cover $\text{Spin}(n) \rightarrow SO(n)$.

A spin structure is a coherent, globally consistent way of taking the “square root” of the rotation group acting on the fibers; the obstruction to assembling the local square roots into a global one is a mod-2 class in degree two, and the content of the next theorem is that this class is $w_2(E)$.

Theorem 3.2.23: Second Stiefel-Whitney Class as Spin Obstruction

Let $E \rightarrow B$ be an oriented real vector bundle of rank $n \geq 3$ over a connected CW complex B . Then E admits a spin structure if and only if $w_2(E) = 0$ in $H^2(B; \mathbb{Z}/2\mathbb{Z})$. When spin structures exist, they form a torsor under $H^1(B; \mathbb{Z}/2\mathbb{Z})$.

Proof:

The criterion is proved in [13, Second Stiefel-Whitney Class is the Obstruction to a Spin Structure], whose hypotheses are the ones stated here, an oriented real bundle of rank $n \geq 3$ over a connected CW complex: $w_2(E)$ is the obstruction to lifting the structure group along $\text{Spin}(n) \rightarrow SO(n)$, and a lift exists exactly when it vanishes. That statement carries no torsor clause, so the count is proved here.

Write $\lambda: \text{Spin}(n) \rightarrow SO(n)$ for the double cover and $\ker \lambda = \{1, \epsilon\}$. This subgroup is central: it is discrete and normal in $\text{Spin}(n)$, which is connected, so for fixed $\eta \in \ker \lambda$ the continuous map $g \mapsto g\eta g^{-1}$ from a connected space to a discrete one is constant, with value η . Suppose $w_2(E) = 0$ and fix a spin structure $(P_{\text{Spin}}, \varphi)$ as in definition 3.2.22.

Step 1: The difference of two spin structures. Let (P', φ') be a second spin structure on E . The

fiber product

$$P_{\text{Spin}} \times_{P_{SO}} P' = \{(p, p') \mid \varphi(p) = \varphi'(p')\}$$

is *not* a principal $\mathbb{Z}/2\mathbb{Z}$ -bundle over B . Projection to the first factor exhibits it as the pullback of the double cover $P' \rightarrow P_{SO}$ along φ , so over $b \in B$ its fiber is the set of pairs (p, p') with $p \in (P_{\text{Spin}})_b$ and p' one of the two points of P' above $\varphi(p)$, that is, a copy of $\text{Spin}(n) \times \mathbb{Z}/2\mathbb{Z}$ rather than of $\mathbb{Z}/2\mathbb{Z}$. (For $B = \text{pt}$ and $n = 3$ that fiber is $\text{Spin}(3) \times \mathbb{Z}/2\mathbb{Z}$, two copies of S^3 .) The structure group has to be divided out. It acts diagonally, $(p, p') \cdot g = (pg, p'g)$, which lands back in the fiber product because $\varphi(pg) = \varphi(p)\lambda(g)$ and $\varphi'(p'g) = \varphi'(p')\lambda(g)$, and the action is free because the principal action on P_{Spin} is. Set

$$Q(P') := (P_{\text{Spin}} \times_{P_{SO}} P') / \text{Spin}(n)$$

Equivalently, the fiber of $Q(P')$ over b is the set of $\text{Spin}(n)$ -equivariant maps $(P_{\text{Spin}})_b \rightarrow (P')_b$ compatible with φ and φ' : the class of (p, p') is the unique such map carrying p to p' . Since $\text{Spin}(n)$ acts simply transitively on the first coordinate within a fiber, each fiber of $Q(P')$ has exactly two elements, represented by (p, p') and $(p, p'\epsilon)$ for any fixed p . The assignment $[(p, p')] \mapsto [(p, p'\epsilon)]$ is well defined because $p'g\epsilon = p'\epsilon g$ by centrality, and it is free and transitive on each fiber. Local triviality is inherited from P_{Spin} : over an open set carrying a section $b \mapsto p(b)$ of P_{Spin} , sending $[(p(b), p')]$ to p' identifies $Q(P')$ with the pullback of the double cover $P' \rightarrow P_{SO}$ along $\varphi \circ p$, which is trivializable after shrinking. So $Q(P') \rightarrow B$ is a principal $\mathbb{Z}/2\mathbb{Z}$ -bundle, that is, a double cover.

Step 2: Twisting. In the other direction, let $Q \rightarrow B$ be a principal $\mathbb{Z}/2\mathbb{Z}$ -bundle and put

$$P_{\text{Spin}} \cdot Q := (P_{\text{Spin}} \times_B Q) / (\mathbb{Z}/2\mathbb{Z}), \quad (p, q) \cdot \epsilon = (p\epsilon, q\epsilon)$$

Centrality of ϵ makes the right $\text{Spin}(n)$ -action $[(p, q)] \cdot g = [(pg, q)]$ well defined, since $(p\epsilon g, q\epsilon) = (pg\epsilon, q\epsilon)$, and that action is free and transitive on fibers, so $P_{\text{Spin}} \cdot Q$ is a principal $\text{Spin}(n)$ -bundle over B . Because $\varphi(p\epsilon) = \varphi(p)\lambda(\epsilon) = \varphi(p)$, the map $\varphi_Q[(p, q)] = \varphi(p)$ is well defined and restricts on each fiber to λ , so $(P_{\text{Spin}} \cdot Q, \varphi_Q)$ is again a spin structure on E .

Step 3: The two constructions are mutually inverse. This is what makes the action free and transitive, and it is the step the torsor claim rests on.

First, $Q(P_{\text{Spin}} \cdot Q) \cong Q$. A point of the source is a class $[p, [(p_1, q)]]$ with $\varphi(p) = \varphi_Q[(p_1, q)] = \varphi(p_1)$. Writing $p_1 = pg$ for the unique g with $p_1 = pg$, the condition $\varphi(p)\lambda(g) = \varphi(p)$ forces $\lambda(g) = 1$, so $g = \delta \in \{1, \epsilon\}$. Send the class to $q\delta \in Q$. Replacing (p_1, q) by $(p_1\epsilon, q\epsilon)$ replaces δ by $\delta\epsilon$ and q by $q\epsilon$, leaving $q\delta$ fixed; applying the outer diagonal g replaces p by pg and p_1 by $p_1g = pg\delta$, leaving δ and q fixed. So the map is well defined, and it is continuous because it is given by these formulas in any local trivialization. It is $\mathbb{Z}/2\mathbb{Z}$ -equivariant, since the ϵ -action on the source replaces $[(p_1, q)]$ by $[(p_1\epsilon, q)] = [(p_1, q\epsilon)]$ and hence δ by $\delta\epsilon$. A $\mathbb{Z}/2\mathbb{Z}$ -equivariant map of principal $\mathbb{Z}/2\mathbb{Z}$ -bundles over B is an isomorphism.

Second, $P_{\text{Spin}} \cdot Q(P') \cong P'$. Because the diagonal action is simply transitive on the first coordinate, every class in $Q(P')$ has a unique representative with prescribed first coordinate p ; write it (p, p') and send $[p, [(p, p')]] \mapsto p'$. Replacing p by $p\epsilon$ replaces $[(p, p')]$ by $[(p, p'\epsilon)] = [(p\epsilon, p')]$, whose representative with first coordinate $p\epsilon$ is $(p\epsilon, p')$, giving the same value p' , so the map descends to the quotient. It sends $[(pg, \cdot)]$ to $p'g$, hence is $\text{Spin}(n)$ -equivariant, and it commutes with the maps to P_{SO} because $\varphi'(p') = \varphi(p)$. A $\text{Spin}(n)$ -equivariant map of principal $\text{Spin}(n)$ -bundles over B commuting with the maps to P_{SO} is an isomorphism of spin structures.

Step 4: Twisting is a group action. Steps 2 and 3 produce mutually inverse maps between double covers of B and spin structures on E , which is a bijection depending on the chosen base point

P_{Spin} . A torsor needs more: a group, acting. The group is the set $\text{Prin}_{\mathbb{Z}/2\mathbb{Z}}(B)$ of isomorphism classes of principal $\mathbb{Z}/2\mathbb{Z}$ -bundles under the *contracted product*

$$Q_1 \otimes Q_2 := (Q_1 \times_B Q_2) / (\mathbb{Z}/2\mathbb{Z}), \quad (q_1, q_2) \cdot \epsilon = (q_1 \epsilon, q_2 \epsilon)$$

with the residual free transitive $\mathbb{Z}/2\mathbb{Z}$ -action $[q_1, q_2] \epsilon = [q_1 \epsilon, q_2] = [q_1, q_2 \epsilon]$. Two identities make Step 2 an action of it. First $P_{\text{Spin}} \cdot (B \times \mathbb{Z}/2\mathbb{Z}) \cong P_{\text{Spin}}$, since every class in $(P_{\text{Spin}} \times \mathbb{Z}/2\mathbb{Z}) / (\mathbb{Z}/2\mathbb{Z})$ has a unique representative with second coordinate 1, and that identification commutes with φ . Second, $(P_{\text{Spin}} \cdot Q_1) \cdot Q_2 \cong P_{\text{Spin}} \cdot (Q_1 \otimes Q_2)$: both sides are quotients of $P_{\text{Spin}} \times_B Q_1 \times_B Q_2$, the left by the subgroup of $(\mathbb{Z}/2\mathbb{Z})^3$ generated by $(\epsilon, \epsilon, 1)$ and $(\epsilon, 1, \epsilon)$, the right by the subgroup generated by $(\epsilon, \epsilon, 1)$ and $(1, \epsilon, \epsilon)$. Each of those two subgroups contains the product of its generators, so both equal the kernel of the multiplication map $(\mathbb{Z}/2\mathbb{Z})^3 \rightarrow \mathbb{Z}/2\mathbb{Z}$, and the two quotients coincide together with their residual $\text{Spin}(n)$ -actions and their maps to P_{SO} .

Step 5: The torsor. Principal $\mathbb{Z}/2\mathbb{Z}$ -bundles over B are classified by $[B, B\mathbb{Z}/2\mathbb{Z}]$ ([13, Classification of Principal G -Bundles]), and $B\mathbb{Z}/2\mathbb{Z} = \mathbb{RP}^\infty$ is a $K(\mathbb{Z}/2\mathbb{Z}, 1)$ ([13, $\mathbb{RP}^\infty$ as $K(\mathbb{Z}/2, 1)$ and the Classifying Space BG]). Since B is a connected CW complex, based homotopy classes of maps $B \rightarrow K(\mathbb{Z}/2\mathbb{Z}, 1)$ are identified with $\tilde{H}^1(B; \mathbb{Z}/2\mathbb{Z}) = H^1(B; \mathbb{Z}/2\mathbb{Z})$ by [13, Ordinary Cohomology as a Represented Theory], and dropping basepoints changes nothing: the target is aspherical, so π_1 of the target acts on that set by conjugation in $\pi_1(K(\mathbb{Z}/2\mathbb{Z}, 1)) = \mathbb{Z}/2\mathbb{Z}$, which is abelian.

This bijection carries $\otimes$ to addition. The contracted product is the bundle associated to the multiplication homomorphism $m: \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$, so if Q_1, Q_2 are classified by f_1, f_2 then $Q_1 \otimes Q_2$ is classified by the composite $B \xrightarrow{(f_1, f_2)} \mathbb{RP}^\infty \times \mathbb{RP}^\infty \xrightarrow{\mu} \mathbb{RP}^\infty$ with $\mu = Bm$, which is the H -space multiplication inducing the group structure on $[B, K(\mathbb{Z}/2\mathbb{Z}, 1)]$ that [13, Ordinary Cohomology as a Represented Theory] matches with addition in $H^1(B; \mathbb{Z}/2\mathbb{Z})$. (This is the mod-2 instance of the device used for line bundles in the proof of proposition 3.3.2, with $m: U(1) \times U(1) \rightarrow U(1)$ there.) So $\text{Prin}_{\mathbb{Z}/2\mathbb{Z}}(B) \cong H^1(B; \mathbb{Z}/2\mathbb{Z})$ as groups, and Step 4 makes the set of spin structures a set with an $H^1(B; \mathbb{Z}/2\mathbb{Z})$ -action.

The action is transitive by the second half of Step 3, which writes an arbitrary P' as a twist of P_{Spin} . It is free: by Step 4 it is enough to check the stabilizer of one point, and if $P_{\text{Spin}} \cdot Q \cong P_{\text{Spin}}$ then the first half of Step 3 gives $Q \cong Q(P_{\text{Spin}} \cdot Q) \cong Q(P_{\text{Spin}})$, which is the trivial double cover because $b \mapsto [(p, p)]$ is a section of it. Hence the spin structures form a torsor under $H^1(B; \mathbb{Z}/2\mathbb{Z})$, as we sought to show.

The pattern is now visible: orientability removes w_1 , a spin structure removes w_2 , and each removal is a reduction of the structure group whose obstruction is the corresponding Stiefel-Whitney class. The Lie-theoretic ceiling on this tower is set by the homotopy of $SO(n)$: since $\pi_2(SO(n)) = 0$ there is no obstruction in degree three, and the next genuine obstruction appears in degree four, governed by $\pi_3(SO(n))$, which for $n \geq 5$ is $\mathbb{Z}$ (it is $\mathbb{Z} \oplus \mathbb{Z}$ at $n = 4$ and 0 at $n = 2$), and which is where the first Pontryagin-type class lives, consistent with the discussion preceding theorem 3.2.4.

Example 3.2.24: Spin Structures on Surfaces and on $\mathbb{CP}^2$

1. Let Σ be a closed connected orientable surface of genus g and $E \rightarrow \Sigma$ an oriented rank- n bundle with $n \geq 3$. Since $H^2(\Sigma; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, the bundle admits a spin structure if and only if the single class $w_2(E) \in \mathbb{Z}/2\mathbb{Z}$ vanishes. For the (stabilized) tangent bundle, $w_2(T\Sigma)$ is the mod-2 reduction of the Euler class, and $\langle e(T\Sigma), [\Sigma] \rangle = \chi(\Sigma) = 2 - 2g$ is even, so

$w_2(T\Sigma) = 0$ and every such surface is spin. The spin structures then form a torsor under $H^1(\Sigma; \mathbb{Z}/2\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})^{2g}$, giving 2^{2g} of them, matching the classical count.

2. Consider $\mathbb{CP}^2$, whose tangent bundle (as a rank-4 real bundle) has total Stiefel-Whitney class computed from the complex structure. The mod-2 reduction of the total Chern class of $T\mathbb{CP}^2$ is $(1+a)^3 = 1 + 3a + 3a^2 = 1 + a + a^2$ in $H^*(\mathbb{CP}^2; \mathbb{Z}/2\mathbb{Z})$, where a is the generator of $H^2(\mathbb{CP}^2; \mathbb{Z}/2\mathbb{Z})$. By proposition 3.2.10, $w_2(T\mathbb{CP}^2)$ is the mod-2 reduction of $c_1(T\mathbb{CP}^2) = 3a$, namely $a \neq 0$. Hence $w_2(T\mathbb{CP}^2) \neq 0$ and $\mathbb{CP}^2$ admits no spin structure on its tangent bundle, a closed orientable 4-manifold that is not spin.

The complex story parallels the real one with the simplification that $\mathrm{GL}_n(\mathbb{C})$, hence $U(n)$, is connected: there is no orientability obstruction, every complex bundle is canonically oriented as a real bundle, and the first interesting obstruction to a section appears at the top. For a complex bundle of rank n the unit sphere bundle has fiber S^{2n-1} , which is $(2n-2)$ -connected with $\pi_{2n-1}(S^{2n-1}) \cong \mathbb{Z}$, so the primary obstruction lives in $H^{2n}(B; \mathbb{Z})$, the degree of the top Chern class.

Theorem 3.2.25: Top Chern Class as Section Obstruction

Let $E \rightarrow B$ be a complex vector bundle of rank n over a connected CW complex B . The primary obstruction to the existence of a nowhere-zero section of E is the top Chern class

$$c_n(E) \in H^{2n}(B; \pi_{2n-1}(S^{2n-1})) \cong H^{2n}(B; \mathbb{Z})$$

In particular, if E admits a nowhere-zero section, then $c_n(E) = 0$. Equivalently, $c_n(E) = e(E_{\mathbb{R}})$, the Euler class of the underlying oriented real rank- $2n$ bundle $E_{\mathbb{R}}$.

Proof:

Proved in [13, Top Class is the Obstruction to a Nowhere-Zero Section] together with [13, Top Class is the Euler Class]: the top Chern class is the primary obstruction to a nowhere-zero section of a complex bundle, equal to the Euler class of the underlying oriented real bundle.

The top Chern class therefore plays, for complex bundles, exactly the role the Euler class plays for oriented real bundles, and the identity $c_n(E) = e(E_{\mathbb{R}})$ is the precise form of the slogan that the top Chern class is the complex Euler class. Lower Chern classes $c_k(E)$ for $k < n$ are, in the same spirit, the obstructions to finding $n - k + 1$ linearly independent sections, governed by the homotopy of complex Stiefel manifolds, though the higher obstructions there are no longer always primary.

Example 3.2.26: The Top Chern Class of a Bundle With a Section

Let $E \rightarrow B$ be a complex rank- n bundle that splits off a trivial line summand, $E \cong E' \oplus \underline{\mathbb{C}}$ for some complex rank- $(n-1)$ bundle E' . The trivial summand supplies a nowhere-zero section (the constant section $b \mapsto (0, 1)$), so by theorem 3.2.25,

$$c_n(E) = 0$$

This also follows directly from the Whitney formula: $c(\underline{\mathbb{C}}) = 1$, so $c(E) = c(E') \smile 1 = c(E')$, and $c(E')$ has no component in degree $2n$ because E' has complex rank $n-1$, forcing $c_n(E) = 0$. The two computations agree, illustrating that the obstruction-theoretic vanishing and the multiplicative vanishing are the same phenomenon.

A concrete instance is the tangent bundle of $\mathbb{CP}^n$ via the Euler sequence $0 \rightarrow \underline{\mathbb{C}} \rightarrow (H^{\vee})^{\oplus(n+1)} \rightarrow$

$TC\mathbb{P}^n \rightarrow 0$, where $H^\vee$ is the dual of the tautological bundle, also called the hyperplane bundle: here $c_n(T\mathbb{CP}^n)$ need not vanish (indeed $\langle c_n(T\mathbb{CP}^n), [\mathbb{CP}^n] \rangle = \chi(\mathbb{CP}^n) = n+1 \neq 0$), and correspondingly $T\mathbb{CP}^n$ has no nowhere-zero section, recovering that $\mathbb{CP}^n$ has nonzero Euler characteristic.

Exercise 3.2.1

1. Let $E \rightarrow B$ be an oriented real vector bundle of odd rank n over a CW complex B of finite type with $H^n(B; \mathbb{Z})$ free of torsion (for instance $B = S^n$ with n odd). Show that $e(E) = 0$.
2. Using the Euler sequence $T\mathbb{CP}^n \oplus \mathbb{C} \cong (H^\vee)^{\oplus(n+1)}$, compute the Euler number $\langle e(T\mathbb{CP}^n), [\mathbb{CP}^n] \rangle$ of the tangent bundle of $\mathbb{CP}^n$, and check that it equals $\chi(\mathbb{CP}^n)$.
3. Let $E \rightarrow B$ be a real vector bundle that is the underlying real bundle of a complex line bundle L . Show that $p_1(E) = c_1(L)^2$.
4. Show that the Pontryagin classes are stable: $p_i(E \oplus \mathbb{R}^k) = p_i(E)$ for the trivial bundle $\mathbb{R}^k$. Deduce that $p_i(TS^n) = 0$ for all $i > 0$, using that $TS^n \oplus NS^n \cong \mathbb{R}^{n+1}$ with NS^n trivial. Then explain why this argument does *not* extend to an arbitrary stably trivial bundle in the sense of definition 1.1.12 (a bundle E with $E \oplus F \cong \mathbb{R}^N$ for *some* F , not necessarily trivial), and exhibit a counterexample using $T\mathbb{CP}^2$ from example 3.2.15.
5. Let $E \rightarrow B$ be an oriented real rank- n bundle over a CW complex with $\dim B < n$. Using theorem 3.2.17, show that E admits a nowhere-zero section and that $e(E) = 0$. Explain why this is consistent with $e(E) \in H^n(B; \mathbb{Z})$.
6. Compute w_1 of the Möbius bundle pulled back along the degree-2 map $f: S^1 \rightarrow S^1$, $z \mapsto z^2$, and decide whether the pullback is orientable. (Use theorem 3.2.20 and naturality.)
7. Let M be a closed oriented 3-manifold. Show that $\chi(M) = 0$ and conclude that M admits a nowhere-zero vector field. Where does the argument use corollary 3.2.18, and where does it use that the only obstruction is primary?
8. Proposition 3.2.8 says the mod-2 reduction of $e(E)$ is the top Stiefel-Whitney class. For an oriented real rank-2 bundle $E \rightarrow B$, deduce that $\rho(e(E)) = w_2(E)$. Then let Σ be a closed connected orientable surface and $E \rightarrow \Sigma$ an oriented rank-2 bundle, and use theorem 3.2.17 to show that E admits a nowhere-zero section if and only if $e(E) = 0$. Deduce that $w_2(E) = 0$ is necessary but not sufficient, and exhibit a bundle witnessing the gap.
9. Let $E \rightarrow B$ be a complex rank- n bundle and suppose $c_n(E) \neq 0$. Prove that E is not isomorphic to a bundle of the form $E' \oplus \mathbb{C}$, and that the underlying real bundle $E_{\mathbb{R}}$ has no nowhere-zero section. Which theorem of this section gives each conclusion?
10. Explain why $\mathbb{CP}^2$ admits no spin structure on its tangent bundle while S^4 does, using theorem 3.2.23. Compute the number of spin structures on S^4 .

3.3 The Chern Character

The computations of section 2.4 leave a coincidence unexplained. For complex projective space, theorem 2.4.3 gives $K^0(\mathbb{CP}^n) \cong \mathbb{Z}[t]/(t^{n+1})$ with $t = [H] - 1$, while ordinary cohomology gives $H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1})$ with $|x| = 2$. Two truncated polynomial rings on one generator, of the

same rank. The same coincidence of ranks appeared for spheres, where $\tilde{K}^0(S^{2k}) \cong \mathbb{Z} \cong \tilde{H}^{2k}(S^{2k}; \mathbb{Z})$. The coincidence is structural, and this section identifies the map that explains it. There is a natural ring homomorphism from $K^0(X)$ to rational cohomology, the Chern character, which becomes an isomorphism after tensoring the source with $\mathbb{Q}$. It is not an isomorphism over $\mathbb{Z}$, and the failure is not a defect of the construction: it is the integrality phenomenon that section 2.4 already stumbled onto in $\mathbb{CP}^2$, and the subject of the next section.

Two conventions before beginning. Throughout, $H^{\text{even}}(X; \mathbb{Q}) = \bigoplus_{k \geq 0} H^{2k}(X; \mathbb{Q})$ and $H^{\text{odd}}(X; \mathbb{Q}) = \bigoplus_{k \geq 0} H^{2k+1}(X; \mathbb{Q})$, both finite sums for a finite complex. And H continues to denote the *tautological* line bundle, with $H^\vee$ its dual, so that $c_1(H) = -x$ where $x = c_1(H^\vee)$ is the positive generator. *Algebraic Topology* [13] makes the opposite choice, writing γ^1 for the tautological bundle and reserving H for the hyperplane bundle with $c_1(H) = +x$. Every sign below is stated in this book's convention, and a reader carrying a formula across should dualize.

Two inputs are needed that the series does not supply in the form required, and both are proved here. The first is a statement about symmetric polynomials, and it is genuinely an integrality statement: the power sums are *integral* polynomial expressions in the elementary symmetric functions, not merely rational ones.

Lemma 3.3.1: Power Sums as Integral Polynomials

For each $k \geq 1$ there is a polynomial $s_k \in \mathbb{Z}[y_1, \dots, y_k]$, independent of n , such that for all n and all $x_1, \dots, x_n$,

$$\sum_{j=1}^n x_j^k = s_k(\sigma_1, \dots, \sigma_k)$$

where σ_i is the i -th elementary symmetric polynomial in $x_1, \dots, x_n$, with the convention $\sigma_i = 0$ for $i > n$. The first three are

$$s_1 = y_1, \quad s_2 = y_1^2 - 2y_2, \quad s_3 = y_1^3 - 3y_1y_2 + 3y_3$$

Proof:

Write $p_k = \sum_j x_j^k$ and let $E(T) = \prod_{j=1}^n (1 + x_j T) = \sum_{i \geq 0} \sigma_i T^i$, a polynomial identity in $\mathbb{Z}[x_1, \dots, x_n][T]$. Differentiating and dividing,

$$\frac{E'(T)}{E(T)} = \sum_{j=1}^n \frac{x_j}{1 + x_j T} = \sum_{j=1}^n \sum_{m \geq 0} (-1)^m x_j^{m+1} T^m = \sum_{m \geq 0} (-1)^m p_{m+1} T^m$$

as formal power series, the middle step being the geometric expansion of $(1 + x_j T)^{-1}$. Multiplying by $E(T)$ and comparing coefficients of T^{k-1} gives Newton's identity

$$k \sigma_k = \sum_{j=1}^k (-1)^{j-1} \sigma_{k-j} p_j \quad (3.1)$$

valid for every $k \geq 1$, with $\sigma_0 = 1$. Solving (3.1) for p_k , whose coefficient is $\sigma_0 = 1$, gives

$$p_k = \sigma_1 p_{k-1} - \sigma_2 p_{k-2} + \dots + (-1)^k \sigma_{k-1} p_1 + (-1)^{k-1} k \sigma_k$$

and induction on k expresses p_k as a polynomial with integer coefficients in $\sigma_1, \dots, \sigma_k$. Since the coefficient of p_k in (3.1) is 1 rather than k , no division occurs at any stage, which is what

makes the expression integral. The recursion does not mention n , so the resulting polynomial s_k is independent of n . The convention $\sigma_i = 0$ for $i > n$ is what makes the same s_k serve every n . Evaluating for $k = 1, 2, 3$ gives $p_1 = \sigma_1$, $p_2 = \sigma_1^2 - 2\sigma_2$ and $p_3 = \sigma_1^3 - 3\sigma_1\sigma_2 + 3\sigma_3$, as displayed.

Core Algebra [9] proves a fundamental theorem on symmetric functions, but in the form that a symmetric *rational function* over a field is a rational function of the elementary symmetric functions. That statement is about a different ring and does not give the integrality above, which is the property the Chern character needs: the denominators appearing in ch must come from the explicit $1/k!$ and from nowhere else.

The second input concerns line bundles. The Chern character is built so as to convert tensor products into products, and that rests on the first Chern class converting tensor products into sums.

Proposition 3.3.2: The First Chern Class is Additive on Tensor Products

Let L, L' be complex line bundles over X .

1. If X is paracompact, then

$$c_1(L \otimes L') = c_1(L) + c_1(L')$$

and in particular $c_1(L^\vee) = -c_1(L)$.

2. If moreover X is a CW complex, c_1 is an isomorphism of groups from $(\text{Vect}_{\mathbb{C}}^1(X), \otimes)$ to $(H^2(X; \mathbb{Z}), +)$.

Proof:

Part (1). Complex line bundles over a paracompact X are classified by homotopy classes of maps into $\text{Grass}_1(\mathbb{C}^\infty) = BU(1) = \mathbb{CP}^\infty$ by theorem 1.2.13, and $c_1(L) = f_L^*x$ for a classifying map f_L by proposition 3.2.1, where x generates $H^2(\mathbb{CP}^\infty; \mathbb{Z}) \cong \mathbb{Z}$ ([13, The First Chern Class of a Line Bundle]).

The identity is proved once on $\mathbb{CP}^\infty \times \mathbb{CP}^\infty$ and transported. Write γ_1, γ_2 for the pullbacks of the tautological line bundle along the two projections, and let $\mu: \mathbb{CP}^\infty \times \mathbb{CP}^\infty \rightarrow \mathbb{CP}^\infty$ be a classifying map for $\gamma_1 \otimes \gamma_2$; such a map exists because $\mathbb{CP}^\infty \times \mathbb{CP}^\infty$ is paracompact, and it is the map Bm induced by the multiplication $m: U(1) \times U(1) \rightarrow U(1)$. Then $L \otimes L' \cong (f_L, f_{L'})^*(\gamma_1 \otimes \gamma_2)$, since pulling γ_1, γ_2 back along $(f_L, f_{L'})$ gives L, L' and tensor product commutes with pullback, so by proposition 3.2.1 it suffices to compute μ^*x . Nothing beyond theorem 1.2.13 and proposition 3.2.1 is used of X .

The tensor product of line bundles is the bundle associated to the multiplication homomorphism $m: U(1) \times U(1) \rightarrow U(1)$, so if L, L' are classified by $f_L, f_{L'}$ then $L \otimes L'$ is classified by the composite

$$X \xrightarrow{(f_L, f_{L'})} \mathbb{CP}^\infty \times \mathbb{CP}^\infty \xrightarrow{\mu} \mathbb{CP}^\infty$$

where $\mu = Bm$. It therefore suffices to compute μ^*x .

Since $H^*(\mathbb{CP}^\infty; \mathbb{Z}) = \mathbb{Z}[x]$ is concentrated in even degrees and is free, the Künneth theorem gives $H^2(\mathbb{CP}^\infty \times \mathbb{CP}^\infty; \mathbb{Z}) = \mathbb{Z}x_1 \oplus \mathbb{Z}x_2$, where x_i is the pullback of x along the i -th projection, with no cross term because $H^1(\mathbb{CP}^\infty; \mathbb{Z}) = 0$. Write $\mu^*x = ax_1 + bx_2$ with $a, b \in \mathbb{Z}$. Restricting along the inclusion $\iota_1: \mathbb{CP}^\infty \rightarrow \mathbb{CP}^\infty \times \mathbb{CP}^\infty$, $z \mapsto (z, *)$, the composite $\mu\iota_1$ classifies the tensor

product of the tautological bundle with the trivial bundle, which is the tautological bundle itself, so μ_1 is homotopic to the identity. Hence $x = \iota_1^* \mu^* x = ax$, giving $a = 1$, and symmetrically $b = 1$. Therefore $\mu^* x = x_1 + x_2$ and

$$c_1(L \otimes L') = (f_L, f_{L'})^* \mu^* x = (f_L, f_{L'})^* (x_1 + x_2) = f_L^* x + f_{L'}^* x = c_1(L) + c_1(L')$$

Part (2). The map c_1 is a bijection $\text{Vect}_{\mathbb{C}}^1(X) \rightarrow H^2(X; \mathbb{Z})$ when X is a CW complex, and this is where that hypothesis is needed and part (1)'s is not enough. It takes two inputs: the classification just quoted gives $\text{Vect}_{\mathbb{C}}^1(X) \cong [X, \mathbb{CP}^\infty]$, and separately $\mathbb{CP}^\infty = BU(1)$ is an Eilenberg-MacLane space $K(\mathbb{Z}, 2)$, so that $[X, \mathbb{CP}^\infty] \cong H^2(X; \mathbb{Z})$ by representability of ordinary cohomology on CW complexes ([13, Ordinary Cohomology as a Represented Theory]), the isomorphism being $f \mapsto f^* x$. Composing the two is exactly c_1 . The displayed identity then makes this bijection a homomorphism, hence an isomorphism of groups. Applying it to $L \otimes L^\vee \cong \underline{\mathbb{C}}$, whose first Chern class vanishes, gives $c_1(L^\vee) = -c_1(L)$, completing the proof.

With both inputs in hand the definition can be made. The guiding formula is that for a sum of line bundles $L_1 \oplus \cdots \oplus L_n$ the Chern character should be $\sum_j e^{c_1(L_j)}$, since that is the unique expression converting $\oplus$ to $+$ and $\otimes$ to $\times$ simultaneously. The splitting principle reduces the general case to that one, and lemma 3.3.1 is what turns the resulting symmetric expression back into a formula in the Chern classes of E itself.

Definition 3.3.3: [Topological] Chern Character

Let X be a finite CW complex and $E \rightarrow X$ a complex vector bundle of rank r . The *Chern character* of E is

$$\text{ch}(E) = r + \sum_{k \geq 1} \frac{1}{k!} s_k(c_1(E), \dots, c_k(E)) \in H^{\text{even}}(X; \mathbb{Q})$$

with s_k the integral polynomials of lemma 3.3.1. The k -th summand lies in $H^{2k}(X; \mathbb{Q})$ and is written $\text{ch}_k(E)$, so $\text{ch}(E) = \sum_{k \geq 0} \text{ch}_k(E)$ with $\text{ch}_0(E) = \text{rk} E$. The sum is finite because X is finite dimensional. Written out,

$$\text{ch}(E) = \text{rk} E + c_1 + \frac{1}{2}(c_1^2 - 2c_2) + \frac{1}{6}(c_1^3 - 3c_1c_2 + 3c_3) + \cdots$$

Equivalently, if $p^*E \cong L_1 \oplus \cdots \oplus L_r$ splits into line bundles with Chern roots $x_j = c_1(L_j)$, then $p^*\text{ch}(E) = \sum_{j=1}^r e^{x_j}$.

The bracket in the title marks the word this book drops. Three books in the series define a Chern character and all three are correct: *Intersection Theory* [6] defines an operator on Chow groups $A_*(X)_{\mathbb{Q}}$, and *Riemannian Geometry* [7] defines a de Rham class $\text{tr} \exp(\frac{i}{2\pi}\Omega)$ built from the curvature of a connection. Neither construction is available here, since a compact space carries neither an algebraic structure nor a connection, and neither book can state the theorem this section is aimed at, which is a statement about K^0 .

The two descriptions in the definition agree by construction: the elementary symmetric polynomials in the Chern roots are the Chern classes, so $\sum_j x_j^k = s_k(c_1, \dots, c_k)$ by lemma 3.3.1, and dividing by $k!$ and summing gives $\sum_j e^{x_j}$. What needs checking is that the formula on the left, which involves no choices, is the one that governs, and the splitting principle supplies the injectivity that makes the identity on $F(E)$ an identity on X .

Proposition 3.3.4: The Chern Character of a Line Bundle

For a complex line bundle L over a finite CW complex,

$$\text{ch}(L) = e^{c_1(L)} = 1 + c_1(L) + \frac{1}{2}c_1(L)^2 + \frac{1}{6}c_1(L)^3 + \cdots$$

and for line bundles L, L' one has $\text{ch}(L \otimes L') = \text{ch}(L) \text{ch}(L')$.

Proof:

A line bundle has $c_i(L) = 0$ for $i \geq 2$ by theorem 3.2.5(3), so $s_k(c_1, 0, \dots, 0) = c_1^k$: the recursion of lemma 3.3.1 reduces to $p_k = \sigma_1 p_{k-1}$ when every σ_i with $i \geq 2$ vanishes, and $p_1 = \sigma_1$. Hence $\text{ch}(L) = 1 + \sum_{k \geq 1} c_1(L)^k / k! = e^{c_1(L)}$. For the product, $c_1(L \otimes L') = c_1(L) + c_1(L')$ by proposition 3.3.2, so

$$\text{ch}(L \otimes L') = e^{c_1(L) + c_1(L')} = e^{c_1(L)} e^{c_1(L')} = \text{ch}(L) \text{ch}(L')$$

the middle equality being the identity $e^{a+b} = e^a e^b$ for commuting elements of the rational cohomology ring, which holds because the exponential series is finite here, completing the proof.

Theorem 3.3.5: The Chern Character is a Natural Ring Homomorphism

Let X be a finite CW complex and let E, F be complex vector bundles over X . Then

1. $\text{ch}(E \oplus F) = \text{ch}(E) + \text{ch}(F)$;
2. $\text{ch}(E \otimes F) = \text{ch}(E) \text{ch}(F)$;
3. $\text{ch}(f^*E) = f^* \text{ch}(E)$ for every cellular map $f: Y \rightarrow X$ of finite CW complexes.

Consequently ch extends uniquely to a natural homomorphism of unital rings

$$\text{ch}: K^0(X) \longrightarrow H^{\text{even}}(X; \mathbb{Q}), \quad \text{ch}([E] - [F]) = \text{ch}(E) - \text{ch}(F)$$

which, for X connected, restricts to $\tilde{K}^0(X) \rightarrow \bigoplus_{k \geq 1} H^{2k}(X; \mathbb{Q})$ on reduced classes. (Connectedness is needed: $\tilde{K}^0$ is the kernel of restriction to the basepoint, which for disconnected X constrains the rank on one component only, so ch_0 need not vanish. See example 2.4.1 at S^0 .)

Proof:

Step 1: Naturality. The Chern classes are natural (theorem 3.2.5(1)) and s_k has constant coefficients, so $\text{ch}(f^*E) = \sum_k \frac{1}{k!} s_k(c_\bullet(f^*E)) = f^* \sum_k \frac{1}{k!} s_k(c_\bullet(E)) = f^* \text{ch}(E)$. This is (3).

Step 2: Reduction to split bundles. Apply proposition 3.2.3(2) to E : there is $q: F(E) \rightarrow X$ with q^*E a sum of line bundles and $q^*: H^*(X; \mathbb{Z}) \hookrightarrow H^*(F(E); \mathbb{Z})$ injective, and then $r: F(q^*F) \rightarrow F(E)$ splitting q^*F with r^* injective as well. Write $p = q \circ r$, so that p^*E and p^*F both split as sums of line bundles and p^* is injective on $H^*(-; \mathbb{Z})$. The two bundles have to be split one after the other in this way: splitting $E \oplus F$ in one step and then refining does not separate the summands, because vector bundles admit no cancellation.

Injectivity persists rationally. Here X is a finite CW complex, so $H^*(X; \mathbb{Z})$ is finitely generated in each degree and $H^*(X; \mathbb{Q}) \cong H^*(X; \mathbb{Z}) \otimes \mathbb{Q}$, and the same holds upstairs since each stage of the tower has fiber a finite projective space; flatness of $\mathbb{Q}$ over $\mathbb{Z}$ then carries the injectivity of p^* across. (Without finite generation this step fails, since $H^*(-; \mathbb{Q})$ is not in general $H^*(-; \mathbb{Z}) \otimes \mathbb{Q}$.) The base X is paracompact, so proposition 3.2.3 applies and its last clause makes each stage paracompact in turn; the Chern classes, and with them ch , are therefore defined upstairs by proposition 3.2.1, and are natural under p^* by the same proposition. It therefore suffices to prove (1) and (2) after pulling back along p , where both bundles split.

Step 3: The split case. Suppose $E \cong L_1 \oplus \cdots \oplus L_r$ and $F \cong M_1 \oplus \cdots \oplus M_s$ with Chern roots $x_j = c_1(L_j)$ and $y_l = c_1(M_l)$. The Whitney formula (theorem 3.2.5(2)) makes the total Chern class of a direct sum the product of the totals, so the Chern roots of $E \oplus F$ are the x_j together with the y_l , and

$$\text{ch}(E \oplus F) = \sum_j e^{x_j} + \sum_l e^{y_l} = \text{ch}(E) + \text{ch}(F)$$

which is (1). For (2), the tensor product $E \otimes F$ splits as $\bigoplus_{j,l} L_j \otimes M_l$, and $c_1(L_j \otimes M_l) = x_j + y_l$ by proposition 3.3.2(1), whose paracompactness hypothesis is met by the last clause of proposition 3.2.3, so its Chern roots are the $x_j + y_l$ and

$$\text{ch}(E \otimes F) = \sum_{j,l} e^{x_j + y_l} = \left(\sum_j e^{x_j} \right) \left(\sum_l e^{y_l} \right) = \text{ch}(E) \text{ch}(F)$$

which is (2). Both identities were proved on $F(q^*F)$ and descend to X by the injectivity of Step 2.

Step 4: Extension to K^0 . By (1), ch is a homomorphism of monoids from $(\text{Vect}(X), \oplus)$ to the additive group of $H^{\text{even}}(X; \mathbb{Q})$. The Grothendieck group is universal among such (definition 2.1.1), so ch extends uniquely to a group homomorphism on $K^0(X)$ with $\text{ch}([E] - [F]) = \text{ch}(E) - \text{ch}(F)$; (2) makes the extension multiplicative for the ring structure of proposition 2.1.10, and $\text{ch}(\underline{\mathbb{C}}) = 1$ makes it unital. For the last clause, X is connected, so a reduced class has rank 0 by proposition 2.1.3, and $\text{ch}_0 = \text{rk}$, so its Chern character has vanishing degree-zero component, completing the proof.

So far ch is a ring map out of K^0 of a space. To compare it with cohomology as a *theory*, it has to be defined on the relative and negative groups and shown to respect the structure maps that tie those groups together. That is the content of the next proposition, and it is what the comparison theorem consumes.

Proposition 3.3.6: Compatibility with Suspension and the Bott Map

Let (X, A) be a finite CW pair.

1. ch is defined on relative groups, $\text{ch}: K^0(X, A) \rightarrow H^{\text{even}}(X, A; \mathbb{Q})$, naturally in the pair, and commutes with the maps i^* and j^* of theorem 2.3.7.
- (1') ch is defined on the odd group. By definition 2.3.1, $K^{-1}(X) = \tilde{K}^0(\Sigma X_+)$, and the suspension isomorphism of reduced rational cohomology gives $\tilde{H}^{\text{even}}(\Sigma X_+; \mathbb{Q}) \cong H^{\text{odd}}(X; \mathbb{Q})$. Define

$$\text{ch}: K^{-1}(X) \longrightarrow H^{\text{odd}}(X; \mathbb{Q})$$

as the composite of ch on $\tilde{K}^0(\Sigma X_+)$ with that isomorphism. This is the map meant whenever ch is applied to an odd class below, in corollary 3.3.7 and in theorem 3.3.8; with it, ch is a map of $\mathbb{Z}/2$ -graded theories.

2. For the Bott generator $\beta = [H] - 1 \in \tilde{K}^0(S^2)$ of definition 2.2.7,

$$\text{ch}(\beta) = c_1(H) = -x$$

a generator of $H^2(S^2; \mathbb{Z})$, where $x = c_1(H^\vee)$.

3. ch intertwines the Bott map with the double suspension: for every finite pointed CW complex X the square formed by $\beta_X: \tilde{K}^0(X) \rightarrow \tilde{K}^0(\Sigma^2 X)$, the suspension isomorphism σ^2 of reduced rational cohomology, and ch on both rows commutes. Consequently ch commutes with the connecting map δ of theorem 2.3.7.

Proof:

(1). By definition 2.3.3, $K^0(X, A) = \tilde{K}^0(X/A)$, and X/A is again a finite CW complex, so definition 3.3.3 applies to it and lands in $\tilde{H}^{\text{even}}(X/A; \mathbb{Q}) \cong H^{\text{even}}(X, A; \mathbb{Q})$, the identification being the one for a CW pair. Naturality and commutation with i^*, j^* are theorem 3.3.5(3) applied to the maps of spaces inducing them, since both i^* and j^* are induced by maps.

(2). By proposition 3.3.4, $\text{ch}(\beta) = \text{ch}(H) - \text{ch}(\mathbb{C}) = e^{c_1(H)} - 1 = c_1(H) + \frac{1}{2}c_1(H)^2 + \dots$. On S^2 every term beyond the first vanishes, since $c_1(H)^2 \in H^4(S^2; \mathbb{Q}) = 0$. So $\text{ch}(\beta) = c_1(H)$, which is $-x$ in this book's convention (example 2.1.12) and generates $H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}$.

(3). The external product in cohomology is $a \times b = \pi_X^* a \smile \pi_Y^* b$, and the same formula defines the external product in K-theory (definition 2.2.6). By theorem 3.3.5(2),(3),

$$\text{ch}(\mu(a \otimes b)) = \text{ch}(\pi_X^* a) \text{ch}(\pi_Y^* b) = \pi_X^* \text{ch}(a) \smile \pi_Y^* \text{ch}(b) = \text{ch}(a) \times \text{ch}(b)$$

so ch carries the external product to the external product. Taking $a = \beta$ and restricting to reduced classes on the smash, which is where both products land by proposition 2.2.5, gives $\text{ch}(\beta_X(\alpha)) = \text{ch}(\beta) \times \text{ch}(\alpha)$. Since $\text{ch}(\beta)$ generates $\tilde{H}^2(S^2; \mathbb{Z})$ by (2), external multiplication by it is precisely the double suspension isomorphism σ^2 on reduced rational cohomology, so the square commutes.

For the last clause, δ in K-theory is induced by the map $\partial: X/A \rightarrow \Sigma A_+$ of the Puppe sequence appearing in Step 1 of theorem 2.3.7, and the connecting map in ordinary cohomology is induced by the same map of spaces. Naturality, which is (1), therefore makes ch commute with both,

completing the proof.

Corollary 3.3.7: The Chern Character on Even Spheres

For $k \geq 1$ the Chern character carries $\tilde{K}^0(S^{2k}) \cong \mathbb{Z}$ isomorphically onto the integral lattice $\tilde{H}^{2k}(S^{2k}; \mathbb{Z})$ inside $\tilde{H}^{2k}(S^{2k}; \mathbb{Q})$, and $\tilde{K}^{-1}(S^{2k}) = 0 = \tilde{H}^{\text{odd}}(S^{2k}; \mathbb{Q})$. On odd spheres the two are interchanged.

Proof:

By example 2.4.1, $\tilde{K}^0(S^{2k})$ is infinite cyclic generated by the k -fold external power $\beta^{\times k}$, under $S^{2k} \cong S^2 \wedge \cdots \wedge S^2$. Applying proposition 3.3.6(3) k times, $\text{ch}(\beta^{\times k}) = \text{ch}(\beta)^{\times k}$, which by (2) is the k -fold external power of a generator of $H^2(S^2; \mathbb{Z})$, hence a generator of $H^{2k}(S^{2k}; \mathbb{Z})$. So ch sends a generator to a generator, which is the claim. The vanishing statements are example 2.4.1 and the cohomology of a sphere. The odd case follows by replacing S^{2k} with its suspension, completing the proof.

The comparison theorem now follows by the standard method for two cohomology theories agreeing on a point: build up the space one cell at a time and force the agreement along the exact sequences.

Theorem 3.3.8: The Chern Character is a Rational Isomorphism

For every finite CW complex X the Chern character induces isomorphisms

$$\text{ch} \otimes \mathbb{Q}: K^0(X) \otimes_{\mathbb{Z}} \mathbb{Q} \xrightarrow{\sim} H^{\text{even}}(X; \mathbb{Q}), \quad K^{-1}(X) \otimes_{\mathbb{Z}} \mathbb{Q} \xrightarrow{\sim} H^{\text{odd}}(X; \mathbb{Q})$$

the first an isomorphism of rings. Equivalently, ch is an isomorphism of $\mathbb{Z}/2$ -graded rings $K^*(X) \otimes \mathbb{Q} \cong H^*(X; \mathbb{Q})$.

Proof:

Step 1: Both sides are cohomology theories, and the map is a transformation. On finite CW pairs the family $\{K^{-n}\}$ with its long exact sequence is a cohomology theory by theorem 2.3.6 and theorem 2.3.7, folded into two degrees by box 2.3.5. Tensoring with $\mathbb{Q}$ preserves exactness, since $\mathbb{Q}$ is flat over $\mathbb{Z}$, so $\{K^*(-) \otimes \mathbb{Q}\}$ is again one. Ordinary rational cohomology, folded by parity into H^{even} and H^{odd} , is a cohomology theory on the same pairs ([13]). By proposition 3.3.6, $\text{ch} \otimes \mathbb{Q}$ is a natural transformation between them commuting with the connecting maps.

Step 2: Agreement on a point. $K^0(\{*\}) = \mathbb{Z}$ by example 2.1.2 and $\text{ch}_0 = \text{rk}$ (proposition 2.1.3), so $\text{ch} \otimes \mathbb{Q}: K^0(\{*\}) \otimes \mathbb{Q} \rightarrow H^0(\{*\}; \mathbb{Q}) = \mathbb{Q}$ is the identity. In odd degree $K^{-1}(\{*\}) = 0 = H^{\text{odd}}(\{*\}; \mathbb{Q})$ by example 2.3.2. So the transformation is an isomorphism on a point, and by corollary 3.3.7 it is an isomorphism on every sphere.

Step 3: Induction over cells. Argue by induction on the number of cells of X . If X is a point, Step 2 applies. Otherwise write $X = X' \cup e^m$ for a subcomplex X' with one fewer cell, so that $X/X' \cong S^m$. The pair (X, X') has a long exact sequence in each theory, and by Step 1 the

Chern character maps one to the other, giving a commutative ladder

$$\begin{array}{ccccc} K^*(X, X') \otimes \mathbb{Q} & \rightarrow & K^*(X) \otimes \mathbb{Q} & \rightarrow & K^*(X') \otimes \mathbb{Q} \rightarrow \\ \downarrow \text{ch} & & \downarrow \text{ch} & & \downarrow \text{ch} \\ H^*(X, X'; \mathbb{Q}) & \longrightarrow & H^*(X; \mathbb{Q}) & \longrightarrow & H^*(X'; \mathbb{Q}) \longrightarrow \end{array}$$

extending in both directions. The relative groups are those of $X/X' \cong S^m$, where the vertical map is an isomorphism by Step 2, and the groups of X' are handled by the inductive hypothesis. Four of every five consecutive vertical maps are therefore isomorphisms, and the five lemma ([13, The Five Lemma]) forces the fifth, which is the map for X . A disjoint union is handled by additivity, Axiom 5 of theorem 2.3.6, on both sides.

Step 4: The ring structure. The map is multiplicative by theorem 3.3.5(2), and tensoring with $\mathbb{Q}$ preserves that, so the isomorphism in even degree is one of rings. The $\mathbb{Z}/2$ -graded statement is the two displayed isomorphisms taken together, completing the proof.

Corollary 3.3.9: Kernel, Ranks, and the Euler Characteristic

Let X be a finite CW complex. The kernel of $\text{ch}: K^0(X) \rightarrow H^{\text{even}}(X; \mathbb{Q})$ is exactly the torsion subgroup of $K^0(X)$, and

$$\text{rk}_{\mathbb{Z}} K^0(X) = \dim_{\mathbb{Q}} H^{\text{even}}(X; \mathbb{Q}), \quad \text{rk}_{\mathbb{Z}} K^{-1}(X) = \dim_{\mathbb{Q}} H^{\text{odd}}(X; \mathbb{Q})$$

Consequently $\text{rk}_{\mathbb{Z}} K^0(X) - \text{rk}_{\mathbb{Z}} K^{-1}(X) = \chi(X)$, the Euler characteristic.

Proof:

For a finitely generated abelian group A the kernel of $A \rightarrow A \otimes \mathbb{Q}$ is the torsion subgroup, and $K^0(X)$ is finitely generated because X is a finite complex. Since $\text{ch} \otimes \mathbb{Q}$ is injective by theorem 3.3.8, a class dies under ch exactly when it dies in $K^0(X) \otimes \mathbb{Q}$, which is exactly when it is torsion. The rank statements are the dimensions of the two sides of the same theorem. For the last, $\chi(X) = \sum_i (-1)^i \dim_{\mathbb{Q}} H^i(X; \mathbb{Q})$ splits by parity into $\dim H^{\text{even}} - \dim H^{\text{odd}}$, completing the proof.

Example 3.3.10: The Chern Character on $\mathbb{CP}^2$

Take $n = 2$, so $K^0(\mathbb{CP}^2) \cong \mathbb{Z}[t]/(t^3)$ with $t = [H] - 1$, free on $1, t, t^2$ (theorem 2.4.3), and $H^*(\mathbb{CP}^2; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^3)$ with $|x| = 2$. By proposition 3.3.4 the Chern character of a line bundle is $\text{ch}(L) = e^{c_1(L)}$, and $c_1(H) = -x$ because H is the tautological bundle and $x = c_1(H^\vee)$, so

$$\text{ch}(H) = e^{-x} = 1 - x + \frac{1}{2}x^2$$

using $x^3 = 0$. Therefore

$$\text{ch}(t) = \text{ch}([H] - 1) = e^{-x} - 1 = -x + \frac{1}{2}x^2$$

and squaring (with $x^3 = 0$),

$$\text{ch}(t^2) = \text{ch}(t)^2 = (-x + \frac{1}{2}x^2)^2 = x^2$$

Thus on the basis $1, t, t^2$ the Chern character is

$$\text{ch}(1) = 1, \quad \text{ch}(t) = -x + \frac{1}{2}x^2, \quad \text{ch}(t^2) = x^2$$

The matrix of ch whose columns express $\text{ch}(1), \text{ch}(t), \text{ch}(t^2)$ in the basis $\{1, x, x^2\}$ is triangular,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & \frac{1}{2} & 1 \end{pmatrix}$$

with determinant -1 , so ch is a rational isomorphism. Because the determinant is a unit, the image $\text{ch}(K^0(\mathbb{CP}^2))$ is a full-rank lattice in $H^{\text{even}}(\mathbb{CP}^2; \mathbb{Q})$ of the same covolume as $H^{\text{even}}(\mathbb{CP}^2; \mathbb{Z})$. The half-integer entry $\frac{1}{2}$ shows that the two lattices are nonetheless different, neither containing the other. Solving $a \cdot 1 + b(-x + \frac{1}{2}x^2) + cx^2 = x$ forces $b = -1$ and $c = \frac{1}{2} \notin \mathbb{Z}$, so the cohomology generator x is not in the image. This is the integrality phenomenon: K^0 and H^{even} are abstractly the same free group, but the Chern character identifies them only after inverting 2, and the half-integral discrepancy is the first appearance of the Todd genus that corrects the Riemann-Roch theorem.

The contrast with corollary 3.3.7 is the point. On an even sphere the Chern character hits the integral lattice exactly. On $\mathbb{CP}^2$ it does not, and the obstruction is a single factor of 2.

3.3.11 Remark: What the Rational Isomorphism Does Not Say Three things are worth separating.

First, theorem 3.3.8 is an isomorphism only after tensoring with $\mathbb{Q}$. Example 3.3.10 exhibits a space where the integral map is not surjective, so the rational hypothesis is not an artifact of the proof.

Second, the theorem says nothing about torsion. By corollary 3.3.9 the whole torsion subgroup of $K^0(X)$ lies in the kernel of ch , so any information carried by torsion classes is invisible to cohomology through this map. For $X = \mathbb{RP}^2$, for instance, $\tilde{K}^0(\mathbb{RP}^2)$ is a nonzero finite group while $\tilde{H}^*(\mathbb{RP}^2; \mathbb{Q}) = 0$, and ch is identically zero on reduced classes.

Third, the failure of integrality is measured rather than repaired. Section 3.4 bounds the denominators that can occur, ?? measures the failure intrinsically through the γ -filtration, whose associated graded is compared to H^{2k} by the Adams operations, and ?? corrects for it with the Todd genus in the Riemann-Roch formula. The half-integer $\frac{1}{2}$ in example 3.3.10 is the first term of that correction.

Exercise 3.3.1

1. The Euler sequence gives $T\mathbb{CP}^n \oplus \mathbb{C} \cong (H^\vee)^{\oplus(n+1)}$. Use theorem 3.3.5 and proposition 3.3.4 to show $\text{ch}(T\mathbb{CP}^n) = (n+1)e^x - 1$ with $x = c_1(H^\vee)$, and write out the terms through degree 6. Check that $\text{ch}_0(T\mathbb{CP}^n) = n$ and that $\text{ch}_1(T\mathbb{CP}^n)$ agrees with the first Chern class computed in exercise 3.2.1 (2).
2. From lemma 3.3.1, derive $\text{ch}_1 = c_1$, $\text{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2)$ and $\text{ch}_3 = \frac{1}{6}(c_1^3 - 3c_1c_2 + 3c_3)$. Deduce that $k! \text{ch}_k(E)$ is an integral class for every bundle E , and exhibit a space and a line bundle for which the denominator of ch_k is exactly $k!$.
3. Using theorem 3.3.8, compute $\text{rk}_{\mathbb{Z}} K^0$ and $\text{rk}_{\mathbb{Z}} K^{-1}$ for $S^2 \times S^4$, for $\mathbb{CP}^n \times \mathbb{CP}^m$, and for a

closed orientable surface Σ_g of genus g . Reconcile the first with what theorem 2.4.6 gives directly.

4. Show that ch carries $\tilde{K}^0(S^{2k})$ onto the integral lattice $\tilde{H}^{2k}(S^{2k}; \mathbb{Z})$, but that $\text{ch}(K^0(\mathbb{CP}^2))$ does not contain $H^{\text{even}}(\mathbb{CP}^2; \mathbb{Z})$. Determine the smallest n for which the cohomology generator $x \in H^2(\mathbb{CP}^n; \mathbb{Z})$ fails to lie in $\text{ch}(K^0(\mathbb{CP}^n))$, and say what goes right for smaller n .
5. Prove that $\ker(\text{ch}: K^0(X) \rightarrow H^{\text{even}}(X; \mathbb{Q}))$ is the torsion subgroup, for X a finite CW complex. Deduce that if $H^{\text{odd}}(X; \mathbb{Q}) = 0$ then $K^{-1}(X)$ is a finite group. Then give a space for which ch is *not* injective on K^0 , and identify the classes it kills.
6. Using proposition 3.2.3(2) and $c_1(L^\vee) = -c_1(L)$, prove that $\text{ch}_k(E^\vee) = (-1)^k \text{ch}_k(E)$ for every complex bundle E over a finite CW complex. Deduce that the degree-2 component of $\text{ch}(E \otimes E^\vee)$ vanishes, and say what that means about $c_1(E \otimes E^\vee)$.
7. From proposition 3.3.6, show that $\text{ch}(\beta)$ generates $H^2(S^2; \mathbb{Z})$ and that ch carries external multiplication by β to the double suspension isomorphism. Use this to recompute ch on $\tilde{K}^0(S^{2k})$ without appealing to corollary 3.3.7, and state which sign your answer carries under this book's convention for H .

3.4 Integrality and the Image of the Chern Character

Theorem 3.3.8 identifies $K^0(X) \otimes \mathbb{Q}$ with $H^{\text{even}}(X; \mathbb{Q})$, and example 3.3.10 shows the identification does not descend to the integral lattices. That failure is the subject here. It is not a defect of the construction: ch is built out of the polynomials s_k of lemma 3.3.1 divided by $k!$, so denominators are present from the definition, and the question is which ones actually occur. The answer bounds them by $n!$ on a complex of dimension $2n$, exhibits a space where that bound is attained, and isolates the first degree in which integrality can fail.

The starting point is that although ch_k is a rational combination of Chern classes, its leading term is controlled: modulo products of lower classes, only one denominator survives.

Proposition 3.4.1: The Leading Term of the Chern Character

Let $E \rightarrow X$ be a complex vector bundle over a finite CW complex and let $k \geq 1$. Then

$$\text{ch}_k(E) \equiv \frac{(-1)^{k-1}}{(k-1)!} c_k(E)$$

modulo the $\mathbb{Q}$ -subspace of $H^{2k}(X; \mathbb{Q})$ spanned by products $c_{i_1}(E) \cdots c_{i_r}(E)$ with $r \geq 2$, each $i_j \geq 1$, and $i_1 + \cdots + i_r = k$. (The span must be taken over $\mathbb{Q}$, not over $\mathbb{Z}$: the error term carries the denominator $k!$, as ch_2 on a bundle with $c_2 = 0$ already shows.) In particular $\text{ch}_1(E) = c_1(E)$ and $\text{ch}_2(E) = \frac{1}{2}c_1(E)^2 - c_2(E)$.

Proof:

By definition 3.3.3, $\text{ch}_k(E) = \frac{1}{k!} s_k(c_1(E), \dots, c_k(E))$ with s_k the integral polynomial of lemma 3.3.1 expressing the k -th power sum p_k in the elementary symmetric functions. Solving Newton's

identity (3.1) for p_k , whose coefficient there is $(-1)^{k-1}\sigma_0 = \pm 1$ and in particular a unit, gives

$$p_k = \sigma_1 p_{k-1} - \sigma_2 p_{k-2} + \cdots + (-1)^k \sigma_{k-1} p_1 + (-1)^{k-1} k \sigma_k$$

Every term but the last carries a factor σ_i with $1 \leq i \leq k-1$, and p_{k-i} is itself an integral polynomial in $\sigma_1, \dots, \sigma_{k-i}$ by lemma 3.3.1, so every such term is a $\mathbb{Z}$ -linear combination of monomials $\sigma_{i_1} \cdots \sigma_{i_r}$ with $r \geq 2$. Hence $s_k = (-1)^{k-1} k y_k + (\text{decomposable monomials})$. Substituting $c_i(E)$ for y_i and dividing by $k!$,

$$\text{ch}_k(E) = \frac{(-1)^{k-1} k c_k(E)}{k!} + (\text{decomposables}) = \frac{(-1)^{k-1}}{(k-1)!} c_k(E) + (\text{decomposables})$$

For $k = 1$ the decomposable part is empty and $s_1 = y_1$, giving $\text{ch}_1(E) = c_1(E)$; for $k = 2$, $s_2 = y_1^2 - 2y_2$ gives $\text{ch}_2(E) = \frac{1}{2}(c_1(E)^2 - 2c_2(E)) = \frac{1}{2}c_1(E)^2 - c_2(E)$, as we sought to show.

The proposition says the Chern character and the Chern classes carry the same information one degree at a time, but with a denominator that grows: in degree $2k$ the comparison costs a factor $(k-1)!$. Since $k! \text{ch}_k(E) = s_k(c_1(E), \dots, c_k(E))$ is an integral class outright, the denominators are bounded by $k!$, and on a finite complex only finitely many degrees occur. That is the content of the next theorem, which also extends the statement from bundles to arbitrary K-theory classes.

Theorem 3.4.2: The Denominators of the Chern Character are Bounded

Let X be a finite CW complex with $H^{2k}(X; \mathbb{Q}) = 0$ for all $k > n$. Then for every $\xi \in K^0(X)$ the class $n! \cdot \text{ch}(\xi)$ lies in the image of the coefficient map $H^{\text{even}}(X; \mathbb{Z}) \rightarrow H^{\text{even}}(X; \mathbb{Q})$. The same holds with n replaced by any $m \geq n$.

Proof:

Step 1: Bundles. Let $E \rightarrow X$ be a complex vector bundle and $1 \leq k \leq n$. By definition 3.3.3, $k! \text{ch}_k(E) = s_k(c_1(E), \dots, c_k(E))$, and s_k has integer coefficients by lemma 3.3.1, so $k! \text{ch}_k(E)$ is the image of an integral class, namely the same polynomial evaluated on the integral Chern classes of theorem 3.2.5. Since $k \leq n$, the integer $n!/k!$ is a positive integer, and

$$n! \text{ch}_k(E) = \frac{n!}{k!} \cdot k! \text{ch}_k(E)$$

is again the image of an integral class. For $k > n$ the group $H^{2k}(X; \mathbb{Q})$ vanishes, so $\text{ch}_k(E) = 0$ there and contributes nothing. Summing over $0 \leq k \leq n$, and noting $\text{ch}_0(E) = \text{rk } E$ is an integer, gives that $n! \text{ch}(E)$ is integral.

Step 2: Arbitrary classes. Every $\xi \in K^0(X)$ is a difference $[E] - [F]$ of bundle classes by the Grothendieck construction of definition 2.1.1, and ch is additive by theorem 3.3.5(1), so $n! \text{ch}(\xi) = n! \text{ch}(E) - n! \text{ch}(F)$ is a difference of two integral classes, hence integral. The final clause holds because $k!$ divides $m!$ for every $k \leq n \leq m$, completing the proof.

Two remarks on the shape of this statement. It is a bound on denominators, not a vanishing statement: it does not say ch is integral after any smaller multiple, and example 3.4.6 shows $n!$ cannot be improved in general. And it is stated over the image of integral cohomology rather than over $H^{\text{even}}(X; \mathbb{Z})$ itself, because ch takes values in rational cohomology, so “integral” can only mean

lying in the image of the coefficient map.

With the denominators bounded, the image of ch becomes a lattice that can be compared with the integral one, and the comparison is by a single finite number.

Definition 3.4.3: Integrality Defect

Let X be a finite CW complex. Write $V = H^{\text{even}}(X; \mathbb{Q})$, let

$$L(X) = \text{ch}(K^0(X)) \subseteq V, \quad L_0(X) = \text{im}(H^{\text{even}}(X; \mathbb{Z}) \rightarrow V) \subseteq V$$

The *integrality defect* of X is the index

$$\delta(X) = [L(X) + L_0(X) : L_0(X)]$$

which proposition 3.4.4 shows is finite. It equals 1 exactly when $L(X) \subseteq L_0(X)$, that is, exactly when the Chern character of every class in $K^0(X)$ is integral.

Proposition 3.4.4: The Image of the Chern Character is a Full-Rank Lattice

Let X be a finite CW complex. Then $L(X)$ and $L_0(X)$ are finitely generated free subgroups of $V = H^{\text{even}}(X; \mathbb{Q})$, both of rank $\dim_{\mathbb{Q}} V$, and $\delta(X)$ is finite.

Proof:

X is a finite complex, so $H^{\text{even}}(X; \mathbb{Z})$ is finitely generated and $L_0(X)$, being its image in a $\mathbb{Q}$ -vector space, is finitely generated and torsion-free, hence free. Its rank is $\dim_{\mathbb{Q}} V$ because $H^{\text{even}}(X; \mathbb{Z}) \otimes \mathbb{Q} \cong V$. For $L(X)$, choose n with $H^{2k}(X; \mathbb{Q}) = 0$ for $k > n$, which is possible since X is finite dimensional. By theorem 3.4.2, $L(X) \subseteq \frac{1}{n!} L_0(X)$, a free abelian group of rank $\dim_{\mathbb{Q}} V$; a subgroup of a free abelian group of finite rank is free of no larger rank, so $L(X)$ is free of rank at most $\dim_{\mathbb{Q}} V$. It has rank exactly $\dim_{\mathbb{Q}} V$ because theorem 3.3.8 makes $\text{ch} \otimes \mathbb{Q}$ surjective, so $L(X)$ spans V over $\mathbb{Q}$. (This route avoids any appeal to finite generation of $K^0(X)$.)

Both lattices have rank equal to $\dim_{\mathbb{Q}} V$, so $L(X) + L_0(X)$ is a finitely generated subgroup of V containing $L_0(X)$ with the same rank. A containment of free abelian groups of equal finite rank has finite index, so $\delta(X) < \infty$, as we sought to show.

The defect is a single integer measuring how far the Chern character is from carrying $K^0(X)$ into integral cohomology. It is finite by the proposition and bounded by the denominators of theorem 3.4.2, and it is a homotopy invariant, since ch is natural (theorem 3.3.5(3)) and both K^0 and H^* are homotopy functors. The two extremes are worth seeing side by side.

Example 3.4.5: The Defect of an Even Sphere

Let $X = S^{2k}$ with $k \geq 1$. Then $H^{\text{even}}(S^{2k}; \mathbb{Q})$ has basis the classes 1 and a generator u of H^{2k} , and $K^0(S^{2k}) \cong \mathbb{Z} \oplus \tilde{K}^0(S^{2k}) \cong \mathbb{Z} \oplus \mathbb{Z}$. By corollary 3.3.7 the Chern character carries $\tilde{K}^0(S^{2k})$ isomorphically onto the integral lattice $\tilde{H}^{2k}(S^{2k}; \mathbb{Z})$, and on the rank summand it is the identity $\text{ch}_0 = \text{rk}$. Hence $L(S^{2k}) = L_0(S^{2k})$ and

$$\delta(S^{2k}) = 1$$

Every K-theory class on an even sphere has integral Chern character. What proposition 3.4.1 contributes is a reduction, not the answer: on S^{2k} every decomposable product $c_{i_1} \cdots c_{i_r}$ with $r \geq 2$ and each $i_j \geq 1$ vanishes, since each factor lies in a group $H^{2i_j}(S^{2k}; \mathbb{Z})$ with $0 < i_j < k$, which is zero. So the leading term is exact, $\text{ch}_k(E) = (-1)^{k-1} c_k(E)/(k-1)!$, and integrality is equivalent to the divisibility $(k-1)! | c_k(E)$. That divisibility is not formal, and for $k \geq 3$ it is a genuine constraint on which bundles exist over S^{2k} ; what supplies it here is corollary 3.3.7, which rests on Bott periodicity.

The projective spaces behave in the opposite way, and they show that the bound of theorem 3.4.2 is attained.

Example 3.4.6: The Defect of $\mathbb{CP}^n$

Let $x = c_1(H^\vee)$ be the generator of $H^2(\mathbb{CP}^n; \mathbb{Z})$ used in theorem 2.4.3, so $H^{\text{even}}(\mathbb{CP}^n; \mathbb{Z}) = \mathbb{Z}[x]/(x^{n+1})$ with basis $1, x, \dots, x^n$, and $K^0(\mathbb{CP}^n) = \mathbb{Z}[t]/(t^{n+1})$ with basis $1, t, \dots, t^n$ for $t = [H] - 1$. Since $c_1(H) = -x$, proposition 3.3.4 gives

$$\text{ch}(t) = e^{-x} - 1 = -x + \frac{x^2}{2} - \frac{x^3}{6} + \cdots + \frac{(-1)^n x^n}{n!}$$

and $\text{ch}(t^m) = (e^{-x} - 1)^m$ by theorem 3.3.5(2).

The two lattices have the same covolume. The lowest-degree term of $(e^{-x} - 1)^m$ is $(-1)^m x^m$, so the matrix of ch from the basis $1, t, \dots, t^n$ to the basis $1, x, \dots, x^n$ is triangular with diagonal entries $(-1)^m$ for $m = 0, \dots, n$. Its determinant is $(-1)^{0+1+\cdots+n} = (-1)^{n(n+1)/2}$, a unit. So $L(\mathbb{CP}^n)$ and $L_0(\mathbb{CP}^n)$ are full-rank lattices of equal covolume, and neither need contain the other. For $n = 2$ this is the matrix computed in example 3.3.10.

The defect is divisible by $n!$. Work in the finite group $V/L_0(\mathbb{CP}^n)$, which is $\bigoplus_{k=0}^n (\mathbb{Q}/\mathbb{Z})x^k$. The class of $\text{ch}(t)$ there has x -component $-1 \equiv 0$ and x^k -component $(-1)^k/k!$ for $2 \leq k \leq n$, so its order is $\text{lcm}(2!, 3!, \dots, n!) = n!$, since $k!$ divides $n!$ for every $k \leq n$. That order divides $\delta(\mathbb{CP}^n)$, so

$$n! \mid \delta(\mathbb{CP}^n)$$

In particular $\text{ch}(t)$ has x^n -coefficient $(-1)^n/n!$, so no integer smaller than $n!$ clears the denominators of every class: the bound in theorem 3.4.2 is sharp.

The divisibility is strict in general. Equality $\delta(\mathbb{CP}^n) = n!$ holds for $n \leq 3$ but fails after that: a direct computation gives $\delta(\mathbb{CP}^4) = 72$ against $4! = 24$, the extra factor of 3 coming from the x^4 -coefficient $\frac{7}{12}$ of $\text{ch}(t^2)$. So $n!$ is a lower bound for the defect and an upper bound for the denominators, and the two need not agree.

Small cases. For $n = 1$, $\text{ch}(t) = -x$ and $\delta(\mathbb{CP}^1) = 1$, consistent with $\mathbb{CP}^1 = S^2$ and example 3.4.5. For $n = 2$, $\text{ch}(t) = -x + x^2/2$ and $\text{ch}(t^2) = x^2$, so $L = \mathbb{Z}\langle 1, -x + \frac{1}{2}x^2, x^2 \rangle$. Modulo L_0 the generator $\text{ch}(t)$ has order 2 and the others are integral, giving $\delta(\mathbb{CP}^2) = 2$. This is the “single factor of 2” recorded in example 3.3.10.

The contrast between the two examples is not about the size of the space but about the shape of its cohomology ring. An even sphere has nothing to multiply; $\mathbb{CP}^n$ has a generator in degree 2 whose powers fill every even degree, and it is exactly the powers of a degree-2 class that proposition 3.3.4 divides by $k!$. The next proposition makes the mildest form of this precise: when there are no degrees to fill, the defect vanishes.

Proposition 3.4.7: Integrality in Low Cohomological Dimension

Let X be a finite CW complex with $H^{2k}(X; \mathbb{Q}) = 0$ for all $k \geq 2$. Then $\delta(X) = 1$: the Chern character of every class in $K^0(X)$ is integral. Explicitly, writing $\xi = [E] - [F]$,

$$\text{ch}(\xi) = (\text{rk} E - \text{rk} F) + (c_1(E) - c_1(F))$$

both terms being images of integral classes.

Proof:

Let $\xi \in K^0(X)$. Its Chern character decomposes as $\text{ch}(\xi) = \sum_{k \geq 0} \text{ch}_k(\xi)$ with $\text{ch}_k(\xi) \in H^{2k}(X; \mathbb{Q})$ by definition 3.3.3. By hypothesis $H^{2k}(X; \mathbb{Q}) = 0$ for $k \geq 2$, so every term with $k \geq 2$ vanishes and $\text{ch}(\xi) = \text{ch}_0(\xi) + \text{ch}_1(\xi)$. Writing $\xi = [E] - [F]$, the degree-zero term is $\text{rk} E - \text{rk} F \in \mathbb{Z}$, and by proposition 3.4.1 the degree-two term is $c_1(E) - c_1(F)$, the image of an integral class under $H^2(X; \mathbb{Z}) \rightarrow H^2(X; \mathbb{Q})$. Hence $\text{ch}(\xi) \in L_0(X)$ for every ξ , so $L(X) \subseteq L_0(X)$ and $\delta(X) = 1$, completing the proof.

Beyond that range the first obstruction appears in degree four, where proposition 3.4.1 gives $\text{ch}_2 = \frac{1}{2}c_1^2 - c_2$ and the only possible denominator is a single factor of 2. It is worth recording exactly when that factor is real, since this is the case that occurs in practice.

Corollary 3.4.8: The Degree-Four Criterion

Let X be a finite CW complex with $H^*(X; \mathbb{Z})$ torsion-free, and let $E \rightarrow X$ be a complex vector bundle. Then $\text{ch}_2(E)$ is integral if and only if

$$c_1(E)^2 \in 2H^4(X; \mathbb{Z})$$

Proof:

Since $H^*(X; \mathbb{Z})$ is torsion-free, the coefficient map $H^4(X; \mathbb{Z}) \rightarrow H^4(X; \mathbb{Q})$ is injective and its image is L_0 in degree four, so “integral” means lying in that image and no torsion ambiguity arises. By proposition 3.4.1, $2\text{ch}_2(E) = c_1(E)^2 - 2c_2(E)$. If $\text{ch}_2(E)$ is integral then $c_1(E)^2 = 2\text{ch}_2(E) + 2c_2(E) \in 2H^4(X; \mathbb{Z})$. Conversely if $c_1(E)^2 = 2\alpha$ for some $\alpha \in H^4(X; \mathbb{Z})$ then $\text{ch}_2(E) = \alpha - c_2(E)$ is integral, as we sought to show.

The criterion separates the two examples above at a glance. On S^4 the group H^2 vanishes, so $c_1(E) = 0$ and the condition holds trivially, recovering example 3.4.5 in the first degree where it could fail. On $\mathbb{CP}^2$ the tautological bundle H has $c_1(H) = -x$ and $c_1(H)^2 = x^2$, which generates $H^4(\mathbb{CP}^2; \mathbb{Z}) \cong \mathbb{Z}$ and so is not twice anything. This is the failure computed in example 3.3.10, now identified as the only way integrality can break in degree four.

Exercise 3.4.1

1. Show that the congruence in proposition 3.4.1 genuinely needs the $\mathbb{Q}$ -span and would be false over the $\mathbb{Z}$ -span: exhibit a space X , a bundle E and a degree k for which $\text{ch}_k(E) - \frac{(-1)^{k-1}}{(k-1)!} c_k(E)$ is not a $\mathbb{Z}$ -linear combination of products of Chern classes. (Try $k = 2$ on $\mathbb{CP}^2$ with a bundle

whose c_2 vanishes.)

2. Compute the integrality defect $\delta(\mathbb{CP}^3)$. (Expand $\text{ch}(t^m)$ for $m = 1, 2, 3$ modulo x^4 , identify the lattice $L(\mathbb{CP}^3)$, and compute the order of its image in V/L_0 .) Compare your answer with the divisibility $n!|\delta(\mathbb{CP}^n)$ established in example 3.4.6.
3. Let X be a finite CW complex with $H^*(X; \mathbb{Z})$ torsion-free and let $L \rightarrow X$ be a complex line bundle. Prove that $\text{ch}(L)$ is integral if and only if $c_1(L)^k \in k! H^{2k}(X; \mathbb{Z})$ for every $k \geq 1$. Deduce that for $n \geq 2$ the tautological bundle $H \rightarrow \mathbb{CP}^n$ fails this at $k = 2$, and say why $n = 1$ is different.
4. Show that $\delta(S^2 \times S^2) = 1$. (Use theorem 3.3.5(2) and corollary 3.3.7 to compute ch on a basis of $K^0(S^2 \times S^2)$, then compare with the basis $1, a, b, ab$ of $H^{\text{even}}(S^2 \times S^2; \mathbb{Z})$.) Contrast this with $\mathbb{CP}^2$: both are closed simply connected 4-manifolds with cohomology in even degrees only, yet the defect is 1 here and 2 there. Explain the difference in terms of how each degree-4 generator is built from degree-2 classes.
5. Prove that the integrality defect is a homotopy invariant: if $f: X \rightarrow Y$ is a homotopy equivalence of finite CW complexes then $\delta(X) = \delta(Y)$. Then show that $\delta(X) = 1$ for every finite wedge of even-dimensional spheres, using lemma 2.2.2.

3.5 Characteristic Numbers and Cobordism

The classes of section 3.2 are cohomology classes, and comparing two bundles means comparing classes in a group that depends on the base. On a closed manifold there is a way to compress that data into integers: evaluate a product of characteristic classes of the tangent bundle against the fundamental class. The resulting numbers are invariants not merely of the diffeomorphism type but of the far coarser relation of cobordism, and Thom's theorems say they are close to a complete set of invariants for it.

Recall that a closed smooth m -manifold M has a fundamental class $[M] \in H_m(M; \mathbb{Z}/2\mathbb{Z})$ always, and $[M] \in H_m(M; \mathbb{Z})$ when M is oriented ([13, Cohomology, Fundamental Class]). A partition of an integer m is a finite unordered collection $I = (i_1, \dots, i_r)$ of integers $i_j \geq 1$ with $i_1 + \dots + i_r = m$; for partitions into the relevant degrees one evaluates the corresponding monomial in characteristic classes on $[M]$.

Definition 3.5.1: Stiefel-Whitney Number

Let M be a closed smooth m -manifold with tangent bundle TM and fundamental class $[M] \in H_m(M; \mathbb{Z}/2\mathbb{Z})$. For a partition $I = (i_1, \dots, i_r)$ of m , the associated *Stiefel-Whitney number* is

$$w_I[M] = \langle w_{i_1}(TM) \smile \dots \smile w_{i_r}(TM), [M] \rangle \in \mathbb{Z}/2\mathbb{Z}$$

using the Stiefel-Whitney classes of definition 3.1.2. Only partitions I of m give possibly-nonzero numbers, since the relevant cup product must land in $H^m(M; \mathbb{Z}/2\mathbb{Z})$.

Definition 3.5.2: Pontryagin Number

Let M be a closed oriented smooth manifold of dimension $m = 4k$ with tangent bundle TM and fundamental class $[M] \in H_m(M; \mathbb{Z})$. For a partition $I = (i_1, \dots, i_r)$ of k , the associated *Pontryagin number* is

$$p_I[M] = \langle p_{i_1}(TM) \smile \dots \smile p_{i_r}(TM), [M] \rangle \in \mathbb{Z}$$

using the Pontryagin classes of definition 3.1.3. Since $p_{i_j}(TM) \in H^{4i_j}(M; \mathbb{Z})$, the cup product lands in $H^{4k}(M; \mathbb{Z}) = H^m(M; \mathbb{Z})$ exactly when I partitions k ; Pontryagin numbers are defined only when $\dim M \equiv 0 \pmod{4}$.

The Pontryagin and Stiefel-Whitney numbers are diffeomorphism invariants of M , being built functorially from the tangent bundle and the fundamental class. Their significance is that they are unchanged under the much coarser equivalence of cobordism: two closed manifolds are unoriented-cobordant if their disjoint union bounds a compact manifold with boundary, and oriented-cobordant if $M \sqcup (-N)$ bounds an oriented one. Characteristic numbers are cobordism invariants because the relevant classes extend over the bounding manifold while $[M] = \partial[W, \partial W]$, which forces the pairing to vanish by exactness. The argument is given inside the proof of theorem 3.5.4 below.

Example 3.5.3: Pontryagin Number of $\mathbb{CP}^2$

The manifold $\mathbb{CP}^2$ is closed, oriented, of real dimension $4 = 4 \cdot 1$, so it has a single Pontryagin number, indexed by the partition $I = (1)$ of $k = 1$. From example 3.2.15, $p_1(T\mathbb{CP}^2) = 3a^2$ with $\langle a^2, [\mathbb{CP}^2] \rangle = 1$. Therefore

$$p_1[\mathbb{CP}^2] = \langle p_1(T\mathbb{CP}^2), [\mathbb{CP}^2] \rangle = \langle 3a^2, [\mathbb{CP}^2] \rangle = 3$$

This is consistent with the Hirzebruch signature theorem ([12]), which in dimension 4 reads $\sigma(M) = \frac{1}{3}p_1[M]$: the signature of $\mathbb{CP}^2$ is $\sigma(\mathbb{CP}^2) = 1$ (its intersection form on H^2 is the rank-one form $\langle a, a \rangle = 1$), and indeed $\frac{1}{3} \cdot 3 = 1$. Since $p_1[\mathbb{CP}^2] = 3 \neq 0$, the manifold $\mathbb{CP}^2$ is not the boundary of any compact oriented 5-manifold: it represents a nonzero class in the oriented cobordism group $\Omega_4^{SO} \cong \mathbb{Z}$.

The two systems of characteristic numbers control the two basic cobordism theories. Thom's theorems make this precise: unoriented cobordism is detected completely by Stiefel-Whitney numbers, and oriented cobordism is detected rationally by Pontryagin (and Stiefel-Whitney) numbers.

Theorem 3.5.4: Characteristic Numbers Determine Cobordism

Let M, N be closed smooth manifolds of the same dimension.

1. (Unoriented cobordism, Thom) M and N are cobordant (over the unoriented cobordism relation) if and only if they have the same Stiefel-Whitney numbers $w_I[M] = w_I[N]$ for every partition I . In particular, M bounds a compact manifold if and only if all its Stiefel-Whitney numbers vanish. The unoriented cobordism ring $\mathfrak{N}_*$ is a polynomial algebra over $\mathbb{Z}/2\mathbb{Z}$.
2. (Oriented cobordism, Thom) If M and N are oriented and all their Pontryagin numbers and Stiefel-Whitney numbers agree, then M and N are cobordant in the rational oriented cobordism group; equivalently, some disjoint union of copies of M is oriented-cobordant to the corresponding union of copies of N . The oriented cobordism ring tensored with $\mathbb{Q}$ is the polynomial algebra $\Omega_*^{SO} \otimes \mathbb{Q} \cong \mathbb{Q}[[\mathbb{CP}^2], [\mathbb{CP}^4], \dots]$ generated by the even complex projective spaces, and Pontryagin numbers give a complete set of rational invariants.

Proof:

The forward implication is proved here; the converse is cited, for the reason stated at the end.

Step 1: A manifold that bounds has vanishing characteristic numbers. Suppose W is a compact manifold with $\partial W = M$, and suppose W is oriented with the induced orientation on M whenever Pontryagin numbers are in play (the unoriented half needs no orientation). Along the boundary the tangent bundle of W splits off the trivial line spanned by the inward normal, $\iota^*TW = TW|_M \cong TM \oplus \mathbb{R}$ for $\iota: M \hookrightarrow W$. Two inputs combine. By naturality (theorem 3.2.4(1)) the classes of ι^*TW are ι^* of the classes of TW ; and by the Whitney formula (theorem 3.2.4(2)) with $w(\mathbb{R}) = 1$, the classes of $TM \oplus \mathbb{R}$ are those of TM . Hence $w_i(TM) = \iota^*w_i(TW)$ for every i .

The Pontryagin case needs the same two inputs but the exact form of stability, which theorem 3.2.13 supplies only modulo 2-torsion. The exact statement is available directly: the trivial real bundle complexifies to the trivial complex bundle, $c(\mathbb{C}^k) = 1$, and complexification commutes with direct sums, so by theorem 3.2.5(2)

$$c((E \oplus \mathbb{R}^k) \otimes \mathbb{C}) = c(E \otimes \mathbb{C}) \smile c(\mathbb{C}^k) = c(E \otimes \mathbb{C})$$

whence $p_i(E \oplus \mathbb{R}^k) = p_i(E)$ exactly, by definition 3.1.3. Combined with naturality of the Chern classes this gives $p_i(TM) = \iota^*p_i(TW)$.

Step 2: The pairing vanishes. Write $x = w_{i_1}(TW) \smile \dots \smile w_{i_r}(TW) \in H^m(W)$, so that $w_I(TM) = \iota^*(x)$ by Step 1. The fundamental class $[M]$ is the image of the relative fundamental class $[W, \partial W]$ under the connecting homomorphism $\partial: H_{m+1}(W, \partial W) \rightarrow H_m(\partial W)$, which uses that M is the *whole* boundary of W . By the duality between the two long exact sequences of the pair, the Kronecker pairing satisfies $\langle \iota^*(x), \partial z \rangle = \pm \langle \delta \iota^*(x), z \rangle$ for $z = [W, \partial W]$, the sign being irrelevant to a vanishing statement. Now $\delta \iota^* = 0$, because

$$H^m(W) \xrightarrow{\iota^*} H^m(\partial W) \xrightarrow{\delta} H^{m+1}(W, \partial W)$$

is exact at the middle term. Therefore $w_I[M] = 0$ for every partition I , and the same argument run with $\mathbb{Z}$ coefficients, legitimate because W was assumed oriented in that case, gives $p_I[M] = 0$.

Step 3: From bounding to cobordant. Characteristic numbers are additive over disjoint unions, since $[M \sqcup N] = [M] + [N]$ under $H_m(M \sqcup N) \cong H_m(M) \oplus H_m(N)$ and the classes restrict componentwise. In the unoriented theory M and N are cobordant exactly when $M \sqcup N$ bounds, so Steps 1 and 2 give $w_I[M] + w_I[N] = 0$, that is $w_I[M] = w_I[N]$ over $\mathbb{Z}/2\mathbb{Z}$. In the oriented theory M and N are cobordant exactly when $M \sqcup (-N)$ bounds an oriented manifold, and reversing orientation negates the fundamental class, so $p_I[-N] = -p_I[N]$ and the same computation gives $p_I[M] = p_I[N]$.

Step 4: The converse is cited. That agreement of characteristic numbers forces cobordism is the substantive half. Its proof identifies the cobordism groups with stable homotopy groups of Thom spectra (MO unoriented, MSO oriented) through the Pontryagin-Thom construction and then computes those homotopy groups. Both are upstream of this book. The unoriented statement is [13, Stiefel-Whitney Numbers Are a Complete Unoriented Cobordism Invariant], resting there on [13, Thom's Theorem]; the oriented statement is [13, Oriented Bordism and the Spectrum MSO], whose rational clause is itself stated with the computation cited to Thom's original paper, since it needs rational-homotopy machinery that volume does not develop. They are cited rather than reproduced, as the de-duplication convention of this series requires. Together with Steps 1 to 3 this gives both directions, completing the proof.

The theorem answers the question opened at the start of the chapter: characteristic classes were introduced as measures of the twisting of a bundle, and characteristic numbers turn that local twisting data into global invariants strong enough to classify manifolds up to the bounding relation. The unoriented statement is a complete invariant; the oriented statement is complete only rationally, because Ω_*^{SO} has torsion (detected by Stiefel-Whitney numbers and by more refined invariants) that Pontryagin numbers cannot see, since they are additive invariants vanishing on torsion classes and so factor through $\Omega_*^{SO} \otimes \mathbb{Q}$. The appearance of $\mathbb{CP}^2, \mathbb{CP}^4, \dots$ as rational generators is why example 3.5.3 matters: $\mathbb{CP}^2$ is the first generator, and its nonzero Pontryagin number $p_1[\mathbb{CP}^2] = 3$ certifies that it is not rationally null-cobordant.

Exercise 3.5.1

1. Let M be a closed oriented 4-manifold all of whose Stiefel-Whitney numbers and Pontryagin numbers vanish. What does theorem 3.5.4 allow you to conclude about M , and what can it not conclude in general? Then show that in dimension 4 the gap closes, and identify what would have to be true in higher dimensions for it to reopen.
2. Compute every Stiefel-Whitney number of $\mathbb{RP}^2$, and deduce that $\mathbb{RP}^2$ does not bound a compact 3-manifold. (Use $T\mathbb{RP}^2 \oplus \underline{\mathbb{R}} \cong (\gamma^1)^{\oplus 3}$ with $w(\gamma^1) = 1 + a$ and $H^*(\mathbb{RP}^2; \mathbb{Z}/2\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}[a]/(a^3)$.)
3. Show that every closed orientable 3-manifold has all Stiefel-Whitney numbers zero, and conclude from theorem 3.5.4(1) that it bounds. Explain why the Pontryagin numbers give no information here.
4. Compute the total Pontryagin class $p(T\mathbb{CP}^3)$ and all Pontryagin numbers of $\mathbb{CP}^3$. Reconcile your answer with the fact that $\mathbb{CP}^3$ has odd complex dimension.